

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: V Tsambalas
Start Date: 11/02/26
Checked by: J Riley
End Date: 12/02/26
Location Meth.: dGPS0.1
Driller: MacGeo
Drill Rig: Comacchio 300
Total Depth: 13.77 m
Hole Diameter: 96-125 mm
Inclination: 90°
Bearing: N/A
Easting: 314472.52
Northing: 6238143.87
Hor. Datum: MGA2020-56
Ver. Datum: AHD
Surface: Fill

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
AD/T	U	SPT: 1.20-1.65 m 10, 11, 9 N=20 D: 1.20-1.65 m	12/02/2026 22:35:37 ▼	0.0	0.0	TOPSOIL - Silty SAND: fine grained, grey; trace rootlets, trace wood fragments.					FILL
		D: 1.65-3.00 m ES: 1.65-3.00 m		-4.0	-4.0	FILL - SAND: fine to medium grained, brown and yellow-brown; trace clay; trace gravel, fine, rounded to sub-rounded, sandstone.					
		SPT: 3.00-3.45 m 4, 5, 6 N=11 D: 3.00-3.45 m		-3.0	-3.0	FILL - SAND: fine to coarse grained, yellow and pale pink, (sandstone); with gravel, fine to coarse grained; with cobbles.					
		D: 4.00-4.50 m ES: 4.00-4.50 m		-2.0	-2.0	FILL - SAND: fine to medium grained, brown; trace clay.			D		
		SPT: 4.50-4.95 m 0, 1, 3 N=4		-1.0	-1.0	SAND: fine to medium grained, yellow-brown; trace clay.	SP				ALLUVIUM
		D: 5.50-5.75 m ES: 5.50-5.75 m		-0.0	-0.0	from 4.00 m: becoming grey			L - MD		
				5.0	5.0	Clayey Silty SAND: fine grained, dark grey; trace gravel, fine, angular (shale).	SC-SM		W	VL	
				-1.0	-1.0	Sandy CLAY: medium to high plasticity, orange-brown and yellow-brown.	CI-CH		w ≈ PI		RESIDUAL SOIL
Log continued on next page.											
				6.0	6.0						
				-2.0	-2.0						
				7.0	7.0						
				-3.0	-3.0						
				8.0	8.0						

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Location Meth.: dGPS0.1
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Hole Diameter: 96-125 mm
Eastings: 314472.52
RL: 4.237 m
Drill Rig: Comacchio 300
Inclination: 90°
Northings: 6238143.87
Ver. Datum: AHD
Total Depth: 13.77 m
Bearing: N/A
Hor. Datum: MGA2020-56
Surface: Fill

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: ○ UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0					SOIL 0-4 VL 2 L 4 M 20 H 60 VH 200 EH	20 60 200 600 2000	
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HQ3	3		Is(50) D=1.4 MPa Is(50) A=1.6 MPa	-4.0			SANDSTONE: medium to coarse grained, red-brown and yellow-brown, indistinctly bedded.	SW	[99]			7.98 m: HB 8.06 m: HB 8.26 m: P, 5°, RF, PR, CN 8.34 m: DB
	4	9.27	Is(50) D=0.45 MPa Is(50) A=0.49 MPa	-5.0				MW - SW	100 (98) [98]			8.84 m: P, 0°, RF, PR, SN Fe 9.00 m: P, 5°, RF, CT Cl 9.27 m: MB 9.55 m: P, 10°, RF, PR, SN Fe
	5	10.80	Is(50) D=0.16 MPa Is(50) A=0.17 MPa	-6.0			from 10.0 m: becoming red-brown and pale grey from 10.12 to 10.22 m: becoming pale grey-white from 10.22 to 10.40 m: pale yellow and red-brown staining from 10.40 m: becoming pale grey with occasional pale yellow and pale red brown staining, medium bedded at 0-20°	HW - MW				10.00 m: MB 10.14-10.17 m: EW, 5°
								MW	100 (98) [80]			10.78 m: HB 10.90 m: P, 5°, RF, PR, SN Fe 10.92-10.95 m: CS, 0-5°, SN Fe 11.00 m: MB 11.05 m: P, 5°, RF, PR, SN Fe, 30 mm spacing, x 2 11.11 m: P, 15°, RF, PR, SN Fe 11.15 m: P, 5°, RF, PR, CN 11.27 m: P, 15°, RF, PR, SN Fe 11.39 m: J, 50°, RF, PR, Fe, Healed 11.48 m: P, 10°, RF, PR, SN Fe, 20 mm spacing, x 2 11.66 m: HB
	6	12.27	Is(50) D=0.68 MPa Is(50) A=0.7 MPa	-7.0			from 11.05 to 11.07 m: ironstone band from 11.16 to 11.21 m: conglomerate band, quartz and ironstone from 11.40 m: trace siltstone rip up clasts					11.97 m: J, 55°, RF, PR, SN Fe 12.03-12.11 m: J, 20-35°, RF, PR, SN Fe, 15 mm spacing, x 4 12.15 m: J, 80-90°, RF, PR, VN Fe, partially healed 12.18-12.31 m: HB, x 5 12.33 m: P, 5°, RF, PR, CN
								HW	100 (100) [97]			12.77 m: P, RF, PR, VN 12.96-13.05 m: HB, x 3 13.22 m: P, 15°, PR
			Is(50) D=0.021 MPa Is(50) A=0.052 MPa	-8.0			from 12.12 m: siltstone rip up clast content increasing from 12.5 m: becoming very thinly bedded at 0-10°, trace thin black carbonaceous laminations	SW - FR				
			Is(50) D=1.6 MPa Is(50) A=1.4 MPa	-9.0								
				-10.0			Terminated at 13.77 m. Target depth.					
				-11.0								
				-12.0								

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

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Driller: MacGeo

Hole Diameter: 110 mm

Easting: 314450.23

RL: 4.89 m

Drill Rig: Comacchio 300

Inclination: 90°

Northing: 6238177.51

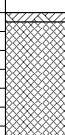
Ver. Datum: AHD

Total Depth: 12.41 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Fill

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AD/T	U	D: 0.05-0.70 m ES: 0.05-0.70 m			0.0			TOPSOIL - Silty SAND: fine to coarse grained, dark brown; trace gravel, fine, angular, igneous, trace rootlets, trace wood fragments. FILL - SAND: fine to medium grained, pale yellow-brown; trace gravel, rounded, sandstone.	D		FILL
		D: 0.70-1.50 m ES: 0.70-1.50 m			4.0		SP	SAND: fine grained, dark brown; trace charcoal, trace wood fragments.	D	VD	ALLUVIUM driller noted increased drilling resistance at 1.20 m, possible boulder/cobble
		SPT: 1.50-1.95 m 8,10.8 N=18 D: 1.50-1.95 m			3.0						
		D: 2.00-2.20 m ES: 2.00-2.20 m			2.0		SP	SAND: fine grained, pale yellow; trace silt; trace gravel, fine to coarse, sandstone.	D	MD	RESIDUAL SOIL
		SPT: 2.20-2.23 m 9/30 mm HB N=R						Log continued on next page.			
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METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • Is(50) UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0							
					4.0							
					1.0							
					3.0							
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							<i>Log continued from previous page.</i>					
	1		Is(50) D=0.49 MPa Is(50) A=0.49 MPa	2.84	2.0		SANDSTONE: medium to coarse grained, pale grey and yellow-brown, very thinly bedded to thickly laminated at 0-5°, trace fine gravel sized quartz inclusions, low strength, iron stained.	MW	100 (100) [86]			HAWKESBURY SANDSTONE 2.37-2.41 m: P, 5°, RF, PR, CN, 40 mm spacing, x 2 2.44 m: J, 45°, RF, PR, CN 2.46 m: P, 5°, RF, PR, CN 2.65 m: P, 5°, RF, PR, CT SC, Fe
	2		Is(50) D=0.45 MPa Is(50) A=0.65 MPa	3.74	3.0		from 2.84 to 3.23 m: pale grey mottled yellow-brown and pale brown		100 (99) [86]			2.84 m: P, 5°, RF, PR, CN 2.98 m: HB
							from 3.32 m: becoming red-brown and pale grey					3.22 m: P, 0-5°, RF, PR, CN, x 3 3.47-3.48 m: EW, 10°, CT CI
					1.0				91 (86) [59]			3.74 m: DB 3.85 m: HB 3.96 m: J, 75°, RF, PR, CT CI, 0.2 m persistence 4.05 m: DB
	3		Is(50) D=0.2 MPa Is(50) A=0.16 MPa	4.86	4.0		NO CORE: 4.05 - 4.15 m.					
							SANDSTONE: fine to medium grained, pale grey and red-brown, very thinly bedded to thickly laminated at 5-10°, iron stained.	XW				4.38 m: P, 10°, RF, PR, SN Fe
							at 4.66m: wood fragments	HW				
							from 4.70 m: becoming pale grey mottled red-brown and yellow-brown	XW				4.73-4.86 m: P, 0-5°, RF, PR, CN, x 2
					5.0			MW	78 (55) [41]			4.98 m: HB 5.08 m: HB
							NO CORE: 5.26 - 5.59 m.					5.25-5.26 m: P, 0°, RF, PR, CN, 10 mm spacing, x 2
	4		Is(50) D=0.61 MPa Is(50) A=0.56 MPa	6.36	6.0		SANDSTONE: fine to medium grained, pale grey mottled yellow-brown, very thinly bedded to thickly laminated at 0-5°.	MW				5.59 m: P, 10°, RF, PR, VN CI 5.66 m: P, 15°, RF, PR, CN 5.74 m: DB 5.87 m: P, 20°, RF, PR, CN 5.94 m: P, 10°, RF, PR, SN Fe 6.00-6.05 m: CS
							SILTSTONE: grey and pale grey, indistinct bedding, multiple irregular healed defects, ironstone inclusions.	HW				6.05-6.12 m: EW, 55°, gravelly clay 6.16 m: P, 35°, RF, PR, SN Fe 6.16-6.29 m: J, 30-90°, RF, PR, CT CI, SN Fe, IR, partially healed 6.30-6.34 m: EW, 10-25°, CI 6.51 m: HB
	5		Is(50) D=0.063 MPa Is(50) A=0.079 MPa	6.87	7.0		NO CORE: 6.87 - 7.00 m.		100 (100) [0]			6.87 m: P, 5°, RF, PR, SN Fe 7.00-7.02 m: EW, 5°, CI 7.10-7.16 m: P, 0-5°, RF, PR, SN Fe, x 2
							SANDSTONE: fine to coarse grained, yellow-brown, pale brown and pale grey, indistinctly bedded, with fine gravel quartz inclusions, iron stained.	HW	87 (85) [71]			7.45 m: P, 0-10°, RF, PR, VN Fe
	6		Is(50) D=0.51 MPa Is(50) A=1.5 MPa C: 7.65-7.85 m UCS=35 MPa	7.87	8.0		from 7.03 to 7.06 m: ironstone	MW				7.66 m: P, 5°, RF, PR, CN 7.86 m: J, 15°, RF, PR, CN
							from 7.80 m: trace quartz inclusions		100			

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Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS(50) A: • IS(50) UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	7		Is(50) D=1.1 MPa Is(50) A=0.91 MPa	4.0	9.0		SANDSTONE: medium to coarse grained, pale grey, very thickly bedded at 0-10°, iron stained .	MW	(100) [100]	VL L M H VH EH	20 60 200 600 2000	8.04 m: HB 8.18 m: P, 10°, RF, PR, SN Fe 8.35 m: P, 5-10°, RF, PR, CN 8.58 m: DB 8.60 m: P, 10°, RF, PR, CN 9.00 m: MB
	8		Is(50) D=2.5 MPa Is(50) A=2.5 MPa	5.0	10.0		NO CORE: 9.28 - 9.38 m. SANDSTONE: medium to coarse grained, pale grey, iron stained .	SW	94 (88) [88]			9.63-9.67 m: J, 40-90°, RF, PR, VN Fe, CU 9.92 m: P, 0°, RF, PR, VN Cl
			Is(50) D=0.75 MPa Is(50) A=0.58 MPa	6.0	11.0		from 10.18 to 10.22 m: ironstone from 10.27 m: no iron staining, trace black flecks	FR	100 (100) [100]			10.18 m: P, 0-5°, RF, PR, CT Fe 10.22 m: P, 0°, RF, PR, VN Fe 10.87 m: DB
	9		Is(50) D=1.1 MPa Is(50) A=1.1 MPa	7.0	12.0		from 11.75 to 11.77 m: becoming coarse grained, with gravel inclusions, fine to coarse grained, siltstone and quartz					11.82 m: HB
			Is(50) D=1.1 MPa Is(50) A=0.91 MPa	8.0	13.0		Terminated at 12.41 m. Target depth.					

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: V Tsambalas
Start Date: 16/02/26

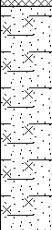
Checked by: J Riley
End Date: 16/02/26
Location Meth.: dGPS0.1

Driller: MacGeo
Drill Rig: Comacchio 300
Total Depth: 13.93 m

Hole Diameter: 110 mm
Inclination: 90°
Bearing: N/A

Easting: 314424.04
Northing: 6238213.60
Hor. Datum: MGA2020-56

RL: 3.248 m
Ver. Datum: AHD
Surface: Pavement

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)		
AD/T	U	D: 0.15-0.75 m ES: 0.15-0.75 m	16/02/2026 21:56:48 ▼	0.0	0.0			TOPSOIL - Silty SAND: fine to coarse grained, dark brown; with gravel, fine, angular, igneous; trace wood, trace plastic, trace root fibres.	D		TOPSOIL		
		-3.0					FILL - SAND: fine to medium grained, pale brown; trace gravel, fine, rounded, sandstone; trace silt.			FILL			
		D: 0.75-1.50 m ES: 0.75-1.50 m			1.0			SM	SAND: fine to medium grained, dark brown.	M	L	ALLUVIUM	
		SPT: 1.50-1.95 m 2.2,1 N=3 D: 1.50-1.95 m		-2.0									
		D: 2.00-2.60 m ES: 2.00-2.60 m		2.0				SP	SAND: medium to coarse grained, yellow-brown; trace clay.	W	MD		
		D: 2.60-3.00 m ES: 2.60-3.00 m		-1.0									
		SPT: 3.00-3.45 m 2.2,3 N=5 D: 3.00-3.45 m		3.0				SP	SAND: medium to coarse grained, dark brown; trace gravel, fine, rounded, quartz; fine gravel sized carbonaceous inclusions.	D			
				-0.0									
		SPT: 4.50-4.93 m 3.2,15/130 mm N=R		4.0									
		D: 4.80-4.93 m		-1.0									
				4.0									
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Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Logged by: V Tsambalas

Start Date: 16/02/26

Checked by: J Riley

End Date: 16/02/26

Location Meth.: dGPS0.1

Driller: MacGeo

Hole Diameter: 110 mm

Drill Rig: Comacchio 300

Inclination: 90°

Total Depth: 13.93 m

Bearing: N/A

Easting: 314424.04

Northing: 6238213.60

Hor. Datum: MGA2020-56

RL: 3.248 m

Ver. Datum: AHD

Surface: Pavement

[illegible]

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: V Tsambalas
Start Date: 16/02/26
Checked by: J Riley
End Date: 16/02/26
Location Meth.: dGPS0.1
RL: 3.248 m
Ver. Datum: AHD
Surface: Pavement

Driller: MacGeo
Drill Rig: Comacchio 300
Total Depth: 13.93 m
Hole Diameter: 110 mm
Inclination: 90°
Bearing: N/A

Easting: 314424.04
Northing: 6238213.60
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS(50) A: • IS(50) D: • UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	5		Is(50) A=0.3 MPa Is(50) D=0.15 MPa C: 9.00-9.18 m UCS=2.2 MPa	9.35	-5.0		NO CORE: 7.94 - 8.05 m.	MW	(82) [66]			8.08 m: DB
							SANDSTONE: fine to medium grained, pale grey mottled red-brown and yellow-brown, very thinly bedded at 5-25°, iron stained. from 8.33 to 8.70 m: indistinctly bedded					8.30 m: P, 0-15°, RF, CN
	6		Is(50) A=0.49 MPa Is(50) D=0.25 MPa C: 9.00-9.18 m UCS=2.2 MPa	10.87	-6.0		from 8.72 to 8.79 m: pale grey, black flecks from 8.88 to 8.95 m: pale grey, black flecks	HW MW HW MW				8.72-8.79 m: EW 8.80 m: DB 8.88-8.95 m: EW, 5°, CI 9.00 m: MB
							SILTSTONE: fine grained, pale grey and grey, highly fractured, healed, iron stained.	HW	100 (100) [74]			9.18 m: DB 9.35-9.48 m: SZ 9.57-9.59 m: EW, 0°, CI
HQ3	7		Is(50) A=0.84 MPa Is(50) D=0.42 MPa C: 12.20-12.43 m UCS=20 MPa Is(50) D=1.3 MPa Is(50) A=1.46 MPa	12.43	-7.0		SANDSTONE: fine to medium grained, orange-brown, red-brown and pale grey, thinly bedded at 0-10° from 9.80 to 9.88 m: fine gravel sized siltstone and quartz inclusions at 9.91 m: void, 20 x 50 mm	MW - SW				9.75-9.79 m: EW, 0-10°, GC 9.84 m: J, 20°, RF, SN Fe 9.93 m: P, 0°, RF, VN Fe 10.00 m: MB 10.09 m: P, 10°, RF, CN 10.15 m: HB
							from 10.87 m: pale grey, indistinctly bedded	SW - FR	100 (100) [100]			10.69 m: P, 0°, RF, CN 10.87 m: DB 11.00 m: MB
	8		Is(50) A=0.91 MPa Is(50) D=0.52 MPa C: 12.20-12.43 m UCS=20 MPa Is(50) D=1.3 MPa Is(50) A=1.46 MPa	12.43	-8.0		from 12.43 to 12.82 m: dark grey laminations at 5-10°		100 (99) [99]			12.00 m: MB 12.43 m: P, 10°, RF, CN
							from 12.82 to 13.14 m: gravel inclusions, medium to coarse grained, rounded, siltstone and quartz from 13.41 to 13.51 m: grey laminations at 5-10°, gravel inclusions, medium to coarse grained, rounded, siltstone					13.00 m: MB 13.12 m: HB 13.47 m: P, 5°, RF, CN 13.48 m: P, 0°, RF, CT X 13.50-13.51 m: EW, 0°, CI
HQ3			Is(50) D=0.29 MPa Is(50) A=0.44 MPa		-10.0		Terminated at 13.93 m. Target depth.					

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: V Tsambalas

Start Date: 18/02/26

Checked by: J Riley

End Date: 19/02/26

Location Meth.: dGPS0.1

Driller: MacGeo

Hole Diameter: 96-125 mm

Easting: 314622.20

RL: 26.21 m

Drill Rig: Comacchio 300

Inclination: 90°

Northing: 6237799.80

Ver. Datum: AHD

Total Depth: 10.63 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Pavement

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
DT		CONCC: 0.00-0.15 m D: 0.20-0.25 m			0.0			ASPHALTIC CONCRETE: dark grey; aggregates up to 15 mm, 150 mm thick.			PAVEMENT
	U				26.0			Sandy GRAVEL: medium to coarse grained, angular to sub-angular, dark grey; sand, fine to coarse grained; heavily bound; basecourse.			FILL
AD/T							SW	SAND: fine to coarse grained, yellow-brown; with clay; trace gravel, fine, rounded, sandstone.	D	D	COLLUVIUM
					1.0		SW	SAND: fine to medium grained, pale grey; with gravel, fine to coarse, sub-rounded to sub-angular, sandstone; potentially VL to L sandstone; extremely weathered material.		VD	EXTREMELY WEATHERED MATERIAL
		SPT: 1.16-1.19 m 5/30 mm HB N=R		25.0				Log continued on next page.			
					2.0						
					24.0						
					3.0						
					23.0						
					4.0						
					22.0						
					5.0						
					21.0						
					6.0						
					20.0						
					7.0						
					19.0						
					8.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: V Tsambalas

Start Date: 18/02/26

Checked by: J Riley

End Date: 19/02/26

Location Meth.: dGPS0.1

Driller: MacGeo

Hole Diameter: 96-125 mm

Easting: 314622.20

RL: 26.21 m

Drill Rig: Comacchio 300

Inclination: 90°

Northing: 6237799.80

Ver. Datum: AHD

Total Depth: 10.63 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Pavement

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ Soil: VL L e M 2s H 6s VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0							
					26.0							
					1.0							
					25.0		<i>Log continued from previous page.</i>					
	1				1.78		SANDSTONE: fine to medium grained, pale grey mottled pale yellow-brown, iron stained.	MW - SW	100 (100) [90]			HAWKESBURY SANDSTONE 1.22 m: P, 10°, RF, PR, VN CI
					2.0		from 1.44 to 1.68 m: liesegang banding		99 (99) [99]			1.78 m: MB 1.93 m: P, 15°, RF, PR, CN 2.00 m: MB
	2				24.0		from 2.66 m: becoming pale grey and pale yellow-brown					2.66 m: P, 5-10°, RF, PR, CN 2.89 m: P, 0°, RF, PR, CN
					3.0		NO CORE: 3.10 to 3.12 m.	SW	100 (90) [87]			3.10-3.82 m: DB, x5
					23.0		SANDSTONE: fine to medium grained, pale grey, very thinly bedded at 0-15°.					
					4.0		from 3.26 to 3.73 m: becoming pale yellow-brown and pale red-brown, iron stained					
	3				22.0				100 (83) [76]			4.00 m: MB 4.57-4.71 m: MB, x2 4.71-4.75 m: DB, x2 4.78-5.00 m: HB
					5.0							5.05 m: P, 5°, RF, PR, SN Fe, healed 5.11 m: J, 15°, RF, PR, CN 5.15 m: DB 5.16-5.17 m: EW, 10°, CI 5.22 m: P, 10°, RF, PR, CN
	4				21.0		from 5.54 to 5.65 m: pale yellow-brown, iron stained					5.68-5.69 m: EW, 5-10°, CI
					6.0		from 6.00 m: indistinct bedding					
					20.0				100 (57) [53]			6.25-6.50 m: J, 60-75°, PR, PR-IR, RF, CN, spaced 60-80mm, x5 6.28 m: DB
	5				19.0		from 7.05 m: yellow-brown, iron stained	MW - SW				7.01 m: HB 7.07 m: J, 60°, RF, PR, SN Fe 7.13 m: J, 65°, PR, healed 7.17 m: J, 70°, PR, healed 7.31 m: DB 7.37-7.51 m: CS 7.45-7.58 m: J, 75°, RF, PR, CT CI
					18.0		from 7.70 m: becoming thickly to thinly bedded at 0-15°		100 (68)			7.76-7.78 m: P, 5°, RF, PR, CN, spaced 20mm, x2 7.92 m: J, 65°, RF, PR, CN

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: V Tsambalas

Start Date: 18/02/26

Checked by: J Riley

End Date: 19/02/26

Location Meth.: dGPS0.1

Driller: MacGeo

Hole Diameter: 96-125 mm

Drill Rig: Comacchio 300

Inclination: 90°

Total Depth: 10.63 m

Bearing: N/A

Easting: 314622.20

Northing: 6237799.80

Hor. Datum: MGA2020-56

RL: 26.21 m

Ver. Datum: AHD

Surface: Pavement

[illegible]

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: V Tsambalas
Start Date: 12/03/26
Checked by: J Riley
End Date: 13/03/26
Location Meth.: dGPS0.1
RL: 3.706 m
Ver. Datum: AHD
Surface: Fill

Driller: MacGeo
Drill Rig: Comacchio 300
Total Depth: 12.42 m
Hole Diameter: 110 mm
Inclination: 90°
Bearing: N/A

Easting: 314461.35
Northing: 6238163.87
Hor. Datum: MGA2020-56

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
AD/T	HWT	D: 0.06-1.50 m ES: 0.06-1.50 m	13/02/2026 04:20:46	3.706	0.0			TOPSOIL - Silty SAND: fine to medium grained, dark brown; trace root fibres, trace wood fragments.	D		TOPSOIL
		SPT: 1.50-1.95 m 13,20,11 N=31			3.0			FILL - SAND: fine to medium grained, pale brown; trace gravel, fine to coarse, angular to sub-angular, sandstone.			FILL
		D: 2.00-3.00 m ES: 2.00-3.00 m			2.0		SP	SAND: fine grained, dark grey; with clay.		MD	ALLUVIUM
		SPT: 3.00-3.45 m 5,2,2 N=4 D: 3.00-3.45 m			1.0			SAND: fine to coarse grained, yellow-brown; trace clay; trace charcoal.			
		D: 3.50-4.50 m ES: 3.50-4.50 m			3.0		SP	SAND: fine to medium grained, dark grey; with clay; with thin bands of carbonaceous material.		W	
		PP=120, 110, 120 kPa	13/02/2026 04:20:46	3.706	0.0		CH	CLAY: high plasticity, pale grey mottled yellow, red and orange-brown; trace sand, fine to medium grained.		St	RESIDUAL SOIL
		PP=190, 220, 210 kPa SPT: 4.50-4.95 m 5,8,10 N=18 D: 4.50-4.95 m PP=220, 330, 340 kPa PP=450, 420, 340 kPa			4.0						
					5.0					VSt	

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: V Tsambalas

Start Date: 12/03/26

Checked by: J Riley

End Date: 13/03/26

Location Meth.: dGPS0.1

Driller: MacGeo

Hole Diameter: 110 mm

Drill Rig: Comacchio 300

Inclination: 90°

Total Depth: 12.42 m

Bearing: N/A

Easting: 314461.35

Northing: 6238163.87

Hor. Datum: MGA2020-56

RL: 3.706 m

Ver. Datum: AHD

Surface: Fill

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
AD/T	HWT	SPT: 5.20-5.23 m 5/30 mm HB N=R					CH	CLAY: high plasticity, pale grey mottled yellow, red and orange-brown; trace sand, fine to medium grained.		VSt	
									Log continued on next page.		

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: V Tsambalas
Start Date: 12/03/26
Checked by: J Riley
End Date: 13/03/26
Location Meth.: dGPS0.1
Driller: MacGeo
Hole Diameter: 110 mm
Easting: 314461.35
Drill Rig: Comacchio 300
Inclination: 90°
Northing: 6238163.87
Total Depth: 12.42 m
Bearing: N/A
Hor. Datum: MGA2020-56
Ver. Datum: AHD
Surface: Fill

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					5.0							
							<i>Log continued from previous page.</i>					
	1		Is(50) D=0.63 MPa Is(50) A=0.5 MPa		5.74		SANDSTONE: medium to coarse grained, yellow-brown and pale grey, indistinctly bedded, with gravel inclusions, fine grained, quartz; iron stained.	XW HW - MW	100 (90) [41]			HAWKESBURY SANDSTONE 5.31 m: J, 5°, RF, IR, CN 5.31-5.56 m: J, 0-10°, IR, 10-120 mm spacing, PR-IR, RF, CN x 5
	2				6.0				100 (100) [40]			5.74 m: DB 5.78-6.09 m: J, 5°, UN, 90-130 mm spacing, PR-IR, RF, CN x 4
							from 6.50 m: thinly bedded at 5-15°; trace gravel inclusions, fine grained, quartz and shale	MW				6.11 m: J, 5°, RF, IR, CN, healed 6.17-6.36 m: MB, x 5
			Is(50) D=1 MPa Is(50) A=1.3 MPa		6.36				100 (100) [95]			6.63 m: P, 5°, RF, IR, CN 6.74-7.00 m: MB, x 5
	3		Is(50) D=0.95 MPa Is(50) A=1.3 MPa		7.0							7.43 m: P, 10°, RF, UN, SN Fe 7.50 m: P, 10°, RF, IR, CN 7.63 m: P, 10°, RF, IR, CN
									100 (93) [60]			7.86 m: P, 5°, RF, CU, VN Cl 7.96-8.00 m: HB, x 2 8.10 m: P, 15°, RF, IR, CN
			Is(50) D=1.3 MPa Is(50) A=1.3 MPa		7.86		SANDSTONE: medium to coarse grained, yellow-brown, thinly bedded at 5-15°, trace gravel inclusions, fine grained quartz and shale; iron stained.	HW				8.36 m: P, 10°, RF, IR, SN Fe 8.42-8.45 m: CS 8.48 m: J, 5°, RF, IR, SN Fe 8.57 m: P, 25°, RF, UN, CT Cl
	4				8.0		from 8.41 to 8.77 m: yellow-brown, pale grey and pale pink-grey	MW - SW				8.73-8.77 m: EW, 0°, IR, Cl 8.78 m: P, 0°, RF, PR, CN 8.86-8.88 m: CS
							from 8.77 m: becoming pale red and pale yellow-brown	XW HW - MW				8.96 m: HB 9.00-9.04 m: J, 85°, RF, UN, SN Fe 9.04 m: P, 15°, RF, UN, CT SC
									100 (100) [91]			9.28 m: P, 5°, RF, IR, SN Fe 9.32-9.39 m: MB, RF, CN 9.45 m: P, 10°, RF, IR, CN
			Is(50) D=1.1 MPa Is(50) A=1.3 MPa C: 9.59-9.80 m UCS=40 MPa		9.36							9.78 m: J, 5°, RF, IR, SN Fe 9.87 m: P, 0°, SM, IR, VN Cl 9.94-10.00 m: HB, x 4
							SANDSTONE: medium grained, pale grey, trace indistinctly bedded carbonaceous flecks.	SW - FR				
					10.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: V Tsambalas

Start Date: 12/03/26

Checked by: J Riley

End Date: 13/03/26

Location Meth.: dGPS0.1

Driller: MacGeo

Hole Diameter: 110 mm

Drill Rig: Comacchio 300

Inclination: 90°

Total Depth: 12.42 m

Bearing: N/A

Easting: 314461.35

Northing: 6238163.87

Hor. Datum: MGA2020-56

RL: 3.706 m

Ver. Datum: AHD

Surface: Fill

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: ○ UCS: □ ▢ ▣ ▤ ▥ ▦ ▧ ▨ ▩	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	5		Is(50) D=0.8 MPa Is(50) A=0.74 MPa		10.0		SANDSTONE: medium grained, pale grey, trace indistinctly bedded carbonaceous flecks.	SW - FR				
	10.86			-7.0					98 (98) [96]			
	1.56		Is(50) D=1.1 MPa Is(50) A=1.2 MPa	-8.0	11.0		from 11.00 to 11.15 m: medium to coarse grained at 11.10: siltstone inclusion, 40 mm from 11.14 to 11.16 m: gravel inclusions, medium grained, rounded, siltstone and quartz at 11.30 m: siltstone inclusion, 40 mm					11.03 m: P, 10°, RF, UN, CN 11.14 m: P, 5°, RF, IR, CN 11.28-11.35 m: DB
			Is(50) D=1.4 MPa Is(50) A=1.4 MPa	-12.0			from 11.69 to 11.83 m: gravel inclusions, medium to coarse grained, rounded, siltstone and quartz					11.70 m: P, 20°, RF, IR, CN 11.74 m: P, 15°, RF, IR, CN
					12.0		from 12.09 to 12.21 m: coarse grained from 12.18 to 12.22 m: gravel inclusions, medium grained, rounded, siltstone and quartz					11.96-12.42 m: HB, x 4
							Terminated at 12.42 m. Target depth.					
				-9.0								
					13.0							
				-10.0								
					14.0							
				-11.0								

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: V Tsambalas

Start Date: 11/02/26

Checked by: J Riley

End Date: 12/02/26

Location Meth.: dGPS0.1

Driller: MacGeo

Hole Diameter: 96 mm

Drill Rig: Comacchio 300

Inclination: 90°

Total Depth: 8.78 m

Bearing: N/A

Easting: 6238048.01

Northing: 314533.35

Hor. Datum: MGA2020-56

RL: 6.443 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
AD/T	U	D: 0.10-1.50 m	Not Observed	0.0			SP	TOPSOIL - SAND: fine to medium grained, grey; trace rootlets, trace wood, trace gravel, fine, sandstone.	D		TOPSOIL
								RESIDUAL SOIL			
		SPT: 1.50-1.61 m 13/110 mm N=R D: 1.50-1.61 m		6.0				from 0.50 m: becoming orange-brown	L		
				1.0							
				5.0				from 1.55 m: red-brown		VD	
								Log continued on next page.			
				2.0							
				4.0							
				3.0							
				4.0							
				2.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: V Tsambalas
Start Date: 11/02/26
Checked by: J Riley
End Date: 12/02/26
Location Meth.: dGPS0.1
RL: 6.443 m
Ver. Datum: AHD
Surface: Topsoil

Driller: MacGeo
Drill Rig: Comacchio 300
Total Depth: 8.78 m
Hole Diameter: 96 mm
Inclination: 90°
Bearing: N/A

Easting: 6238048.01
Northing: 314533.35
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: o UCS: □ SOIL CLASSIFICATION VL L e M 2 ₁ H 6 ₀ VH 200 EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0							
					6.0							
					1.0							
					5.0							
							<i>Log continued from previous page.</i>					
							SANDSTONE: medium to coarse grained, yellow-brown and red-brown, very thinly bedded at 0-20°, trace gravel, fine grained, quartz; iron stained.	MW - SW	100 (92) [92]			HAWKESBURY SANDSTONE
					2.0							1.79 m: P, 10°, RF, PR, CN
												1.92 m: HB
												2.00 m: MB
							from 2.19 to 2.29 m: red-brown					2.19 m: P, 5°, RF, PR, CN
					4.0		at 2.35 m: ironstone lamination, dark red-brown	XW				2.32-2.36 m: EW
							from 2.48 to 2.67 m: red-brown	MW - SW				2.37-2.44 m: EW
												2.47 m: P, 0°, RF, PR, CN
												2.66 m: P, 5°, RF, PR, CN
					3.0							3.08 m: DB
							from 3.30 to 4.16 m: indistinctly bedded		100 (99) [99]			3.23 m: P, 0°, RF, PR, CN
												3.39 m: MB
												3.85 m: HB
					4.0							4.00 m: MB
							from 4.16 m: becoming pale grey, thinly bedded at 0-20°					4.23-4.24 m: CS, 5°
							from 4.38 to 4.46 m: red-brown					4.36 m: P, 5°, RF, PR, CN
					2.0							4.58 m: P, 20°, RF, PR, CN
												4.58-4.64 m: J, 80°, UN, healed, discontinuous
												4.77 m: P, 20°, RF, PR, CN
									100 (87) [87]			4.78 m: HB
					5.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: V Tsambalas
Start Date: 11/02/26
Checked by: J Riley
End Date: 12/02/26
Location Meth.: dGPS0.1
RL: 6.443 m
Ver. Datum: AHD
Surface: Topsoil

Driller: MacGeo
Drill Rig: Comacchio 300
Total Depth: 8.78 m
Hole Diameter: 96 mm
Inclination: 90°
Bearing: N/A

Easting: 6238048.01
Northing: 314533.35
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS(50) A: • D: o UCS: □ SOIL: VL 2 L 6 M 20 H 60 VH 200 EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	3	Not Observed	Is(50) A=0.53 MPa Is(50) D=0.17 MPa		1.0		SANDSTONE: medium to coarse grained, yellow-brown and red-brown, very thinly bedded at 0-20°, trace gravel, fine grained, quartz; iron stained.	MW - SW				4.97-5.00 m: MB
												5.18 m: P, 10°, RF, PR, CN
												5.44 m: P, 10°, PR
	6.28		Is(50) A=0.3 MPa Is(50) D=0.25 MPa		6.0		from 6.15 m: becoming red-brown and yellow-brown, thickly to medium bedded at 0-15°	XW				5.80-6.00 m: EW
								MW - SW				6.00 m: MB
					0.0				100 (100) [100]			6.23 m: DB 6.26 m: MB
	7.78		Is(50) A=1.37 MPa Is(50) D=0.64 MPa Is(50) A=1.36 MPa Is(50) D=0.6 MPa		7.0		from 7.75 to 7.86 m: pale grey					7.00 m: MB
					8.0		SANDSTONE: medium to coarse grained, red-brown and yellow-brown mottled pale grey, thinly bedded to very thinly bedded at 0-15°, trace gravel, fine grained, quartz.		100 (100) [95]			7.80 m: P, 5°, RF, PR, SN Fe 7.86 m: P, 0°, RF, UN, CN
	5				8.0							8.00 m: MB
					2.0		at 8.63 m: carbonaceous inclusion					8.15 m: P, 15°
					9.0		Terminated at 8.78 m. Target depth.					
					3.0							
					4.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: V Tsambalas

Start Date: 08/02/26

Checked by:

End Date: 20/02/26

Location Meth.: dGPS0.1

Driller: MacGeo

Hole Diameter: 240-110 mm

Easting: 314571.23

RL: 36.851 m

Drill Rig: Comacchio 300

Inclination: 90°

Northing: 6237606.65

Ver. Datum: AHD

Total Depth: 9 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Pavement

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC		CONCC: 0.00-0.19 m D: 0.19-1.30 m			0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 10 mm, 190 mm thick.			FILL
AD/T	U	D: 1.30-1.50 m SPT: 1.50-1.66 m 11.3/10 mm HB N=R D: 1.50-1.66 m			36.0 1.0			FILL - Gravelly SAND: fine to coarse grained, brown and yellow-brown; gravel, fine to coarse, angular to sub-angular, igneous; trace clay; trace wood fragments.	M		
							SP	Gravelly SAND: fine to medium grained, yellow-brown; gravel, fine to medium, rounded, sandstone.	D	MD	RESIDUAL SOIL
					35.0 2.0 34.0 3.0 33.0 4.0 32.0 5.0 31.0 6.0 30.0 7.0 29.0 8.0			Log continued on next page.			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: V Tsambalas

Start Date: 08/02/26

Checked by:

End Date: 20/02/26

Location Meth.: dGPS0.1

Driller: MacGeo

Hole Diameter: 240-110 mm

Easting: 314571.23

RL: 36.851 m

Drill Rig: Comacchio 300

Inclination: 90°

Northing: 6237606.65

Ver. Datum: AHD

Total Depth: 9 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Pavement

[illegible]

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Driller: MacGeo

Drill Rig: Comacchio 300

Total Depth: 9 m

Hole Diameter: 240-110 mm

Inclination: 90°

Bearing: N/A

Project No: 60731324

Logged by: V Tsambalas

Start Date: 08/02/26

Easting: 314571.23

Northing: 6237606.65

Hor. Datum: MGA2020-56

Checked by:

End Date: 20/02/26

Location Meth.: dGPS0.1

RL: 36.851 m

Ver. Datum: AHD

Surface: Pavement

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: o UCS: □ SOIL: VL L M H VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	9		D=0.81 MPa Is(50) A=1.62 MPa Is(50) D=0.99 MPa		28.0		SANDSTONE: medium to coarse grained, pale grey and pale yellow-brown, thickly laminated at 0-10°; iron stained. from 8.30 m: becoming thinly to medium bedded at 0-20°	MW - SW	(100) [97]			8.22 m: DB 8.26 m: P, 5°, RF, CN 8.30 m: P, 5°, RF, CT CI 8.75 m: J, 0°, RF, CN 8.90 m: J, 15°, RF, CN
					9.0		Terminated at 9.00 m. Target depth.					
					27.0							
					10.0							
					26.0							
					11.0							
					25.0							
					12.0							
					24.0							
					13.0							
					23.0							
					14.0							
					22.0							
					15.0							
					21.0							
					16.0							

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: V Tsambalas

Start Date: 09/02/26

Checked by: J Riley

End Date: 09/02/26

Location Meth.: dGPS0.1

Driller: MacGeo

Drill Rig: Comacchio 300

Total Depth: 9.3 m

Hole Diameter: 240-110 mm

Inclination: 90°

Bearing: N/A

Easting: 314584.80

Northing: 6237814.90

Hor. Datum: MGA2020-56

RL: 19.356 m

Ver. Datum: AHD

Surface: Pavement

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC		CONCC: 0.00-0.19 m D: 0.20-0.30 m D: 0.30-1.50 m			0.0			ASPHALTIC CONCRETE: dark grey; aggregates up to 10 mm, 190 mm thick.			PAVEMENT
AD/T	U	SPT: 1.50-1.93 m 15.20, 12/130 mm N=R D: 1.80-1.93 m			19.0		SP	FILL - Clayey SAND: fine to coarse grained, brown; trace gravel, fine grained, sub-angular to sub-rounded, igneous, sandstone. SAND: fine to medium grained, yellow-brown; trace clay; trace gravel, fine grained, rounded, sandstone. at 1.00 m: possible cobble	D	L - D	COLLUVIUM
					18.0		SP	Gravelly SAND: fine to medium grained, pale yellow-brown; gravel, fine to coarse grained, sub-rounded, sandstone. from 1.90 m: becoming pale grey <i>Log continued on next page.</i>			RESIDUAL SOIL
					2.0						
					17.0						
					3.0						
					16.0						
					4.0						
					15.0						
					5.0						
					14.0						
					6.0						
					13.0						
					7.0						
					12.0						
					8.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: V Tsambalas
Start Date: 09/02/26
Checked by: J Riley
End Date: 09/02/26
Location Meth.: dGPS0.1
RL: 19.356 m
Ver. Datum: AHD
Surface: Pavement

Driller: MacGeo
Drill Rig: Comacchio 300
Total Depth: 9.3 m
Hole Diameter: 240-110 mm
Inclination: 90°
Bearing: N/A

Easting: 314584.80
Northing: 6237814.90
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: o UCS: □ SOIL ₀₋₆ VL ₂ L ₆ M ₂₀ H ₆₀ VH ₂₀₀ EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0							
					19.0							
					1.0							
					18.0							
					2.0		<i>Log continued from previous page.</i>					
	1				2.43		SANDSTONE: medium to coarse grained, pale grey mottled pale yellow-brown, medium to thinly bedded at 0-5°, trace fine gravel sized quartz inclusions, iron stained.	MW - SW	100 (100) [100]			HAWKESBURY SANDSTONE
	2		Is(50) A=0.4 MPa Is(50) D=0.23 MPa		3.00		from 2.47 to 2.88 m: red-brown, liesegang banding		100 (100) [100]			1.93-2.43 m: DB, x 3
	3		Is(50) A=0.52 MPa Is(50) D=0.41 MPa		4.00		from 3.52 to 3.90 m: liesegang banding		100 (99) [99]			3.00 m: MB
	4		Is(50) A=1.01 MPa Is(50) D=0.4 MPa		5.00		from 4.69 to 5.00 m: very thinly bedded at 10-35°		100 (100) [100]			3.25 m: P, 5°, RF, CN root fibres 3.29-3.30 m: DB 3.34-3.35 m: EW, 0-5°, CI
	5		Is(50) A=1.19 MPa Is(50) D=0.48 MPa		6.00		from 5.00 m: becoming medium to thickly bedded at 20°		100 (100) [100]			4.00 m: MB
			Is(50) A=1.16 MPa Is(50) D=0.5 MPa C: 6.41-6.67 m UCS=24 MPa		7.00		from 6.66 to 6.74 m: medium to coarse grained, red-brown		100 (100) [96]			4.54 m: P, 5°, RF, CN 4.78 m: DB 5.00 m: MB
			Is(50) A=1.24 MPa Is(50) D=0.72 MPa		8.0		at 7.90 m: SANDSTONE: Medium to coarse grained, pale grey mottled pale yellow brown, with siltstone rip up clasts ranging in size from over 100mm to 60mm	HW	100 (85)			5.57 m: P, 30°, SM, CN 6.00 m: MB 6.15 m: P, 10°, SM, SN Fe 6.30 m: DB 6.67 m: P, 0°, RF, SN Fe 6.71 m: P, 0°, SM, CT Fe
												7.60-7.76 m: J, 70°, RF, PR, CN 7.91 m: P, 10°, RF, CN

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: V Tsambalas

Start Date: 09/02/26

Checked by: J Riley

End Date: 09/02/26

Location Meth.: dGPS0.1

Driller: MacGeo

Hole Diameter: 240-110 mm

Easting: 314584.80

RL: 19.356 m

Drill Rig: Comacchio 300

Inclination: 90°

Northing: 6237814.90

Ver. Datum: AHD

Total Depth: 9.3 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Pavement

[illegible]

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Driller: Pointe Engineering

Drill Rig: Atlas Copco R Cobra TTe

Total Depth: 1.8 m

Hole Diameter: 50 mm

Inclination: 90°

Bearing: N/A

Project No: 60731324

Logged by: A Sharma

Start Date: 25/02/26

Easting: 314496.91

Northing: 6238099.14

Hor. Datum: MGA2020-56

Checked by: J Riley

End Date: 25/02/26

Location Meth.: dGPS0.1

RL: 0.278 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
WS	U	ES: 0.00-0.10 m D: 0.10-0.65 m	Not Observed	0.0	0.0		SP	TOPSOIL - Clayey SAND: fine to coarse grained, black; roots.	M		TOPSOIL ALLUVIUM
		D: 0.65-1.17 m		-0.0			SP	SAND: fine to coarse grained, dark grey brown; gravel; roots.			
		D: 1.17-1.50 m		-1.0			SP	SAND: coarse grained, orange-brown; gravel, coarse grained, sub-angular gravel .			
		ES: 1.50-1.60 m		-1.0			SM	Silty SAND: coarse grained, dark grey.	W		
					2.0			Terminated at 1.80 m. Obstruction in the hole.			
					-2.0						
					-3.0						
					-4.0						
					-5.0						
					-6.0						
					-7.0						
					-8.0						

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Hutchins

Start Date: 19/03/26

Checked by: J Riley

End Date: 20/03/26

Location Meth.: dGPS0.1

Driller: Terratest

Drill Rig: Commachio GEO 205

Total Depth: 18.42 m

Hole Diameter: 76-100 mm

Inclination: 90°

Bearing: N/A

Easting: 314473.83

Northing: 6238141.06

Hor. Datum: MGA2020-56

RL: 4.325 m

Ver. Datum: AHD

Surface: TOPSOIL

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
WS	U	ES: 0.00-0.10 m D: 0.10-1.00 m	20/03/2026 07:34:05	▼	0.0			TOPSOIL - Silty SAND: fine to coarse grained, brown; trace root fibres.	D		TOPSOIL
					4.0			FILL - GRAVEL: fine to coarse grained, dark brown; trace root fibres.			FILL
		SPT: 1.50-1.95 m 5,6.6 N=12 D: 1.50-1.95 m			1.0			FILL - Silty SAND: fine to coarse grained, brown with black spots; with gravel, medium to coarse grained, angular, sandstone and ironstone; trace root fibres.	D		
		B: 2.00-3.00 m			2.0						
ADV	U		20/03/2026 07:34:05	▼	3.0			FILL - Silty SAND: medium to coarse grained, orange.			
					2.0						
		SPT: 3.00-3.45 m 6,6.6 N=12 D: 3.00-3.20 m			3.0						
		B: 3.50-4.00 m			4.0				M		
ADV	U		20/03/2026 07:34:05	▼	4.0			Silty SAND: fine to medium grained, grey; minor organic odour.			ALLUVIUM
					0.0						
		SPT: 4.50-4.95 m RW/450 mm N=0 D: 4.50-4.95 m			5.0				W	VL	hole collapsed from 4.80 to 5.33 m prior to casing off

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Driller: Terratest
Drill Rig: Commachio GEO 205
Total Depth: 18.42 m

Hole Diameter: 76-100 mm
Inclination: 90°
Bearing: N/A

Project No: 60731324
Logged by: R Hutchins
Start Date: 19/03/26

Checked by: J Riley
End Date: 20/03/26
Location Meth.: dGPS0.1
RL: 4.325 m
Ver. Datum: AHD
Surface: TOPSOIL

Easting: 314473.83
Northing: 6238141.06
Hor. Datum: MGA2020-56

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
AD/V	U		19/03/2026 09:39:36			SM	Silty SAND: fine to medium grained, grey; minor organic odour.		VL	
		SPT: 5-30 5-33 m 7/30 mm HB N=R		1.0		SP	SAND: medium to coarse grained, orange-brown mottled red-brown.	W	VD	EXTREMELY WEATHERED MATERIAL
							Log continued on next page.			
				6.0						
				2.0						
				7.0						
				3.0						
				8.0						
				4.0						
				9.0						
				5.0						
				10.0						

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: R Hutchins
Start Date: 19/03/26

Checked by: J Riley
End Date: 20/03/26
Location Meth.: dGPS0.1
RL: 4.325 m
Ver. Datum: AHD
Surface: TOPSOIL

Driller: Terratest
Drill Rig: Commachio GEO 205
Total Depth: 18.42 m
Hole Diameter: 76-100 mm
Inclination: 90°
Bearing: N/A

Easting: 314473.83
Northing: 6238141.06
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • Is(50) A: • D: o UCS: □ Soil: 0-4 VL 2 L 6 M 20 H 60 VH 200 EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					5.0							
					-1.0		<i>Log continued from previous page.</i>					
	5.44						NO CORE: 5.33 - 5.44 m.		0 (0)			HAWKESBURY SANDSTONE
	2		Is(50) A=1.11 MPa Is(50) D=2.22 MPa				SANDSTONE: medium to coarse grained, brown mottled orange-brown, indistinctly bedded.	MW	100 (100) [88]			5.49 m: P, 0°, VR, PR, CN
	5.84								100 (100) [100]			5.84-6.00 m: HB, x 3
					6.0		from 6.1 m: red-brown mottled orange-brown, thinly to very thinly bedded at 0-20°, trace pale brown-grey laminations					6.11 m: P, 5-10°, RF, PR, CN
					-2.0							
			C: 6.50-6.65 m UCS=33 MPa Is(50) A=2.23 MPa Is(50) D=1.78 MPa									6.85-7.38 m: HB, x 4
					7.0							
					-3.0		from 7.25 to 7.51 m: cross-bedding visible					
	7.38						from 7.51 - 7.70 m: pale grey	HW	100 (100) [78]			7.49 m: P, 5°, VR, PR, CN 7.53 m: P, 0°, RF, PR, CT sand
			Is(50) A=0.61 MPa Is(50) D=0.56 MPa				from 7.73 - 7.87 m: pale grey	MW HW MW				7.67 m: P, 0-5°, RF, PR, CN 7.81 m: DB
					8.0		SANDSTONE: medium to coarse grained, red-brown mottled orange-brown, cross-bedded at 0-20°, trace pale brown-grey laminations.					7.99 m: HB 8.04 m: P, 20°, RF, PR, CN
					-4.0		lower contact with fresher rock inclined at 50°, intact SANDSTONE: medium to coarse grained, pale grey, cross-bedded 0-10°.	FR				8.22 m: DB
			Is(50) A=0.54 MPa Is(50) D=0.45 MPa									8.59 m: HB
	8.87								100 (24) [0]			9.00 m: J, 80°, RF, PR, CT CL
			PP=70, 70, 80 kPa		9.0		CH CLAY: high plasticity, pale grey; sand, sand is fine to coarse grained.	XW				
					-5.0							
			Is(50) A=1.68 MPa Is(50) D=1.5 MPa				SANDSTONE: coarse grained, pale grey, indistinctly bedded.	HW				9.87 m: P, 20°, VR, PR, CN 9.92 m: HB
					10.0		CH CLAY: high plasticity, pale grey; with sand, fine to coarse grained.					

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Hutchins

Start Date: 19/03/26

Checked by: J Riley

End Date: 20/03/26

Location Meth.: dGPS0.1

Driller: Terratest

Hole Diameter: 76-100 mm

Easting: 314473.83

RL: 4.325 m

Drill Rig: Commachio GEO 205

Inclination: 90°

Northing: 6238141.06

Ver. Datum: AHD

Total Depth: 18.42 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: TOPSOIL

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ Soil: VL L M H VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)		
NMLC	10.37	6	PP=100, 100, 110 kPa Is(50) A=2.11 MPa Is(50) D=1.78 MPa	10.0	-6.0		CH CLAY: high plasticity, pale grey; with sand, fine to coarse grained. from 10.15 m: sand content increasing	XW	95 (95) [0]			10.48 m: P, 20°, RF, PR, CN 10.68 m: HB 10.75 m: HB 10.85 m: P, 10°, RF, CU, SN Fe 11.00-11.24 m: HB, x 4 At 11.45 m, 40% water loss 11.46 m: P, 0-20°, RF, CU, SN Fe, x 2 11.57 m: P, 0°, RF, PR, SN X 11.67 m: DB, x 4		
							NO CORE: 10.27 - 10.37 m.	HW - MW						
	11.89	7	Is(50) A=0.15 MPa Is(50) D=0.58 MPa Is(50) A=1.64 MPa Is(50) D=1.54 MPa	11.0	-7.0		BRECCIA: dark grey and pale grey, shale clasts 2 - 40 mm in medium to coarse grained sandstone matrix, indistinctly bedded. NO CORE: 11.81 - 11.89 m.	HW	100 (100) [79]			12.00-12.06 m: J, 85°, RF, CU, SN Fe, persists 0.06m 12.07 m: P, 0-20°, RF, UN, SN Fe, 2 mm aperture 12.15 m: P, 20°, RF, PR, SN Fe 12.19 m: J, 45°, RF, PR, SN Fe 12.77 m: HB 13.00 m: P, 0°, VR, PR, CN 13.34 m: DB 13.47 m: P, 25°, RF, PR, CT Ca 13.51 m: P, 25°, RF, PR, SN Fe		
							BRECCIA: dark grey and pale grey, shale clasts 2 - 40 mm in medium to coarse grained sandstone matrix, indistinctly bedded. from 12.00 m: iron stained, orange-brown	HW						
	13.34	8	C: 12.50-12.65 m UCS=19 MPa Is(50) A=1.46 MPa Is(50) D=1.3 MPa Is(50) A=0.44 MPa Is(50) D=0.29 MPa	12.0	-8.0		SANDSTONE: medium to coarse grained, pale grey mottled pale orange-brown, thickly bedded at 0-20°, trace black laminations, iron stained.	SW	100 (100) [95]			13.97 m: P, 0°, RF, PR, SN Fe 14.00 m: P, 0°, RF, PR, SN Fe		
							from 13.77 m: becoming very thinly bedded	FR						
	14.81		Is(50) A=0.45 MPa Is(50)	13.0	-9.0		from 14.02 m: becomes pale grey	100 (100) [99]				14.81-15.00 m: HB, x 3		

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Hutchins

Start Date: 19/03/26

Checked by: J Riley

End Date: 20/03/26

Location Meth.: dGPS0.1

Driller: Terratest

Hole Diameter: 76-100 mm

Easting: 314473.83

RL: 4.325 m

Drill Rig: Commachio GEO 205

Inclination: 90°

Northing: 6238141.06

Ver. Datum: AHD

Total Depth: 18.42 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: TOPSOIL

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: o UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC	9		D=0.38 MPa Is(50) A=1.04 MPa Is(50) D=0.8 MPa C: 15.30-15.45 m UCS=28 MPa	15.0			SANDSTONE: medium to coarse grained, pale grey mottled pale orange-brown, thickly bedded at 0-20°, trace black laminations, iron stained. from 15.06 m: trace gravel inclusions, fine grained, black	FR				15.02 m: P, 0°, RF, PR, CN 15.04 m: P, 0-5°, PR, healed 15.18 m: P, 15°, RF, PR, CN
	10		Is(50) A=0.4 MPa Is(50) D=0.28 MPa	16.0			SANDSTONE: medium to coarse grained, pale grey mottled orange-brown, very thinly bedded to thickly laminated at 0-20°, trace black laminations. from 16.35 - 16.68 m: coarse grained at 16.36 m: shale clast, 10 mm from 16.68 - 16.75 m: dark grey at 16.75 m: fine to medium sized shale gravel clasts	SW - FR FR	100 (100) [83]			16.00 m: P, 0-5°, SM, PR, VN CL 16.31-16.35 m: DB 16.40 m: J, 80-85°, RF, UN, CN 16.75 m: P, 0-10°, RF, PR, CN 16.85 m: HB 17.00-17.25 m: P, 5°, RF, PR, VN ML
	11		Is(50) A=0.87 MPa Is(50) D=0.63 MPa Is(50) A=0.88 MPa Is(50) D=0.7 MPa	17.0					100 (100) [100]			17.67 m: P, 10°, RF, PR, CN 17.83-18.42 m: HB, x 4
							Terminated at 18.42 m. Target depth.					

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A Sharma / V

Start Date: Tsambalas

03/03/26

Checked by: J Riley

End Date: 05/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering

Hole Diameter: 76 mm

Easting: 314519.59

RL: 4.478 m

Drill Rig: Atlas Copco

Inclination: 90°

Northing: 6238075.62

Ver. Datum: AHD

Total Depth: 15.8 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
WS		D: 0.00-0.20 m			0.0			TOPSOIL - Sandy GRAVEL: fine to coarse grained, sub-angular, brown; sand, fine to coarse grained; with roots.	D		TOPSOIL
		ES: 0.20-0.30 m D: 0.30-1.10 m			4.0			FILL - Sandy GRAVEL: medium to coarse grained, sub-angular, pale brown; sand, fine to coarse grained. at 0.4 m: asphalt			FILL
WB					1.0			FILL - MIXTURE OF COBBLES AND Gravelly SAND: COBBLES, pink, grey and black; SAND, fine to coarse grained; Gravel, coarse grained; sandstone, asphalt and other lithologies; with clay.	M		
					3.0						
					2.0						
					2.0						
					3.0						
					1.0			FILL - Clayey SAND: fine to medium grained, pale grey mottled pale pink.			
					4.0		SP	SAND: fine to medium grained, pale grey, pale yellow.		W	ALLUVIUM
					0.0						
					5.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: A Sharma / V
Start Date: Tsambalas
03/03/26

Checked by: J Riley
End Date: 05/03/26
Location Meth.: dGPS0.1

Driller: Pointe Engineering	Hole Diameter: 76 mm
Drill Rig: Atlas Copco	Inclination: 90°
Total Depth: 15.8 m	Bearing: N/A

Easting: 314519.59
Northing: 6238075.62
Hor. Datum: MGA2020-56

RL: 4.478 m
Ver. Datum: AHD
Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
WB		D: 6.00-6.18 m ES: 6.00-6.18 m			-1.0		SP	SAND: fine to medium grained, pale grey, pale yellow.			
					6.0		ML	Sandy SILT: low plasticity, black; sand, fine grained; with charcoal; organic odour.	w > PL		
		Is(50) A=0.45 MPa			-2.0			Log continued on next page.			
					7.0						
					-3.0						
					8.0						
					-4.0						
					9.0						
					-5.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

A Sharma / V

Logged by: Tsambalas

Start Date: 03/03/26

Checked by: J Riley

End Date: 05/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering

Hole Diameter: 76 mm

Easting: 314519.59

RL: 4.478 m

Drill Rig: Atlas Copco

Inclination: 90°

Northing: 6238075.62

Ver. Datum: AHD

Total Depth: 15.8 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ Soil: 0-4 VL 2 L 4 M 20 H 60 VH 200 EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					5.0							
					-1.0							
					6.0							
							<i>Log continued from previous page.</i>					
	1		Is(50) D=0.38 MPa		-2.0		SANDSTONE: fine to coarse grained, yellow-brown mottled dark yellow-brown, indistinctly bedded, iron stained.	HW - MW	100 (100) [77]			HAWKESBURY SANDSTONE 6.19 m: J, 40°, RF, PR, VN X 6.24 m: DB 6.27 m: DB 6.32 m: P, 0-5°, SM, UN, CN 6.44 m: DB 6.58 m: P, 0°, RF, UN, CT 6.65 m: J, 20°, RF, PR, VN CI 6.70 m: J, 20°, RF, PR, VN CI
	2		Is(50) A=1.04 MPa Is(50) D=0.8 MPa Is(50) D=0.28 MPa Is(50) A=0.4 MPa Is(50) A=0.87 MPa Is(50) D=0.63 MPa		7.0		from 7.11: trace gravel inclusions, medium to coarse grained, quartz		94 (72) [72]			6.85 m: J, 10°, RF, UN, CN 6.93-7.02 m: MB, x 4 7.10 m: P, 10°, RF, PR, CN
					-3.0							7.46 m: J, 10°, RF, PR, CN
					8.0		NO CORE: 7.90 to 8.00 m.					
							SAND: fine to medium grained, pale grey-white, trace carbonaceous inclusions.	XW				
					-4.0		from 8.31 - 8.42 m: red-brown SANDSTONE: fine to medium grained, red-brown mottled orange-brown, indistinctly bedded, ironstone inclusions; iron staining.	HW	100 (81) [81]			
			C: 8.40-8.55 m UCS=0.005 9 MPa				SAND: fine to coarse grained, pale grey-white, trace carbonaceous inclusions.	XW				
			Is(50) A=0.88 MPa Is(50) D=0.7 MPa		9.0		SANDSTONE: fine to coarse grained, red-brown, medium bedded at 5°, iron stained.	MW				9.00 m: MB
			Is(50) A=1.57 MPa Is(50) D=1.09 MPa				from 8.85 to 9.30 m: orange-brown, indistinctly bedded					
					-5.0		from 9.28 m: very thinly bedded at 5-10°, orange-brown mottled red-brown					9.28 m: P, 5°, RF, PR, SN Fe 9.38 m: P, 5°, RF, PR, SN Fe
			C: 9.60-9.75 m UCS=15 MPa				from 9.55 to 9.58 m: thin carbonaceous laminations, <1 mm					9.79 m: P, 5°, SM, PR, VN CI
			Is(50) A=1.69 MPa Is(50) D=1.19 MPa		10.0							9.95-10.00 m: DB, x 2

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

A Sharma / V

Logged by: Tsambalas

Start Date: 03/03/26

Checked by: J Riley

End Date: 05/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering

Hole Diameter: 76 mm

Easting: 314519.59

RL: 4.478 m

Drill Rig: Atlas Copco

Inclination: 90°

Northing: 6238075.62

Ver. Datum: AHD

Total Depth: 15.8 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS(50) A: • D: o UCS: □ SOIL: VL L M H EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC	4		Is(50) A=1.51 MPa Is(50) D=1.75 MPa	-6.0	10.0		SANDSTONE: fine to coarse grained, red-brown, medium bedded at 5°, iron stained. from 10.09 to 10.16 m: pale grey from 10.16 m: thinly bedded at 15°	MW	100 (98) [76]			10.03 m: P, 5°, RF, PR, SN Fe 10.10 m: P, 0°, RF, PR, VN X 10.17 m: DB 10.38 m: P, 15°, RF, UN, CN 10.49 m: P, 5°, RF, PR, CN 10.57 m: J, 5°, RF, CU, CN 10.65 m: P, 10°, RF, PR, CN 10.75 m: DB 10.83 m: DB
			Is(50) A=1.74 MPa Is(50) D=1.88 MPa									10.96-10.98 m: CS, GC 10.98-11.00 m: EW, CI 11.00-11.06 m: DB, x 2
	5		Is(50) A=2.06 MPa Is(50) D=2.09 MPa	-7.0	11.0		SANDSTONE: fine to medium grained, pale grey, laminated to very thinly bedded at 0-5°; trace iron staining.	SW	100 (96) [78]			11.33 m: P, 0°, RF, PR, CN 11.41 m: P, 0°, RF, PR, SN Fe 11.48-11.51 m: EW, 5°, CI 11.56 m: MB 11.73 m: DB 11.94 m: P, 0°, RF, PR, CN 12.00 m: MB
			Is(50) A=1.17 MPa Is(50) D=1.11 MPa									12.46 m: P, 5°, RF, ST, CN 12.51-12.52 m: CS, 0° 12.66-12.67 m: P, 5°, SC 12.75 m: DB
	6		Is(50) A=1.71 MPa Is(50) D=2.13 MPa Is(50) A=0.71 MPa Is(50) D=0.88 MPa Is(50) A=1.3 MPa Is(50) D=1.06 MPa Is(50) A=0.31 MPa Is(50) D=0.3 MPa	-8.0	12.0		NO CORE: POSSIBLE SOIL MATERIAL 13.16 - 15.20 m.		0 (0) [0]			13.02 m: HB 13.12 m: DB Anomaly encountered. Run 5 drilled to 13.16 m but measured to be 14.30 m with weight tape. Rods for Run 6 then sat at 15.20 m.
				-9.0	13.0							
					14.0							
					10.0							
					15.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Driller: Pointe Engineering

Drill Rig: Atlas Copco

Total Depth: 15.8 m

Hole Diameter: 76 mm

Inclination: 90°

Bearing: N/A

Project No: 60731324

Logged by: A Sharma / V Tsambalas

Start Date: 03/03/26

Easting: 314519.59

Northing: 6238075.62

Hor. Datum: MGA2020-56

Checked by: J Riley

End Date: 05/03/26

Location Meth.: dGPS0.1

RL: 4.478 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • IS ₍₅₀₎ D: o UCS: □ EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC	15.20			15.0			NO CORE: POSSIBLE SOIL MATERIAL 13.16 - 15.20 m.					
							NO CORE: 15.2 - 15.44 m.		60 (60) [60]			
	7		Is(50) A=0.59 MPa Is(50) A=0.65 MPa Is(50) D=0.15 MPa Is(50) D=0.68 MPa	-11.0			SANDSTONE: fine to medium grained, pale grey, laminated to very thinly bedded at 0-5°.	SW				
							Terminated at 15.80 m. Target depth.					
				16.0								
				-12.0								
				17.0								
				-13.0								
				18.0								
				-14.0								
				19.0								
				-15.0								
				20.0								



Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Driller: Pointe Engineering

Drill Rig: Speedy Drill

Total Depth: 14.58 m

Hole Diameter: 50-90-51 mm

Inclination: 90°

Bearing: N/A

Project No: 60731324

Logged by: AL

Start Date: 23/03/26

Easting: 314151.96

Northing: 6238460.66

Hor. Datum: MGA2020-56

Checked by: C Jayasuriya

End Date: 25/03/26

Location Meth.: dGPS0.1

RL: 2.65 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
NMLC	U	D: 0.00-0.25 m		2.65	0.0			TOPSOIL - Silty SAND: fine to coarse grained, brown; trace rootlets .	D		TOPSOIL
		D: 0.25-0.80 m					SM	Silty SAND: coarse grained, dark brown.	M	L	ALLUVIUM
		D: 0.80-1.15 m			2.0		CH	Sandy CLAY: high plasticity, dark brown; sand, fine grained ; trace rootlets.	w ≈ PL	St	
		PP=150, 120, 150 kPa D: 1.15-1.60 m			1.0		SM	Silty SAND: fine to coarse grained, brown; trace rootlets .	W	L	
	PVC	SPT: 1.50-1.95 m 1,0,4 N=4 D: 1.50-1.95 m			1.0		SM	Silty SAND: fine to medium grained, dark brown.			At 1.5m, sand caved in, potential non representative SPT value RESIDUAL SOIL
NMLC	PVC	SPT: 3.00-3.45 m 1,0,0 N=0	02/03/2026 11:30:00		2.0						From 3.00 to 3.45m, no SPT recovery
		SPT: 4.50-4.95 m 1,1,0 N=1			3.0						
					4.0						
NMLC	PVC				1.0						From 4.50 to 4.95m, no SPT recovery
					2.0						



Engineering Log

FINAL

Borehole No. DBH63
Sheet 2 of 4

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Driller: Pointe Engineering

Drill Rig: Speedy Drill

Total Depth: 14.58 m

Hole Diameter: 50-90-51 mm

Inclination: 90°

Bearing: N/A

Project No: 60731324

Logged by: AL

Start Date: 23/03/26

Easting: 314151.96

Northing: 6238460.66

Hor. Datum: MGA2020-56

Checked by: C Jayasuriya

End Date: 25/03/26

Location Meth.: dGPS0.1

RL: 2.65 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
NMLC	PVC					SM	Silty SAND: fine to medium grained, dark brown.		VL	
		Is(50) D=0.2 MPa		3.0 6.0 4.0 7.0 5.0 8.0 6.0 9.0 7.0 10.0			Log continued on next page.			HAWKESBURY SANDSTONE

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: AL

Start Date: 23/03/26

Checked by: C Jayasuriya

End Date: 25/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering

Hole Diameter: 50-90-51 mm

Easting: 314151.96

RL: 2.65 m

Drill Rig: Speedy Drill

Inclination: 90°

Northing: 6238460.66

Ver. Datum: AHD

Total Depth: 14.58 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Topsoil

[illegible]

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: AL

Start Date: 23/03/26

Checked by: C Jayasuriya

End Date: 25/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering

Hole Diameter: 50-90-51 mm

Easting: 314151.96

RL: 2.65 m

Drill Rig: Speedy Drill

Inclination: 90°

Northing: 6238460.66

Ver. Datum: AHD

Total Depth: 14.58 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: ○ UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)			
NMLC	5		Is(50) D=0.8 MPa Is(50) A=0.8 MPa	10.0		SANDSTONE: medium to coarse grained, pale grey, massive.	SW	69 (69) [65]			10.00 m: DB 10.08 m: P, 5°, RF, CN				
				NO CORE: 10.54 - 11.20 m.		10.50-10.54 m: CS, RF, CT CL									
	7		Is(50) D=0.9 MPa Is(50) A=1 MPa	11.0	SANDSTONE: medium to coarse grained, pale grey, massive.		100 (100) [94]	11.58 m: DB							
				12.0	SANDSTONE: medium to coarse grained, pale grey, thinly laminated at 0 - 15°.			12.20 m: P, 10°, RF, VN CL 12.34 m: DB 12.71 m: HB 13.00 m: DB 13.19 m: P, 5°, RF, VN CL							
	8		Is(50) D=1.1 MPa Is(50) A=0.9 MPa	13.0	from 14.40m: shale clusters		100 (100) [91]	13.54 m: DB 13.81 m: HB 13.90 m: DB 14.03 m: P, 5°, RF, CN							
				14.0				14.48 m: P, 30°, RF, CT Qz, CU 14.51 m: P, 15°, healed							
					Is(50) D=1 MPa Is(50) A=1.3 MPa	14.5		Terminated at 14.58 m. Target depth.							
						15.0									

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: RAH

Start Date: 20/03/26

Checked by: C Jayasuriya

End Date: 20/03/26

Location Meth.: dGPS0.1

Driller: Terratest

Drill Rig: Comacchio GEO 205

Total Depth: 14.95 m

Hole Diameter: 100-96 mm

Inclination: 90°

Bearing: N/A

Easting: 314483.89

Northing: 6238164.59

Hor. Datum: MGA2020-56

RL: 2.717 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PT		D: 0.00-0.20 m			0.0		SM	Silty SAND: fine to coarse grained, brown; trace rootlets .	M		TOPSOIL
		ES: 0.20-0.40 m			SM	Silty SAND: fine to coarse grained, orange.	M	FILL			
		D: 0.40-1.00 m			SC	Clayey SAND: fine to coarse grained, brown.					
		ES: 1.00-1.20 m		1.0	SM	Silty SAND: fine to coarse grained, brown; trace rootlets.					
		ES: 1.20-1.60 m									
ADV	U	SPT: 1.50-1.95 m 6,8,8 N=16 D: 1.50-1.95 m	Not Encountered		1.0						
				2.0							
		SPT: 3.00-3.45 m RW/450 mm N=0 D: 3.00-3.45 m			0.0		SP-SC	Clayey SAND: fine to medium grained, grey to dark grey; trace wood fibres.			ALLUVIUM from 2.60 m: organic odour
		B: 3.45-4.00 m			3.0						
					-1.0						
					4.0						
		SPT: 4.50-4.95 m RW/0 mm, HW/450 mm N=0 D: 4.50-4.95 m			-2.0		SP	SAND: fine to medium grained, grey; with silt.	W		from 4.50 m: minor organic odour

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: RAH

Start Date: 20/03/26

Checked by: C Jayasuriya

End Date: 20/03/26

Location Meth.: dGPS0.1

Driller: Terratest

Hole Diameter: 100-96 mm

Easting: 314483.89

RL: 2.717 m

Drill Rig: Comacchio GEO 205

Inclination: 90°

Northing: 6238164.59

Ver. Datum: AHD

Total Depth: 14.95 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
ADV	U	SPT: 6.00-6.45 m RW/450 mm N=0 D: 6.00-6.45 m PP=10, 20, 20 kPa D: 7.00-7.50 m	Not Encountered		3.0 6.0		SP	SAND: fine to medium grained, grey; with silt.		VL	from 6.0 - 7.0 m, auger was pushed in, no rotation required to advance
							CI	Sandy CLAY: medium plasticity, dark brown-grey; sand, fine grained.	w > PL	VS	
							CI	Silty CLAY: medium plasticity, orange-brown mottled orange and pale grey; with sand, fine to medium grained.			RESIDUAL SOIL
WB	HQ	SPT: 7.50-7.95 m 1,1,4 N=5 D: 7.90-7.95 m PP=150, 150, 160 kPa	Not Encountered		5.0 8.0				w ≈ PL	F	SPT result at 7.5 m unreliable. R=700 mm. Sand washed in from above. Pocket penetrometer undertaken in RS in base of split spoon.
		SPT: 8.80-9.25 m 7,11,12 N=23 D: 8.80-9.25 m PP=280, 310, 310 kPa	Not Encountered		9.0		CL	Sandy CLAY: low plasticity, red-brown mottled orange-brown and pale grey, trace black specks; sand, fine to medium grained.		VSt	At 9.7 m, drill mud returns change colour
								from 9.7 m, predominantly pale grey	w ≈ PL	St	

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: RAH

Start Date: 20/03/26

Checked by: C Jayasuriya

End Date: 20/03/26

Location Meth.: dGPS0.1

Driller: Terratest

Hole Diameter: 100-96 mm

Easting: 314483.89

RL: 2.717 m

Drill Rig: Comacchio GEO 205

Inclination: 90°

Northing: 6238164.59

Ver. Datum: AHD

Total Depth: 14.95 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
WB	HQ	SPT: 10.30-10.75 m 4,5,5 N=10 D: 10.30-10.75 m PP=110, 220, 340 kPa SPT: 11.80-11.95 m 25/150 mm N=R	Not Encountered	-8.0	11.0		CL	Sandy CLAY: low plasticity, red-brown mottled orange-brown and pale grey, trace black specks; sand, fine to medium grained.		St	at 11.0 m, driller reports significant increase in drilling resistance
										H	
				12.0				Log continued on next page.			HAWKESBURY SANDSTONE

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: RAH

Start Date: 20/03/26

Checked by: C Jayasuriya

End Date: 20/03/26

Location Meth.: dGPS0.1

Driller: Terratest

Hole Diameter: 100-96 mm

Drill Rig: Comacchio GEO 205

Inclination: 90°

Total Depth: 14.95 m

Bearing: N/A

Easting: 314483.89

Northing: 6238164.59

Hor. Datum: MGA2020-56

RL: 2.717 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: ○ UCS: □ ▢ ▣ ▤ ▥ ▦ ▧ ▨ ▩ Soil: 0-6 V.L. 7-19 M 20-29 H 30-39 VH 40-49 EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					10.0							
					8.0							
					11.0							
					9.0							
NMLC	1	Not Encountered	PP=60, 100, 160 kPa Is(50) A=0.15 MPa Is(50) D=0.16 MPa	12.0			<i>Log continued from previous page.</i> SANDSTONE: fine to coarse grained, dark orange-brown. SANDSTONE: fine to medium grained, pale grey with red and orange-brown bands, indistinctly to cross-bedded at 0-15°. from 12.22m: SANDSTONE: fine to medium grained, pale grey mottled red, thinly laminated at 0-5° from 12.40m: becoming pale grey and indistinctly laminated	HW XW XW MW	100 (100) [82]			11.97 m: P, 0°, VR, PR, SN Fe HAWKESBURY SANDSTONE 12.00 m: HB 12.37 m: HB 12.55 m: J, 15-75°, RF, IR, SN clay , healed 12.75 m: J, 75-85°, RF, UN, SN Fe 13.07 m: J, 15°, RF, IR, SN clay, healed 13.16-13.17 m: EW, 5°, RF, PR, CN 13.41 m: HB 13.56 m: P, 10-15°, VR, PR, CN 13.70 m: DB 13.80 m: P, 20°, VR, PR, CN
	2		Is(50) A=0.41 MPa Is(50) D=0.21 MPa C: 13.80-13.95 m UCS=22 MPa	13.0			SANDSTONE: medium to coarse grained, pale grey, cross-laminated at 0-20° .	SW to FR	100 (100) [77]			14.32 m: P, 5°, SM, PR, VN X 14.41 m: P, 5°, SM, PR, VN X 14.52 m: P, RF, UN, CT CL 14.58 m: P, 10°, SM, PR, VN X 14.62-14.67 m: DB, from 14.62 - 14.67 m, multiple 14.70 m: J, 60°, RF, PR, CN 14.89 m: P, 0-10°, RF, UN, VN X
			Is(50) A=2.11 MPa Is(50) D=2.87 MPa	14.0			SANDSTONE: medium to coarse grained, pale grey, cross-bedded at 0-20°, conglomeritic, with fine gravel sized sub-rounded quartz and fine to medium gravel sized elongated shale inclusions, with black carbonaceous laminations.	SW MW SW				
				15.0			Terminated at 14.95 m. Target depth.					

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A Sharma

Start Date: 19/03/26

Checked by: C Jayasuriya

End Date: 19/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering

Hole Diameter: 76-50 mm

Easting: 313777.52

RL: 42.94 m

Drill Rig: Speedy Drill

Inclination: 60°

Northing: 6238842.54

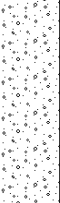
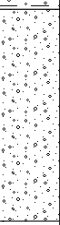
Ver. Datum: AHD

Total Depth: 8 m

Bearing: 230°

Hor. Datum: MGA2020-56

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
NMLC	U	D: 0.00-0.10 m D: 0.10-1.00 m PP=>600 kPa		42.0	0.0		CH	TOPSOIL - Sandy CLAY: medium plasticity, dark brown; sand, fine grained sand; with gravel, fine to coarse grained angular gravel ; with roots and organic matter. CLAY: high plasticity, red-brown and yellow-brown; trace gravel, fine grained sub-angular gravel ; trace root fibres.	w ≈ PL	H	TOPSOIL
					1.0		CH	CLAY: high plasticity, orange-brown.	w < PL		RESIDUAL SOIL
	PVC	D: 1.00-1.20 m			2.0			Sandy GRAVEL: fine to coarse grained, sub-angular to angular, grey mottled pale brown; sand, fine grained; with root fibers .	M	VD	EXTREMELY WEATHERED MATERIAL
					41.0		CH	Gravelly CLAY: high plasticity, brown; gravel, fine to coarse angular grained gravel.	w > PL	S	
								Sandy GRAVEL: fine to coarse grained, sub-angular, pale grey; sand, fine grained .	D	D	
					3.0	Log continued on next page.					HAWKESBURY SANDSTONE
					40.0						
					4.0						
					39.0						
					5.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: A Sharma
Start Date: 19/03/26
Checked by: C Jayasuriya
End Date: 19/03/26
Location Meth.: dGPS0.1
RL: 42.94 m
Ver. Datum: AHD
Surface: Topsoil

Driller: Pointe Engineering
Drill Rig: Speedy Drill
Total Depth: 8 m
Hole Diameter: 76-50 mm
Inclination: 60°
Bearing: 230°

Easting: 313777.52
Northing: 6238842.54
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ Soil: VL L M H VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0							
					1.0							
					42.0							
					2.0							
					41.0							
					3.0							
							Log continued from previous page.					
							SANDSTONE: coarse grained, red-brown, massive.	MW	78 (78) [40]			
							CORE LOSS: 3.11 - 3.29 m.	XW				
					40.0		SANDSTONE: coarse grained, pale grey with dark brown, indistinctly bedded .	HW				3.32-3.48 m: EW, RF, SN, Clay
								XW				3.60-3.65 m: SS, RF, IR, SN, Fe
								MW				3.69 m: HB
					4.0				100 (93) [70]			3.95 m: J, 15°, RF, PR, SN fe , healed
												4.14 m: J, 15°, RF, UN, SN fe
												4.29 m: J, 40°, RF, UN, SN clay
												4.32 m: DB
												4.43 m: DB
					39.0							4.55 m: P, 30°, RF, UN, CN
												4.68 m: P, 10°, RF, PR, SN, Fe
												4.74 m: P, 10°, RF, PR, root fibers
												4.85 m: P, 0°, RF, UN, CN
					5.0							4.93 m: HB

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: A Sharma
Start Date: 19/03/26

Checked by: C Jayasuriya
End Date: 19/03/26
Location Meth.: dGPS0.1
RL: 42.94 m
Ver. Datum: AHD
Surface: Topsoil

Driller: Pointe Engineering
Drill Rig: Speedy Drill
Total Depth: 8 m
Hole Diameter: 76-50 mm
Inclination: 60°
Bearing: 230°

Easting: 313777.52
Northing: 6238842.54
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: ○ UCS: □ SOIL: VL L M H VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC	5.45		C: 5.05-5.19 m UCS=6.1 MPa				SANDSTONE: coarse grained, dark-brown mottled pale-brown, cross at 0-30° with brown bands.	MW	100 (95) [58]			5.05 m: HB
								HW				5.18 m: J, 25°, RF, PR, CN
								MW				5.25 m: HB
			Is(50) A=0.8 MPa Is(50) D=0.76 MPa									5.45 m: DB
												5.46 m: EW, 10°, RF, IR
	7.00		Is(50) A=0.11 MPa Is(50) D=0.63 MPa				SANDSTONE: coarse grained, red-brown, cross-bedded at 0°.	HW	5.49 m: DB			
									5.58 m: J, 5°, VR, PR, SN clay			
									5.72 m: IS, 5°, RF, PR, VN clay			
									5.83 m: J, 15°, RF, PR, SN, Fe			
									6.00 m: DB			
	4		Is(50) A=0.15 MPa Is(50) D=0.09 MPa				SANDSTONE: coarse grained, pale grey with grey and brown bands, thinly laminated at 0-25°. from 7.05 m, grey mottled brown, crystalline	XW	100 (94) [64]			6.18 m: P, 5°, RF, PR, SN clay, Fe
								HW				6.27 m: P, 5°, RF, PR, SN, Fe
												6.35 m: P, 5°, RF, UN, clay
												6.36-6.50 m: J, 85°, RF, UN, SN, Fe
												6.60 m: J, 5°, RF, PR, SN, Fe
			Is(50) A=0.34 MPa Is(50) D=0.39 MPa				SANDSTONE: coarse grained, pale grey with grey and brown bands, thinly laminated at 0-25°. from 7.88 m, black mottled grey	XW	6.66-6.70 m: IS, 0°, VR, PR, VN clay			
								HW	6.74 m: HB			
									6.84 m: HB			
									6.93 m: IS, 0°, SM, PR, clay			
									7.00-7.05 m: EW, 20°, SM, PR, clay			
		Is(50) A=0.17 MPa Is(50) D=0.02 MPa				SANDSTONE: coarse grained, pale grey with grey and brown bands, thinly laminated at 0-25°. from 7.88 m, black mottled grey	XW	7.23 m: HB				
							HW	7.32 m: P, 0°, RF, PR, CN				
								7.43 m: P, 20°, RF, CU, VN clay				
								7.66 m: P, 20°, RF, PR, SN, Fe				
								7.79 m: P, 60°, RF, UN, CN				
						Terminated at 8.00 m. Target depth.					7.87 m: DB, 0°	

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Driller: Pointe Engineering
Drill Rig: Atlas Copco
Total Depth: 0.1 m

Hole Diameter: 50 mm
Inclination: 90°
Bearing: N/A

Project No: 60731324
Logged by: A Sharma
Start Date: 24/02/26

Checked by:
End Date: 24/02/26
Location Meth.: dGPS0.1
RL: 9.146 m
Ver. Datum: AHD
Surface: Topsoil

Easting: 314539.15
Northing: 6237967.20
Hor. Datum: MGA2020-56

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PT	U	D: 0.00-0.10 m	Not Observed		0.0			Sandy GRAVEL: fine to medium grained, poorly graded, sub-rounded, yellow-brown; sand, fine grained ; with rootlets . Terminated at 0.10 m. Refusal on inferred boulder.	D		TOPSOIL
				-9.0							
					1.0						
				-8.0							
					2.0						
				-7.0							
					3.0						
				-6.0							
					4.0						
				-5.0							
					5.0						

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A Sharma

Start Date: 25/02/26

Checked by: C Jayasuriya

End Date: 25/02/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering

Drill Rig: Atlas Copco R Cobra

Total Depth: 0.6 m

Hole Diameter: 50 mm

Inclination: 90°

Bearing: N/A

Easting: 314524.38

Northing: 6238006.47

Hor. Datum: MGA2020-56

RL: 4.69 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PT	U	D: 0.00-0.25 m	Not Encountered	0.0		SP	TOPSOIL - Gravelly SAND: fine medium grained, poorly graded, dark brown; gravel, fine to coarse grained angular gravel ; rootlets .	D	D	TOPSOIL
		D: 0.25-0.40 m				GP-GM	GRAVEL: fine coarse grained, poorly graded, sub-angular, brown and grey; with cobbles, coarse grained, angular, 300 mm maximum particle size; cobbles .		VD	RESIDUAL SOIL
		D: 0.40-0.60 m								
				4.0			Terminated at 0.60 m. Refusal on inferred boulder.			
				1.0						
				3.0						
				2.0						
				3.0						
				2.0						
				1.0						
				4.0						
				0.0						
				5.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A Sharma

Start Date: 24/02/26

Checked by: C Jayasuriya

End Date: 24/02/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering Pty Ltd

Drill Rig: Atlas Copco R Cobra

Drill Rig: TTe

Total Depth: 1.65 m

Hole Diameter: 50 mm

Inclination: 90°

Bearing: N/A

Easting: 314498.07

Northing: 6238050.95

Hor. Datum: MGA2020-56

RL: 2.002 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PT	U	D: 0.00-0.15 m	Not Observed	2.0	0.0			TOPSOIL - SILT: low plasticity, dark brown; with rootlets .	w > LL		TOPSOIL
		ES: 0.15-0.30 m					FILL - SILT: low plasticity, dark brown; with rootlets.			FILL	
		D: 0.30-1.00 m									
		ES: 1.00-1.10 m D: 1.10-1.65 m PP=3.0, 3.0, 2.5 kPa		1.0	1.0		ML	Sandy SILT: low plasticity, dark brown; sand, fine grained sand ; trace rootlets.		VSt	ALLUVIUM
								Terminated at 1.65 m. Target depth.			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: AS/AT

Start Date: 06/03/26

Checked by: J Riley

End Date: 09/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering

Hole Diameter: 90 mm

Easting: 314408.57

RL: 1.25 m

Drill Rig: Hard Access Drilling Rig

Inclination: 90°

Northing: 6238248.25

Ver. Datum: AHD

Total Depth: 16.5 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: TOPSOIL

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PT		D: 0.00-0.20 m	23/02/2026 11:47:28	0.0	0.0		CL	TOPSOIL - Silty CLAY: low plasticity, dark brown; trace root fibres.			TOPSOIL
		D: 0.20-0.55 m PP=100, 90, 80 kPa		-1.0	-1.0		CL	Silty SANDY CLAY: low plasticity, dark brown; sand, fine grained.	w ≈ PL	F	ALLUVIUM
PVC		D: 0.55-0.80 m		-0.0	-0.0		SC	Clayey SAND: fine to medium grained, grey brown; with silt.	w ≈ PL		At 6.1 m: Possible groundwater level
		D: 1.00-1.50 m		-1.0	-1.0						
WB		SPT: 1.50-1.95 m 0,0,0 N=0		-2.0	-2.0		CH	Silty CLAY: high plasticity, dark grey; trace sand.	w > PL		
		PP=20, 30, 20 kPa		-3.0	-3.0						
		SPT: 3.00-3.45 m 0,0,0 N=0		-4.0	-4.0		CI	Sandy CLAY: medium plasticity, orange-brown and yellow-brown occasionally pale grey; sand, fine to medium grained; with silt.	w ≈ PL	S	
				-5.0	-5.0		SP	SAND: fine to medium grained, pale brown grey.	w ≈ PL	VL	
		SPT: 6.10-6.55 m RW/450 mm N=0		-6.0	-6.0		CI	Sandy CLAY: medium plasticity, orange-brown and occasionally yellow-brown; sand, fine to medium grained; trace silt.		VS - S	
				-7.0	-7.0			from 7.5 m: becoming pale grey speckled yellow brown	w ≈ PL	St	
		SPT: 7.50-7.95 m 3,6,9 N=15 PP=120, 150, 160 kPa		-8.0	-8.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Driller: Pointe Engineering

Drill Rig: Hard Access Drilling Rig

Total Depth: 16.5 m

Hole Diameter: 90 mm

Inclination: 90°

Bearing: N/A

Project No: 60731324

Logged by: AS/AT

Start Date: 06/03/26

Easting: 314408.57

Northing: 6238248.25

Hor. Datum: MGA2020-56

Checked by: J Riley

End Date: 09/03/26

Location Meth.: dGPS0.1

RL: 1.25 m

Ver. Datum: AHD

Surface: TOPSOIL

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
WB				-7.0			CI	Sandy CLAY: medium plasticity, orange-brown and occasionally yellow-brown; sand, fine to medium grained; trace silt.		St	
							CI	Log continued on next page.			
					9.0						
					-8.0						
					10.0						
					-9.0						
					11.0						
					-10.0						
					12.0						
					-11.0						
					13.0						
					-12.0						
					14.0						
					-13.0						
					15.0						
					-14.0						
					16.0						

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: AS/AT
Start Date: 06/03/26

Checked by: J Riley
End Date: 09/03/26
Location Meth.: dGPS0.1
RL: 1.25 m
Ver. Datum: AHD
Surface: TOPSOIL

Driller: Pointe Engineering
Drill Rig: Hard Access Drilling Rig
Total Depth: 16.5 m
Hole Diameter: 90 mm
Inclination: 90°
Bearing: N/A

Easting: 314408.57
Northing: 6238248.25
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: o UCS: □ SOIL: VL L M H VH EH 2 4 6 8 10 20 40 60 80 100	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					8.0							
					-7.0		<i>Log continued from previous page.</i>					
	1				8.90		CI Sandy Silty CLAY: medium plasticity, pale grey brown mottled pale yellow-brown and pale red-brown; sand, fine to medium grained.	XW	100 (0) [0]			EXTREMELY WEATHERED MATERIAL
					9.0		from 8.9 m: becoming dark grey		37 (0) [0]			
					-8.0		NO CORE: 9.19 - 10.87 m.					
	2											
					-9.0							
					-10.0							
					-11.0		CI Sandy Silty CLAY: medium plasticity, dark grey; sand, fine to medium grained.	XW				
					-10.0		from 11.00 m: becoming pale grey and pale yellow-brown					
					-11.0		from 11.50 m: becoming pale grey		67 (10) [10]			
					-12.0		from 11.5-12.5 m: becoming pale grey					
	3											
					-11.0		NO CORE: 12.35 - 12.84 m.					
					-12.0							
					-13.0		SANDSTONE: medium grained, pale grey and pale brown, indistinct bedding.	HW	86 (85) [85]			HAWKESBURY SANDSTONE
					-12.0		from 13.03-13.07 m: gravel inclusions					12.84-12.90 m: HB
					-13.0		NO CORE: 13.50 - 13.72 m.					13.00-13.50 m: MB, 100-200 mm spacing, x 4
					-14.0		SANDSTONE: medium grained, pale grey speckled pale brown with occasional dark grey laminations, indistinct bedding, iron stained.	MW - SW				13.07-13.25 m: J, 60-70°, RF, UN, clay, 20 mm aperture, 150 mm spacing, x 2
	4											
					-13.0		NO CORE: 14.83 - 14.93 m.					13.96-14.36 m: MB, 30-250 mm spacing, x 6
					-14.0		SANDSTONE: medium grained, pale yellow-brown, indistinct bedding, trace gravel inclusions, iron stained.					
					-15.0		from 15.1 m: becoming fine to medium grained, pale grey, trace black laminations, rare shale inclusions	SW - FR				
					-16.0							
	5											
					-14.0							14.60 m: DL
					-15.0				94 (94) [89]			14.75 m: P, 10°, RF, UN, CN
					-16.0							14.94-15.20 m: MB, 50-150 mm spacing, x 4
					-17.0							15.07 m: P, 7°, RF, UN, CN

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Driller: Pointe Engineering

Drill Rig: Hard Access Drilling Rig

Total Depth: 16.5 m

Hole Diameter: 90 mm

Inclination: 90°

Bearing: N/A

Project No: 60731324

Logged by: AS/AT

Start Date: 06/03/26

Easting: 314408.57

Northing: 6238248.25

Hor. Datum: MGA2020-56

Checked by: J Riley

End Date: 09/03/26

Location Meth.: dGPS0.1

RL: 1.25 m

Ver. Datum: AHD

Surface: TOPSOIL

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: o UCS: □ SOIL: VL J M H EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC	16.20		Is(50) A=0.5 MPa Is(50) D=0.5 MPa Is(50) A=0.4 MPa	16.0	16.0		SANDSTONE: medium grained, pale yellow-brown, indistinct bedding, trace gravel inclusions, iron stained.	SW - FR	93 (93) [93]			16.00 m: MB 16.20 m: DL 16.25 m: J, 20°, RF, UN, SN, 20 mm aperture 16.30-16.50 m: MB, 50 mm spacing, x 4
	6			-15.0	-15.0		from 16.00-16.20 m: with shale clast inclusions					
					17.0		Terminated at 16.50 m. Target depth.					
					-16.0							
					-18.0							
					-17.0							
					-19.0							
					-18.0							
					-20.0							
					-19.0							
					-21.0							
					-20.0							
					-22.0							
					-21.0							
					-23.0							
					-22.0							
					-24.0							

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A.To

Start Date: 19/02/26

Checked by: J Riley

End Date: 19/02/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering

Hole Diameter: 90 mm

Easting: 314222.38

RL: 4.236 m

Drill Rig: Ripamonti Skid Drill

Inclination: 90°

Northing: 6238437.46

Ver. Datum: AHD

Total Depth: 12.07 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PT		D: 0.15-0.35 m	10/03/2026 11:39:45 ▼	4.0	0.0			TOPSOIL - Sandy Silty CLAY: low plasticity, dark brown; sand, fine to medium grained, sub-rounded; trace root fibres.	w ≈ PL	F	TOPSOIL
		D: 0.35-0.55 m					CI	Sandy CLAY: medium plasticity, brown; sand, fine to medium grained, sub-rounded.	w < PL	St - VSt	ALLUVIUM
		PP=450, 300, >600 kPa D: 0.55-2.00 m					CH	Sandy Silty CLAY: high plasticity, brown grey; sand, fine grained, sub-rounded.			
PVC		SPT: 1.50-1.95 m HW/300 mm, 0.0, 1 N=1			1.0						
					2.0			at 1.5 m: becoming dark grey	w > PL	VS	
WB		D: 3.00-3.45 m			3.0						
					4.0						
		SPT: 4.50-4.58 m 15/65 mm HB N=R			5.0			From 3.5-3.8 m: possible boulder	W	L	
								Log continued on next page.			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: A.To
Start Date: 19/02/26

Checked by: J Riley
End Date: 19/02/26
Location Meth.: dGPS0.1
RL: 4.236 m
Ver. Datum: AHD
Surface: Topsoil

Driller: Pointe Engineering	Hole Diameter: 90 mm
Drill Rig: Ripamonti Skid Drill	Inclination: 90°
Total Depth: 12.07 m	Bearing: N/A

Easting: 314222.38
Northing: 6238437.46
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: ○ UCS: □ ▢ ▣ ▤ ▥ ▦ ▧ ▨ ▩	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0					SOL 0-6 VL 2 L 6 LL 8 EH 20 VH 60 EH 200	20 60 200 600 2000	
NMLC			Is(50)		-2.0		<i>Log continued from previous page.</i> SANDSTONE: medium grained, pale grey, red-brown and yellow-brown, indistinct bedding, with fine gravel inclusions, iron-stained..	MW	100 (87) [80]	● c		4.64-4.74 m: EW, clay, 100 mm thick

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: A.To
Start Date: 19/02/26

Checked by: J Riley
End Date: 19/02/26
Location Meth.: dGPS0.1
RL: 4.236 m
Ver. Datum: AHD
Surface: Topsoil

Driller: Pointe Engineering
Drill Rig: Ripamonti Skid Drill
Total Depth: 12.07 m
Hole Diameter: 90 mm
Inclination: 90°
Bearing: N/A

Easting: 314222.38
Northing: 6238437.46
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC	1		A=0.28 MPa Is(50) D=0.89 MPa	-1.0			SANDSTONE: medium grained, pale grey, red-brown and yellow-brown, indistinct bedding, with fine gravel inclusions, iron-stained..	MW				5.06-5.09 m: EW, clay, 30 mm thick
	5.55						from 5.3 m: becoming medium to coarse grained		100 (99) [96]			5.55-6.55 m: MB, 100-200 mm spacing, x 8 5.62 m: J, 13°, healed 5.65-5.68 m: J, 25°, RF, IR, SN, 30 mm spacing, x 2 5.69 m: J, 12°, healed 5.77-5.78 m: EW, Qz, 10 mm thick
	2		Is(50) A=2.49 MPa Is(50) D=0.64 MPa	6.0								6.08-6.37 m: J, 15-20°, 80-120 mm spacing, healed, x 4
	6.55		Is(50) A=1.01 MPa Is(50) D=1.48 MPa	-2.0			at 6.18 m: shale clast inclusion		100 (93) [87]			6.55-7.54 m: MB, 100-300 mm spacing, x 5
	3		C: 6.60-6.75 m UCS=21 MPa	7.0			from 6.86-7.20 m: becoming pale grey and pale red-brown, trace dark brown laminations					7.02-7.05 m: EW, clay, 30 mm thick
	7.54		Is(50) A=0.34 MPa Is(50) D=0.38 MPa	-3.0			from 7.20-7.27 m: pale grey-white from 7.29-7.54 m: becoming brown, with fine gravel inclusions	SW	100 (95) [95]			7.22-7.26 m: EW, clay, 40 mm thick 7.34-7.40 m: J, RF, IR, 10-20 mm aperture, 20 mm spacing, gravel inclusions, x 3 7.54-8.50 m: MB, 100-300 mm spacing, x 8
5	4		Is(50) A=0.31 MPa Is(50) D=0.38 MPa	8.0			SANDSTONE: medium to coarse grained, pale grey with grey laminations, thickly laminated at 0-15, trace iron staining.		99 (98) [85]			7.80 m: P, 17°, RF, UN, SN 7.94 m: J, 60-70°, VR, IR, SN, healed 8.04-8.09 m: EW, clay, 50 mm thick
	8.50			-4.0			from 8.79-9.02 m: trace carbonaceous laminations					8.50-9.46 m: MB, 60-200 mm spacing, x 5 8.79-8.91 m: P, 15-20°, RF, IR, Qz, 5-10 mm aperture, 60 mm spacing, x 3 8.95 m: J, 70°, RF, UN, SN
	5		C: 9.20-9.35 m UCS=4.3 MPa	9.0					100 (97) [90]			9.43-9.45 m: EW, clay, 20 mm thick 9.46-10.46 m: MB, 20-70 mm spacing, x 5 9.47-9.50 m: EW, clay, 30 mm thick
6	9.46		Is(50) A=0.32 MPa Is(50) D=0.27 MPa	-5.0								9.83 m: J, 60°, RF, UN, CN
				10.0				MW				

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A.To

Start Date: 19/02/26

Checked by: J Riley

End Date: 19/02/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering

Hole Diameter: 90 mm

Drill Rig: Ripamonti Skid Drill

Inclination: 90°

Total Depth: 12.07 m

Bearing: N/A

Easting: 314222.38

Northing: 6238437.46

Hor. Datum: MGA2020-56

RL: 4.236 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: ○ UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC	10.46	7	Is(50) A=0.77 MPa Is(50) D=0.99 MPa Is(50) A=0.32 MPa Is(50) D=0.36 MPa	-6.0	11.0	[GRAPHIC LOG]	SANDSTONE: medium to coarse grained, pale grey with grey laminations, thickly laminated at 0-15, trace iron staining. from 10.21 to 10.75 m: becoming very thinly bedded at 50-60, pale yellow-brown, iron-stained from 10.75 m: becoming very thinly bedded at 0-15	MW	100 (100) [100]	[INFERRED STRENGTH]	[DEFECT SPACING]	10.00-10.02 m: EW, clay, 20 mm thick 10.07 m: J, 87°, VR, ST, SN
								SW				10.37 m: J, 65°, RF, UN, SN 10.46-12.07 m: MB, 50-400 mm spacing, x 8
			Is(50) A=0.98 MPa Is(50) D=1.27 MPa	-8.0	13.0		Terminated at 12.07 m. Target depth.					11.45 m: J, 22°, RF, IR, CN

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: R Rees/ A To
Start Date: 06/03/26
Checked by: J Riley
End Date: 09/03/26
Location Meth.: dGPS0.1
RL: 1.173 m
Ver. Datum: AHD
Surface: Topsoil

Driller: Pointe Engineering
Drill Rig: Ripamonti Skid Drill
Total Depth: 17.46 m
Hole Diameter: 90 mm
Inclination: 90°
Bearing: N/A

Easting: 314185.74
Northing: 6238490.88
Hor. Datum: MGA2020-56

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
WS	PVC	D: 0.00-0.35 m	06/09/2026 10:00:00	0.0	0.0		CI	TOPSOIL - Clayey Silty SAND: fine to medium grained, dark brown; trace root fibres.	M		TOPSOIL
		D: 0.35-0.65 m PP=40, 50, 60 kPa		1.0	1.0		CI	Sandy Silty CLAY: medium plasticity, dark brown mottled yellow; sand, fine to medium grained.	w > PL	VS	ALLUVIUM
WB	HWT	D: 0.65-1.40 m	06/09/2026 10:00:00	1.0	1.0		SC	Clayey SAND: fine to medium grained, grey brown; trace organic material and root fibres; strong organic odour.	W	VL	
		SPT: 1.50-1.95 m 1,1.0 N=1 D: 1.50-1.88 m		2.0	2.0		CI	Sandy Silty CLAY: medium plasticity, dark grey; sand, fine to medium grained; trace charcoal.			
		D: 1.88-1.95 m PP=20, 40, 60 kPa		3.0	3.0		CH	Silty CLAY: high plasticity, grey-brown; trace charcoal.			
		SPT: 3.00-3.45 m HW/350 mm, 0.0,1 N=1 D: 3.00-3.50 m PP=20, 30, 60 kPa		4.0	4.0		SC	from 4.00 m: with minor sand; sand is fine-grained	w > PL	VS	
			06/09/2026 10:00:00	5.0	5.0		SC	Clayey Silty SAND: fine to medium grained, dark grey.	W	VL	

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Driller: Pointe Engineering

Drill Rig: Ripamonti Skid Drill

Total Depth: 17.46 m

Hole Diameter: 90 mm

Inclination: 90°

Bearing: N/A

Project No: 60731324

Logged by: R Rees/ A To

Start Date: 06/03/26

Easting: 314185.74

Northing: 6238490.88

Hor. Datum: MGA2020-56

Checked by: J Riley

End Date: 09/03/26

Location Meth.: dGPS0.1

RL: 1.173 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
WB	HWT			-7.0			SC	Clayey Silty SAND: fine to medium grained, dark grey.	W	VL	
				-9.0				Log continued on next page.			
				-8.0							
				-10.0							
				-9.0							
				-11.0							
				-10.0							
				-12.0							
				-11.0							
				-13.0							
				-12.0							
				-14.0							
				-13.0							
				-15.0							
				-14.0							
				-16.0							

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: R Rees/ A To
Start Date: 06/03/26
Checked by: J Riley
End Date: 09/03/26
Location Meth.: dGPS0.1
RL: 1.173 m
Ver. Datum: AHD
Surface: Topsoil

Driller: Pointe Engineering
Drill Rig: Ripamonti Skid Drill
Total Depth: 17.46 m
Hole Diameter: 90 mm
Inclination: 90°
Bearing: N/A

Easting: 314185.74
Northing: 6238490.88
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ Soil: VL L M H VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					8.0							
					-7.0							
					9.0		<i>Log continued from previous page.</i>					
					-8.0		NO CORE: 9.03 - 9.43 m.		76 (76) [71]			HAWKESBURY SANDSTONE
					10.0		SANDSTONE: medium grained, pale grey and pale yellow-brown, indistinctly bedded; occasional iron staining. from 9.60 m: Becoming orange-brown, iron stained	MW				9.47-10.42 m: MB, 100-300 mm spacing, x 7 From 9.60 m: Core appears to be rotating within the barrel causing mechanical breakages.
					-9.0		from 10.24 to 10.66 m: very thinly bedded to thickly laminated at 10-15°, trace iron stained					
					11.0		from 10.66 to 10.92 m: pale grey		98 (98) [95]			10.73 m: DL
					-10.0		from 11.22 to 11.72 m: pale grey	MW - SW				10.94-10.96 m: P, 0-5°, RF, UN, SN, x 2 11.00-12.00 m: MB, 200-400 mm spacing, x 6
					12.0		from 11.72 to 11.79 m: very thinly bedded at 5-20°					
					-11.0		from 12.27-12.40 m: cross-bedded, typically 15-20 mm spacing from 12.50 to 12.80 m: very thinly bedded		84 (81) [77]			12.03 m: DL 12.13 m: P, 15°, SM, PR, SN 12.16-12.87 m: MB, average 500 mm spacing, x 4
					13.0		at 12.94 m: gravel inclusions					12.93-12.97 m: P, 15°, RF, UN, Fe 13.00 m: MB
					-12.0		NO CORE: 13.10 - 13.31 m.					
					14.0		SANDSTONE: medium grained, pale brown speckled orange brown, indistinct bedding. NO CORE: 13.48 - 14.06 m.	MW	56 (36) [36]			13.35 m: MB 13.38 m: DL 13.44 m: J, 75°, RF, UN, SN
					-13.0		SANDSTONE: medium grained, pale grey and pale brown, indistinct bedding, trace carbonaceous laminations . from 14.35 to 14.47 m: iron-stained	HW MW				14.07 m: J, 75°, RF, UN, SN 14.25-14.30 m: EW, 50 mm thick 14.38 m: J, 87°, RF, UN, SN 14.48-15.10 m: MB, 60-100 mm spacing, x 8 14.63 m: J, 17°, RF, UN, CN 14.69 m: DL 14.81 m: P, 7°, RF, UN, CN
					15.0		NO CORE: 15.13 - 15.85 m.		46 (33) [33]			15.08 m: P, 0°, RF, CU, CN 15.10 m: J, 45°, RF, PR, CN
					-14.0		SANDSTONE: fine to medium grained, pale grey, indistinct bedding.	HW				15.91-15.95 m: CS, 40 mm thick
					16.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Driller: Pointe Engineering

Drill Rig: Ripamonti Skid Drill

Total Depth: 17.46 m

Hole Diameter: 90 mm

Inclination: 90°

Bearing: N/A

Project No: 60731324

Logged by: R Rees/ A To

Start Date: 06/03/26

Easting: 314185.74

Northing: 6238490.88

Hor. Datum: MGA2020-56

Checked by: J Riley

End Date: 09/03/26

Location Meth.: dGPS0.1

RL: 1.173 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: o UCS: □ SOIL: VL L M H EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
	16.00		Is(50) A=1.23 MPa Is(50) D=1.59 MPa C: 16.40-16.55 m UCS=27 MPa Is(50) A=1.43 MPa Is(50) D=1.28 MPa	16.0 15.0	16.0 15.0		SANDSTONE: fine to medium grained, pale grey, indistinct bedding.	FR	94 (94) [93]			16.00 m: DL
	6				17.0 16.0		NO CORE: 16.97 -17.06 m. SANDSTONE: medium grained, pale brown, indistinct bedding , trace carbonaceous laminations .	MW				17.38 m: P, 10°, RF, UN, Qz
					18.0 17.0 19.0 18.0 20.0 19.0 21.0 20.0 22.0 21.0 23.0 22.0 24.0		Terminated at 17.46 m.					

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: R Rees
Start Date: 03/03/26
Checked by: J Riley
End Date: 04/03/26
Location Meth.: dGPS0.1
Driller: Pointe Engineering Pty Ltd
Hole Diameter: 76 mm
Eastings: 314148.35
Northings: 6238530.59
Hor. Datum: MGA2020-56
Ver. Datum: AHD
Surface: TOPSOIL

Drill Rig: Ripamonti Skid Drill
Inclination: 90°
Bearing: N/A
Total Depth: 7.99 m

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
W	S	U	D: 0.00-0.06 m D: 0.06-0.13 m		0.0			Silty SAND: fine to medium grained, brown; trace gravel, gravel is fine to coarse, sub-angular, sandstone . Gravelly SAND: fine to medium grained, orange-brown; gravel, fine to coarse, sub-angular, sandstone . <i>Log continued on next page.</i>	D		TOPSOIL
					4.0						
					1.0						
					3.0						
					2.0						
					2.0						
					3.0						
					1.0						
					4.0						
					0.0						
					5.0						
					1.0						
					6.0						
					2.0						
					7.0						
					3.0						
					8.0						

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 03/03/26

Checked by: J Riley

End Date: 04/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering Pty Ltd

Hole Diameter: 76 mm

Easting: 314148.35

RL: 4.22 m

Drill Rig: Ripamonti Skid Drill

Inclination: 90°

Northing: 6238530.59

Ver. Datum: AHD

Total Depth: 7.99 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: TOPSOIL

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ Soil: 0-4 L 2 M 2 H 2 VH 2 EH 2	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0		<i>Log continued from previous page.</i>					
	1		Is(50) A=0.38 MPa Is(50) D=0.27 MPa Is(50) A=0.7 MPa Is(50) D=0.69 MPa	1.40	4.0		NO CORE: 0.13 - 0.60 m (470mm).		63 (63) [63]			
					1.0		SANDSTONE: fine to medium grained, pale grey with black flecks and mottled orange-brown, indistinctly bedded, iron stained. from 0.83 m: becoming laminated to very thinly bedded at 0-5deg from 1.07 m: becoming pale red-brown with black flecks	MW				HAWKESBURY SANDSTONE 0.62 m: P, 25°, RF, ST, CT rootlets 0.67-0.79 m: DB, x 2 0.83 m: P, 5°, RF, UN, CT rootlets and silt, 1 - 3 mm aperture 0.96 m: P, 15°, SM, UN, CN 1.00-1.44 m: MB, x 4
	2			2.92	2.0		from 1.37 m: becoming indistinctly bedded from 1.55 m: becoming red-brown; distinctly iron stained from 1.64 - 1.86 m: becoming pale grey; occasional carbonaceous material.	SW	95 (92) [81]			From 1.5 - 5.5 m: Drill rig having feed and rotation issue causing it to build up pressure and then 'jolt' resulting in mechanical breakages at similar intervals 1.50 m: P, 0°, SM, PR, CN 2.00-3.00 m: MB, x 6 2.34-2.38 m: EW, SP
			Is(50) A=0.38 MPa Is(50) D=0.42 MPa		3.0		NO CORE: 2.38 - 2.45 m (70 mm). SANDSTONE: fine to medium grained, pale red brown with black flecks, indistinctly bedded, occasional carbonaceous material; occasional iron staining.	SW	100 (97) [79]			
	3			3.69	1.0		from 3.23 - 3.55 m: becoming cross-bedded at 5 - 10 mm at 50-60deg from 3.57 - 3.61 m: becoming pale grey	MW to HW SW SW				3.50 m: J, 0-5°, UN 3.56 m: P, 0°, PR-IR, RF 3.69 m: DB
			Is(50) A=0.33 MPa Is(50) D=0.76 MPa		4.0		from 4.07 - 4.33 m: becoming red-brown with orange-brown banding; laminated to very thinly bedded at 20deg; iron stained from 4.45 to 4.53 m: cross bedding visible from 4.70 m: trace inclusions of silt, up to 5 mm					3.86 m: P, 12°, SM, UN, CT Fe 3.91 m: P, 0°, SM, PR, CN, mechanically rounded 3.97 m: P, 10°, RF, UN, CT Fe, Carbonaceous material 4.00 m: HB 4.06 m: P, 25°, RF, PR, CT Fe 4.14 m: P, 0 - 5°, SM, PR, CN, mechanical rounding on parting 4.20 m: HB 4.32 m: P, 0°, SM, PR, CN, mechanically rounded 4.43 m: J, 25°, RF, UN, CT Fe 4.49 m: DB 4.70 m: J, 80°, RF, UN, CN 5.00-5.04 m: HB, x 2
	4		Is(50) A=1.76 MPa Is(50) D=1.78 MPa C: 4.40-4.55 m UCS=6.9 MPa	5.31	0.0		NO CORE: 5.11 - 5.31 m (200 mm).					5.31 m: MB 5.44 m: P, 15°, SM, PR, CT clay
			Is(50) A=0.91 MPa Is(50) D=0.76 MPa		1.0		SANDSTONE: fine to medium grained, pale grey mottled brown with black flecks, medium to thickly bedded at 0-20°, trace siltstone inclusions up to 3 mm wide; trace iron staining. from 5.52 m: becoming pale grey, grey and red-brown	SW	94 (94) [82]			
	5			6.68	6.0		NO CORE: 5.82 - 5.90 m (80 mm). SANDSTONE: fine to medium grained, pale grey, grey and red-brown, thinly to very thinly bedded at 0-20°, trace cross-bedding, iron stained.	MW SW				5.94 m: P, 0°, SM, PR, CN, mechanically rounded 6.00-6.10 m: HB, x 3 6.30 m: P, 0°, SM, PR, CN, mechanically rounded
			Is(50) A=0.9 MPa Is(50) D=1.09 MPa		2.0		from 6.68 m: iron stained defects	MW SW MW	93 (93) [50]			6.47 m: 0 - 15°, RF, UN, CT X 6.53-6.61 m: P, 0 - 5°, SM, UN, CT X 6.68 m: DB 6.73 m: J, 40°, RF, UN, CT Fe, 2 - 3 mm aperture 6.91-7.04 m: P, 0 - 5°, RF, UN, CT X, 4x partings 7.02-7.21 m: MB, 65 - 70°, RF, UN, CT Fe, 50 - 90 mm spacing 7.20 m: P, 0°, SM, CU, CN 7.23 m: P, 0°, SM, PR, CN 7.30 m: P, 0°, SM, CU, CN
	6		C: 7.20-7.35 m UCS=17 MPa Is(50) D=0.75 MPa Is(50) A=0.49 MPa		3.0		NO CORE: 7.34 - 7.43 m (90 mm). SANDSTONE: fine to medium grained, pale grey with black flecks and mottled red-brown, very thinly bedded at 15-20° and cross-bedded at 30-40°, trace gravel inclusions, fine, subrounded, quartz; occasional iron staining. from 7.48 - 7.78 m: trace carbonaceous material	HW to XW XW FR to SW				7.66-7.71 m: P, 10°, RF, UN, CT
					8.0							

Terminated at 7.99 m. Target depth.

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 27/02/26

Checked by: J Riley

End Date: 02/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering Pty Ltd

Hole Diameter: 85-76 mm

Easting: 314108.46

RL: 14.314 m

Drill Rig: Ripamonti Skid Drill

Inclination: 90°

Northing: 6238562.07

Ver. Datum: AHD

Total Depth: 8.41 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
WS	U	D: 0.00-0.15 m D: 0.15-0.20 m			0.0			TOPSOIL - Silty SAND: fine to medium grained, pale brown; trace gravel, fine to medium grained, sub-rounded, sandstone.	D		TOPSOIL
					14.0		SM	Silty SAND: fine to medium grained, yellow-brown; trace gravel, fine to medium grained, sub-angular, sandstone.			at 0.20 m: refusal on sandstone
								Log continued on next page.			
					1.0						
					13.0						
					2.0						
					12.0						
					3.0						
					11.0						
					4.0						
					10.0						
					5.0						
					9.0						
					6.0						
					8.0						
					7.0						
					8.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 27/02/26

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Location Meth.: dGPS0.1

Driller: Pointe Engineering Pty Ltd

Hole Diameter: 85-76 mm

Easting: 314108.46

Drill Rig: Ripamonti Skid Drill

Inclination: 90°

Northing: 6238562.07

Total Depth: 8.41 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: x Is(50) A: x Is(50) UCS: x Is(50) UCS: x Is(50)	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0							
							<i>Log continued from previous page.</i>					
					14.0		NO CORE: 0.20 - 0.48 m.		77 (77) [77]			HAWKESBURY SANDSTONE
							SANDSTONE: medium to coarse grained, pale grey to pale yellow-brown, laminated to very thinly bedded at 0-15°, trace gravel inclusions; fine, quartz; iron stained.	MW - SW				0.48-0.52 m: MB 0.59 m: P, 0°, SM, UN, CN 0.69 m: P, 15°, SM, UN, CN 0.72 m: DB
					1.0							1.00 m: HB
					1.40		from 1.36: becoming very thinly bedded at 0-5°		77 (74) [69]			1.40 m: DB 1.46 m: P, 5°, SM, UN, CN
							NO CORE: 1.80 - 2.04 m.	XW				1.77-1.80 m: EW
					2.0		SANDSTONE: medium to coarse grained, pale grey, very thinly bedded at 0-15°.	MW - SW				2.25-2.26 m: P, 10-15°, SM, UN, CN, 10 mm spacing 2.30 m: P, 0°, SM, UN, CN 2.43 m: DB
					2.43		from 2.32 m: becoming pale grey, yellow-brown and pale red-grey; trace black flecks; iron stained		95 (95) [93]			2.66 m: P, 0°, SM, UN, CN 2.73 m: HB
							NO CORE: 2.51 - 2.60 m.					3.00 m: P, 0°, SM, UN, CT Fe 3.07 m: HB 3.10 m: HB 3.29 m: P, 0°, SM, PR, CN 3.43 m: P, 5°, SM, UN, CN 3.48-3.51 m: J, 35°, Fe, 2 mm aperture
							SANDSTONE: medium to coarse grained, red-grey, indistinctly bedded, trace gravel inclusions, fine, sub-angular to sub-rounded, quartz; iron stained; iron staining.	MW - SW				3.83 m: P, 5°, SM, PR, CT ML 3.96-4.11 m: HB, x 3
					3.0		from 2.66 m: becoming pale grey from 2.75 m: becoming pale grey mottled yellow-brown with dark brown staining					4.79 m: J, 10°, SM, UN, CT Fe 4.83-4.86 m: SZ, 0-5° 4.92 m: P, 0°, SM, CU, CN 5.00 m: HB 5.18-5.20 m: EW At 5.3 m: 90% water loss and drill head descending quickly. Back to 90% flow at 5.5 m 5.46 m: MB
					3.50		from 3.50 m: becoming yellow-brown					5.78 m: P, 0°, SM, CU, CN 5.87 m: DB 5.94 m: J, 35°, SM, PR, CN 6.00 m: HB 6.02 m: HB 6.04 m: P, 5°, SM, UN, CN 6.32 m: P, 25°, SM, PR, CN 6.37 m: P, 5°, RF, UN, CN
					4.0		from 4.24 - 4.38 m: pale grey from 4.38 m: becoming very thinly bedded at 0-20°		88 (85) [85]			6.73 m: P, 20°, SM, PR, CN 6.91 m: P, 5°, RF, UN, CN 7.00 m: HB 7.13 m: P, 5°, SM, UN, SN Fe, CT ML
					4.11		from 4.81 - 4.91 m: pale grey from 4.91 m: becoming red-brown, pale grey and yellow-brown; thinly laminated at 10°					7.45 m: DB From 7.6 m - 7.7 m: 100% flush loss until approx 7.9 m 7.76 m: P, 15°, RF, UN, CN 7.80-7.85 m: P, 15°, SM, UN, CT Fe, 15 spacing, x 3
					5.0		NO CORE: 5.20 - 5.41 m.					
							SANDSTONE: medium to coarse grained, pale grey with black flecks, thinly bedded to very thinly bedded 0-25°, trace gravel inclusions, fine, sub-rounded, quartz, iron stained.	MW - SW				
					5.87		from 5.46 m: becoming red-brown to orange-brown with black flecks		100 (100) [87]			
							from 6.46 - 6.49 m: increase in gravel inclusions, fine, sub-angular, quartz					
					6.0							
					7.0							
					7.45							
					8.0							
					8.41							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Driller: Pointe Engineering Pty Ltd

Drill Rig: Ripamonti Skid Drill

Total Depth: 8.41 m

Hole Diameter: 85-76 mm

Inclination: 90°

Bearing: N/A

Project No: 60731324

Logged by: R Rees

Start Date: 27/02/26

Easting: 314108.46

Northing: 6238562.07

Hor. Datum: MGA2020-56

Checked by: J Riley

End Date: 02/03/26

Location Meth.: dGPS0.1

RL: 14.314 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: o UCS: □ SOIL: VL L M H VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC			A=1.38 MPa Is(50) D=0.46 MPa Is(50) A=0.62 MPa Is(50) D=0.18 MPa	6.0			SANDSTONE: medium to coarse grained, pale grey with black flecks, thinly bedded to very thinly bedded 0-25°, trace gravel inclusions, fine, sub-rounded, quartz, iron stained.	MW - SW				8.00 m: HB 8.06 m: P, 25°, RF, UN, CT Fe 8.16 m: P, 10°, RF, UN, CT Fe 8.34-8.41 m: MB
							Terminated at 8.41 m.					
					9.0							
					5.0							
					10.0							
					4.0							
					11.0							
					3.0							
					12.0							
					2.0							
					13.0							
					1.0							
					14.0							
					0.0							
					15.0							
					1.0							
					16.0							



Engineering Log

FINAL

Borehole No. DBH21
Sheet 1 of 3

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: A Sharma
Start Date: 05/03/26
Checked by: J Riley
End Date: 06/03/26
Location Meth.: dGPS0.1
RL: 13.8 m
Ver. Datum: AHD
Surface: TOPSOIL

Driller: Pointe Engineering
Drill Rig: Atlas Copco R Cobra
Total Depth: 12.76 m
Hole Diameter: 90 mm
Inclination: 90°
Bearing: N/A

Easting: 314564.64
Northing: 6238027.64
Hor. Datum: MGA2020-56

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
NMLC	U	D: 0.20-0.70 m			0.0		SM	Silty SAND: fine grained, brown, with roots; silt, low plasticity.	D		TOPSOIL
								Silty SAND: fine to coarse grained, orange-brown, with gravel, fine to medium grained, angular.			RESIDUAL SOIL
								Log continued on next page.			
					13.0						
					1.0						
					12.0						
					2.0						
					11.0						
					3.0						
					10.0						
					4.0						
					9.0						
					5.0						
					8.0						
					6.0						
					7.0						
					7.0						
					6.0						
					6.0						

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Logged by: A Sharma

Start Date: 05/03/26

Checked by: J Riley

End Date: 06/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering
Drill Rig: Atlas Copco R Cobra
TTe

Hole Diameter: 90 mm

Inclination: 90°

Bearing: N/A

Easting: 314564.64

Northing: 6238027.64

Hor. Datum: MGA2020-56

RL: 13.8 m

Ver. Datum: AHD

Surface: TOPSOIL

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: ○ UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
										SOIL 0-4 VL 2 L 6 E 8 H 10 VN 12 EH 14	20 60 200 600 2000	
					0.0							
							<i>Log continued from previous page.</i>					
	1				13.0		Cobbles GRAVEL: coarse grained, angular to sub-angular, red-brown, ironstone; trace roots.	XW	100 (23) [0]			EXTREMELY WEATHERED MATERIAL
	1.20		Is(50) A=2.59 MPa Is(50) D=0.8 MPa		1.0		Sandy CLAY: medium plasticity, orange-brown; sand, fine to medium grained; trace gravel, fine, sub-rounded, quartz.					
			Is(50) A=1.18 MPa Is(50) D=2.38 MPa				Clayey Gravelly SAND: fine to coarse grained, yellow-brown mottled pale grey; gravel, fine to coarse, sub-angular to sub-rounded, sandstone; trace roots, iron stained.	MW				0.96 m: P, 10°, RF, PR, SN, Fe 1.07 m: P, 10°, RF, PR, CN 1.15 m: P, 5°, RF, CU, VN roots 1.20 m: DB
	2						SANDSTONE: medium to coarse grained, red-brown, thinly bedded at 0-10°, trace fine gravel sized quartz inclusions, iron stained.		61 (46) [0]			1.47-1.57 m: CS
							NO CORE: 1.20 - 1.47 m.					1.62 m: DB 1.64 m: P, 0°, CT Cl, roots 1.67 m: P, 0°, CT Cl, roots 1.77 m: J, 75°, RF, PR, CN 1.83-2.00 m: DB, x 3
	1.90				12.0		SANDSTONE: medium to coarse grained, red-brown, thinly bedded at 0-5°, trace fine gravel sized quartz inclusions, iron stained.	MW				2.35-2.37 m: CS, sandy Gravel
					2.0		from 1.60 m: pale grey and orange-brown		78 (72) [70]			2.70 m: J, 70°, RF, PR, VN Cl, roots 2.75-4.02 m: DB, x 6
							NO CORE: 1.90 - 2.35 m.					
							SANDSTONE: medium to coarse grained, pale grey and orange-brown, indistinctly bedded, with fine gravel sized quartz inclusions, iron stained.	MW				
	3		Is(50) A=1.8 MPa Is(50) D=1.38 MPa C: 3.25-3.40 m UCS=42 MPa		3.0							
							from 3.94 m: very thinly to thinly bedded at 5-20° with thin brown laminations, 1-5 mm thick		100 (91) [91]			4.38 m: P, 5°, RF, PR, CN 4.51 m: P, 10°, RF, CU, CN 4.67 m: P, 5°, RF, PR, CN partially healed 4.76 m: P, 5°, RF, PR, CN
	3.90		Is(50) A=1.45 MPa Is(50) D=1.04 MPa		4.0		from 4.32 m: pale grey, laminations are grey					5.00 m: HB 5.06 m: P, 5°, RF, PR, VN Cl
							from 5.10 to 5.20 m: cross bedding visible					5.37-5.38 m: EW, 5°, SC 5.41 m: DB 5.45 m: J, 75°, RF, PR, CN 5.51 m: DB 5.56 m: DB 5.61 m: J, 75°, RF, PR, VN Cl, roots 5.99-6.00 m: MB, x 2
	5.51		Is(50) A=0.49 MPa Is(50) D=0.75 MPa		5.0				100 (80) [80]			6.17 m: P, 10°, RF, PR, SN Fe 6.30-6.33 m: DB, multiple 6.33 m: J, 75°, RF, PR, VN Cl, roots, Fe
					6.0							6.75 m: P, 5°, RF, ST, CN
			Is(50) A=1.16 MPa Is(50) D=1.15 MPa		6.0							7.00 m: MB 7.03-7.04 m: EW, 5°, Cl
	7.03		C: 6.80-6.95 m UCS=15 MPa		7.0		from 7.04 m: very thinly bedded to laminated at 80°, orange-brown and pale grey	SW	100 (71) [68]			7.89 m: J, 10°, RF, PR, VN Fe
			Is(50) A=0.8 MPa Is(50) D=0.74 MPa		7.0							
	6				6.0		from 7.95 m: thinly laminated to laminated at 5-20°					

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A Sharma

Start Date: 05/03/26

Checked by: J Riley

End Date: 06/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering

Drill Rig: Atlas Copco R Cobra
TTG

Total Depth: 12.76 m

Hole Diameter: 90 mm

Inclination: 90°

Bearing: N/A

Easting: 314564.64

Northing: 6238027.64

Hor. Datum: MGA2020-56

RL: 13.8 m

Ver. Datum: AHD

Surface: TOPSOIL

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ □ □ □ SOL: 0-6 VL 7 R 8 M 11 VH 200 EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC	8.58	7	Is(50) A=2.26 MPa Is(50) D=2.39 MPa Is(50) A=1.24 MPa Is(50) D=1.27 MPa C: 10.30-10.45 m UCS=48 MPa				SANDSTONE: medium to coarse grained, pale grey and orange-brown, indistinctly bedded, with fine gravel sized quartz inclusions, iron stained.	SW				8.00 m: HB 8.04 m: J, 5°, RF, PR, SN Fe 8.19 m: J, 85-90°, RF, IR, SN Fe, partially healed 8.33 m: P, 5°, RF, PR, CN 8.40 m: P, 5°, RF, PR, CN 8.56 m: P, 5-10°, CU, VN Cl 8.58-9.00 m: HB, x 3
								SANDSTONE: medium to coarse grained, dark brown, brown and pale grey, very thinly bedded at 15°, trace fine to medium gravel sized quartz inclusions, iron stained.	100 (100) [100]			9.47 m: P, 15°, RF, PR, VN SC 9.69-10.00 m: DB, x 3
	10.19							from 9.06 m: indistinctly bedded, pale grey mottled orange-brown from 9.40 m: very thinly bedded at 15° with thin grey laminations from 9.59 m: pale grey mottled brown and dark brown from 10.13 m: pale grey mottled orange-brown	100 (100) [100]			10.19 m: P, 15°, RF, PR, CN 10.49 m: P, 15°, RF, PR, CN 10.89-11.76 m: HB, x 3
				8				from 11.30 m: thinly laminated to laminated at 0-5°, pale grey	FR			11.79-12.75 m: P, 0-20°, RF, PR, CN, x 5, spaced 100-490 mm
	11.76	9					from 11.82 m: very thinly to thinly bedded with thin grey siltstone laminations, at 15-20°		100 (100) [77]			
							Terminated at 12.76 m.					
				-1.0	13.0							
				-0.0	14.0							
				-1.0	15.0							
				-2.0								

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: AL
Start Date: 26/03/26

Checked by: C Jayasuriya
End Date: 26/03/26
Location Meth.: dGPS0.1

Driller: Pointe Engineering
Drill Rig: Speedy Drill
Total Depth: 1.4 m

Hole Diameter: 50 mm
Inclination: 90°
Bearing: N/A

Easting: 314176.21
Northing: 6238441.94
Hor. Datum: MGA2020-56

RL: 3.924 m
Ver. Datum: AHD
Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
WS	U	D: 0.00-0.35 m	Not Observed		0.0			TOPSOIL - Silty SAND: fine to coarse grained, dark brown; trace rootlets .	D		TOPSOIL
		D: 0.35-0.75 m					SM	Silty SAND: coarse grained, dark brown; trace fine angular gravel .	M		RESIDUAL SOIL
		D: 0.75-1.10 m					CH	Sandy CLAY: high plasticity, yellow-brown; sand, fine grained; with rootlets .			
		D: 1.10-1.40 m						Terminated at 1.40 m. Refusal on inferred top of rock.	w < PL		
					1.0						
					2.0						
					3.0						
					4.0						
					5.0						

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Driller: Pointe Engineering Pty Ltd

Drill Rig: Ripamonti Speedydrill

Total Depth: 8.1 m

Hole Diameter: 76-90 mm

Inclination: 60°

Bearing: 230°

Project No: 60731324

Logged by: A Liang

Start Date: 25/03/26

Checked by: C Jayasuriya

End Date: 26/03/26

Location Meth.: dGPS0.1

Easting: 314175.81

Northing: 6238441.70

Hor. Datum: MGA2020-56

RL: 3.97 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PT	U				0.0			TOPSOIL - Silty SAND: medium to coarse grained, dark brown.	M		TOPSOIL
					1.0			Log continued on next page.			
					3.0						
					2.0						
					2.0						
					3.0						
					1.0						
					4.0						
					0.0						
					5.0						

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: A Liang
Start Date: 25/03/26

Checked by: C Jayasuriya
End Date: 26/03/26
Location Meth.: dGPS0.1
RL: 3.97 m
Ver. Datum: AHD
Surface: Topsoil

Driller: Pointe Engineering Pty Ltd
Drill Rig: Ripamonti Speedydrill
Total Depth: 8.1 m
Hole Diameter: 76-90 mm
Inclination: 60°
Bearing: 230°

Easting: 314175.81
Northing: 6238441.70
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: o UCS: □ SOIL: 0-4 VL 2 L 6 M 20 H 60 VH 200 EH	DEFECT SPACING (mm) 20 60 200 600 2000	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0							
							<i>Log continued from previous page.</i>					
							SANDSTONE: coarse grained, orange-brown mottled red, indistinctly bedded .	HW	21 (4)			HAWKSBURO SANDSTONE
							CH Sandy Silty CLAY: high plasticity, pale grey mottled brown; sand, fine to medium grained ; trace root fibers .	XW	[4]			
							SANDSTONE: fine to coarse grained, red-brown, indistinctly bedded .	MW				
							NO CORE: 0.80 - 2.70 m .					
					1.0							
					2.0							
					2.70							
					3.0		SANDSTONE: medium to coarse grained, red-brown mottled grey, massive.	MW	47 (47)			2.79 m: P, 5°, RF, UN, SN Fe
							from 2.80m: SANDSTONE: medium to coarse grained, orange brown indistinctly laminated		[33]			2.96 m: J, 75°, RF, PR, CN
							from 2.90m: becoming pale brown					3.00 m: P, 5°, RF, PR, SN Fe
					3.0		SANDSTONE: medium to coarse grained, pale grey mottled yellow-brown, conglomerated, indistinctly cross laminated.	SW				3.04 m: P, 5°, RF, CU, SN Fe
												3.08 m: P, 10°, RF, PR, SN Fe
												3.16 m: DB
					1.0							3.32 m: DB
					4.0							
							NO CORE: 3.50 - 4.40 m.					
					0.0		SANDSTONE: medium to coarse grained, pale grey mottled brown and red, with fine to medium gravel, conglomerated, indistinctly cross laminated .	SW	86 (86)			4.60 m: P, 10°, RF, PR, SN CL
							SANDSTONE: medium to coarse grained, brown-grey, thinly laminated at 30°.	MW	[55]			4.67 m: P, 20°, RF, PR, SN Fe
												4.70 m: P, 20°, RF, PR, SN Fe
												4.81 m: DB
												4.86 m: DB
												4.93 m: DB

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: A Liang
Start Date: 25/03/26

Checked by: C Jayasuriya
End Date: 26/03/26
Location Meth.: dGPS0.1
RL: 3.97 m
Ver. Datum: AHD
Surface: Topsoil

Driller: Pointe Engineering Pty Ltd
Drill Rig: Ripamonti Speedydrill
Total Depth: 8.1 m
Hole Diameter: 76-90 mm
Inclination: 60°
Bearing: 230°

Easting: 314175.81
Northing: 6238441.70
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS(50) A: • D: o UCS: □ SOIL CLASSIFICATION VL L M H VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
	5.35		C: 5.44-5.62 m UCS=13 MPa Is(50) D=0.6 MPa Is(50) A=1.4 MPa				SANDSTONE: medium to coarse grained, brown-grey, thinly laminated at 30°.	MW				5.15 m: P, 15°, RF, PR, CT CL 5.24 m: J, 30°, RF, IR, SN, Fe and clay 5.34 m: DB
							SANDSTONE: fine to coarse grained, pale grey, massive.	SW to FR	100 (83) [49]			
							SANDSTONE: fine to coarse grained, brown to orange brown, cross-bedded at 30-40 with dark-brown laminations.					5.70 m: J, 60°, PR, healed 5.80 m: J, 75°, PR 5.88 m: DB
							SANDSTONE: fine to coarse grained, pale grey, mottled red-brown and orange brown, indistinctly bedded.	MW				6.09 m: J, 80°, RF, PR, SN, Fe 6.22 m: J, 30°, RF, PR, SN
							SANDSTONE: fine to coarse grained, red and orange-brown, very thinly cross bedded.	HW				6.49 m: J, 15°, RF, CU, CN 6.53 m: J, 30°, RF, PR, CN 6.73 m: DB 6.86 m: P, 10°, RF, UN, SN, Fe 6.95 m: P, 20°, RF, CU, SN, Fe 7.05 m: P, 30°, RF, PR, SN, Fe 7.10 m: P, 5°, RF, PR, Fe 7.15 m: P, 10°, RF, PR, SN, Fe 7.25 m: P, 20°, RF, PR, SN, Fe 7.31 m: P, 20°, RF, UN, SN, Fe 7.34 m: P, 20°, RF, PR, SN, Fe 7.45 m: P, 5°, RF, UN, SN, Fe 7.56 m: P, 30°, RF, UN, SN, Fe 7.62 m: P, 60°, RF, PR, SN, Fe 7.75-7.82 m: EW, RF, IR, SN, Fe
							NO CORE: 7.82 - 8.10 m.					
							Terminated at 8.10 m. Target depth.					

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project No: 60731324

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Logged by: A Sharma

Checked by: C Jayasuriya

Location: Heathcote Road, Holsworthy, NSW 2173

Start Date: 30/03/26

End Date: 30/03/26

Driller: Pointe Engineering Pty Ltd

Hole Diameter: 90 mm

Easting: 314096.66

Location Meth.: dGPS0.1

Drill Rig: Ripamonti Speedydrill

Inclination: 60°

Northing: 6238513.60

Ver. Datum: AHD

Total Depth: 9.89 m

Bearing: 230°

Hor. Datum: MGA2020-56

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PT	PVC	D: 0.00-0.25 m	Not Observed		0.0		SP	TOPSOIL - Gravelly SAND: fine to coarse grained, poorly graded, orange-brown; gravel, coarse grained sub-angular gravel ; with rootlets .	D		TOPSOIL
		D: 0.25-0.35 m					GP	Sandy GRAVEL: fine to coarse grained, poorly graded, sub-angular to angular, orange-brown; sand, coarse grained sand ; trace rootlets .	M		RESIDUAL SOIL
		D: 0.35-1.00 m						Log continued on next page.			
					20.0						
					1.0						
					19.0						
					2.0						
					18.0						
					3.0						
					17.0						
					4.0						
					5.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A Sharma

Start Date: 30/03/26

Checked by: C Jayasuriya

End Date: 30/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering Pty Ltd

Drill Rig: Ripamonti Speedydrill

Total Depth: 180

9.89 m

Hole Diameter: 50 mm

Inclination: 60°

Bearing: 230°

Easting: 314096.66

Northing: 6238513.60

Hor. Datum: MGA2020-56

RL: 20.486 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ Soil: VL L M H VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0							
							<i>Log continued from previous page.</i>					
					20.0		SANDSTONE: fine to coarse grained, yellow brown and grey, indistinctly bedded.	HW	100 (90) [79]			0.53 m: P, 5°, RF, PR, trace rootlets 0.58 m: J, 50°, RF, PR, CN 0.60 m: P, 10°, RF, PR, CN trace rootlets, Fe 0.66 m: J, 50°, RF, PR, SN, Fe
			Is(50) D=0.5 MPa Is(50) A=0.7 MPa D: 1.00-1.40 m				from 0.70 m: indistinctly laminated	MW				0.81 m: P, 20°, RF, PR, SN, Fe
					1.0							1.09 m: DB
					19.0							1.74 m: J, 60°, RF, PR, SN
			Is(50) D=0.9 MPa Is(50) A=0.4 MPa		2.0							
							from 2.41 to 2.51 m: pale grey		100 (92) [79]			2.38 m: DB
							at 2.51 m: SANDSTONE: fine to coarse grained, yellow-brown, thinly laminated at 0-5°	FR				2.47 m: P, 5°, RF, PR, SN, Fe
			Is(50) D=0.9 MPa Is(50) A=0.6 MPa				from 2.80 to 3.00 m: pale grey mottled red					2.59 m: DB
					18.0							3.08 m: DB
							at 3.30 m: SANDSTONE: fine to coarse grained, pale grey, indistinctly laminated		100 (98) [85]			3.43 m: P, 5°, RF, PR, SN Fe
			Is(50) D=0.9 MPa Is(50) A=0.8 MPa									3.53-3.82 m: J, 80°, RF, UN, SN
								SW				3.81 m: J, 20°, RF, PR, CN
					17.0							4.18 m: P, 30°, RF, UN, SN Fe
			Is(50) D=1.1 MPa Is(50) A=1.5 MPa				from 4.40 m: yellow-brown	MW				4.88 m: DB
					5.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Location: Heathcote Road, Holsworthy, NSW 2173

Start Date: 30/03/26

Location Meth.: dGPS0.1

Bearing: 230°

Hor. Datum: MGA2020-56

Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: ○ UCS: □ ▢ ▣ ▤ ▥ ▦ ▧ ▨ ▩	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC	3		Is(50) D=0.6 MPa Is(50) A=0.7 MPa		16.0		SANDSTONE: fine to coarse grained, yellow brown and grey, indistinctly bedded.	MW				5.00 m: HB
	6.78		Is(50) D=0.9 MPa Is(50) A=0.7 MPa		6.0		at 5.52 m: SANDSTONE: fine to coarse grained, pale grey, massive	HW XW SW				5.19 m: P, 5°, RF, PR, SN, Fe
		Not Observed	C: 7.51-7.71 m UCS=15 MPa Is(50) D=0.7 MPa Is(50) A=0.9 MPa		15.0							5.46 m: P, 20°, RF, PR, SN, Fe 5.50-5.52 m: IS, 20°, SM, PR, SN clay, Fe
					7.0							5.69 m: HB
					14.0							6.09 m: J, 50°, RF, UN, CN
	4		Is(50) D=0.5 MPa Is(50) A=0.5 MPa		8.0		from 8.00 m: SANDSTONE: fine to coarse grained, yellow-brown and red-brown, thinly laminated at 0-80°	MW				6.78 m: DB
			Is(50) D=0.7 MPa Is(50) A=0.5 MPa		13.0							7.93 m: HB
					9.0							8.03 m: CS, 20°, SM, PR, VN clay, Fe 8.08 m: P, 20°, RF, PR, SN sand 8.17-8.23 m: J, 70°, RF, UN, SN, Fe
					12.0							8.31 m: DB
							Terminated at 9.89 m. Target depth.					9.51 m: DB

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project No: 60731324

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Logged by: A Sharma

Checked by: C Jayasuriya

Location: Heathcote Road, Holsworthy, NSW 2173

Start Date: 24/03/26

End Date: 25/03/26

Driller: Pointe Engineering

Hole Diameter: 90 mm

Easting: 314046.64

Location Meth.: dGPS0.1

Drill Rig: Atlas Copco R Cobra

Inclination: 60°

Northing: 6238573.90

Ver. Datum: AHD

Total Depth: 11.6 m

Bearing: 230°

Hor. Datum: MGA2020-56

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
NMLC	PVC	D: 0.00-0.20 m	Not Encountered	24.0	0.0			TOPSOIL - Silty SAND: fine to coarse grained, poorly graded, brown; silt, low plasticity ; with rootlets .	D	M	TOPSOIL
						SM	Silty SAND: coarse grained, dark brown; silt, low plasticity ; with root fibers.	RESIDUAL SOIL			
						SP	Gravelly SAND: fine to coarse grained, sub-angular, orange-brown; gravel, fine to coarse grained angular.				
								Log continued on next page.			
					1.0						
					23.0						
					2.0						
					22.0						
					3.0						
					21.0						
					4.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Project No: 60731324
Logged by: A Sharma
Start Date: 24/03/26

Checked by: C Jayasuriya
End Date: 25/03/26
Location Meth.: dGPS0.1
RL: 24.401 m
Ver. Datum: AHD
Surface: Topsoil

Driller: Pointe Engineering	Hole Diameter: 90 mm
Drill Rig: Atlas Copco R Cobra TTe	Inclination: 60°
Total Depth: 11.6 m	Bearing: 230°

Easting: 314046.64
Northing: 6238573.90
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x 15 ₍₅₀₎ A: ● D: ○ UCS: □ σ ₃₀	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)					
					0.0												
					24.0												
NMLC	1	Not Encountered	Is(50) D=1 MPa Is(50) A=0.9 MPa	23.0	1.0		SANDSTONE: coarse grained, red-brown mottled grey with red-brown bands, cross-bedded at 40-90°.	HW	100 (92) [84]			HAWKSBURY SANDSTONE					
							0.91 m: P, 10°, RF, UN, SN, Fe										
							1.07 m: J, 50°, RF, UN, SN, Fe										
							1.27 m: DB										
							1.33 m: P, 20°, RF, PR, SN clay, Fe										
							1.82 m: DB										
							2.00 m: HB										
							2.05-2.11 m: J, 90°, RF, PR, Fe										
							2.16 m: DB										
							2.19 m: P, 10°, RF, PR, SN sand, Fe										
NMLC	2	Not Encountered	Is(50) D=0.3 MPa Is(50) A=0.4 MPa	2.0	2.0		At 1.40 m: SANDSTONE: coarse grained, pale grey mottled red-brown, cross laminated at 10-90°	MW									
							At 2.00 m: SANDSTONE: coarse grained, brown and yellow-brown mottled grey, indistinctly bedded										
							2.16 m: DB										
							2.19 m: P, 10°, RF, PR, SN sand, Fe										
							2.22 m: P, 20°, RF, PR, sand, Fe										
							2.25-2.53 m: IS, 20°, SM, PR, CT clay										
							2.27 m: P, RF, PR, SN gravel , Fe										
							2.33 m: P, 20°, RF, PR, SN, Fe										
							2.39 m: DB										
							2.50 m: P, 20°, RF, PR, SN, Fe										
NMLC	3	Not Encountered	Is(50) D=0.3 MPa Is(50) A=0.6 MPa	3.0	22.0		SANDSTONE: coarse grained, brown mottled pale grey, cross-laminated.	XW				2.80 m: P, 20°, RF, PR, Fe					
							2.83 m: DB										
							2.90 m: J, 10°, RF, PR, SN, Fe										
							2.95 m: J, 40°, RF, PR, SN, Fe										
							3.15 m: P, 30°, RF, PR, trace rootlets , Fe										
							3.38 m: P, 30°, RF, UN, SN, Fe										
							3.45 m: DB										
							3.50-3.59 m: CS, RF, IR, fine to coarse grained gravel, Fe										
							3.66 m: DB										
							3.72 m: P, 20°, RF, PR, CN, x2										
NMLC	4	Not Encountered	Is(50) D=1.6 MPa Is(50) A=1.9 MPa	4.0	21.0		SANDSTONE: fine to medium grained, grey mottled brown, indistinctly bedded .	MW	100 (54) [34]			3.87 m: P, 30°, RF, UN, SN, Fe					
							4.00-4.08 m: EW, SM, IR, sand										
							4.93 m: DB										
NMLC	5	Not Encountered	Is(50) D=1.8 MPa Is(50) A=1.5 MPa	4.21	20.0		NO CORE: 4.04-4.21 m.	XW									
							SANDSTONE: fine to coarse grained, pale brown with red-brown bands with partings, cross-bedded at 10-90° with black laminations .										
NMLC	6	Not Encountered	Is(50) D=1.8 MPa Is(50) A=1.5 MPa	4.21	20.0		At 4.50 m: SANDSTONE: coarse grained, pale grey mottled red, thinly laminated at 0-30°	SW	100 (100) [89]								

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: A Sharma
Start Date: 24/03/26

Checked by: C Jayasuriya
End Date: 25/03/26
Location Meth.: dGPS0.1
RL: 24.401 m
Ver. Datum: AHD
Surface: Topsoil

Driller: Pointe Engineering
Drill Rig: Atlas Copco R Cobra
Total Depth: 11.6 m
Hole Diameter: 90 mm
Inclination: 60°
Bearing: 230°

Easting: 314046.64
Northing: 6238573.90
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • Is(50) UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC	4		Is(50) D=1.6 MPa Is(50) A=2.2 MPa	20.0			SANDSTONE: fine to coarse grained, pale brown with red-brown bands with partings, cross-bedded at 10-90° with black laminations . From 5.20 m: thinly laminated at 70-90°	FR				5.09 m: DB
	5.82								100 (100) [99]			5.82 m: DB
	5		Is(50) D=1.3 MPa Is(50) A=1.5 MPa C: 6.40-6.60 m UCS=14 MPa	19.0								
NMLC	7.39	Not Encountered	Is(50) D=1.1 MPa Is(50) A=1.4 MPa	18.0			From 7.30 m: becoming yellow-brown	HW	100 (96) [75]			7.28 m: DB
							SANDSTONE: coarse grained, pale brown with red and yellow bands, cross-laminated at 0-90° .	XW MW SW				7.66-7.73 m: IS, 30°, RF, PR, sand, Fe
			Is(50) D=0.6 MPa Is(50) A=0.7 MPa	8.0								7.96 m: P, 0°, RF, UN, SN, Fe and black laminations 8.11 m: DB
NMLC	6		Is(50) D=0.3 MPa Is(50) A=0.4 MPa	17.0				HW				8.55 m: J, 60°, RF, PR, VN clay , Fe 8.67 m: HB 8.71 m: EW, RF, IR, CT sand , Fe 8.76 m: P, 5°, RF, PR, SN, Fe 8.83 m: J, 20°, RF, UN, SN, Fe 8.96 m: P, 20°, RF, PR, VN sand, Fe
							SANDSTONE: coarse grained, brown with pale grey and red bands, cross-bedded at 0-30° .	XW MW				9.21 m: P, 30°, RF, PR, SN, Fe 9.23 m: HB 9.46 m: J, 50°, RF, PR, SN sand, Fe-stained 9.58 m: J, 30°, RF, PR, SN, Fe-stained 9.78 m: J, 20°, RF, PR, SN, Fe-stained 9.89 m: DB
				16.0								
				10.0								

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Logged by: A Sharma

Start Date: 24/03/26

Checked by: C Jayasuriya

End Date: 25/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering
Drill Rig: Atlas Copco R Cobra
TTe

Hole Diameter: 90 mm

Inclination: 60°

Bearing: 230°

Easting: 314046.64

Northing: 6238573.90

Hor. Datum: MGA2020-56

RL: 24.401 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: ○ UCS: □ ▢ ▣ ▤ ▥ ▦ ▧ ▨ ▩	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC	7	Not Encountered	Is(50) D=0.5 MPa Is(50) A=0.5 MPa				SANDSTONE: coarse grained, brown with pale grey and red bands, cross-bedded at 0-30° .	MW XW HW	100 (100) [84]			10.04 m: DB 10.07 m: SZ, 20°, RF, PR, CT clay 10.13 m: P, 40°, RF, PR, SN, Fe-stained 10.17 m: HB 10.20-10.23 m: EW, 10°, SM, PR, CT clay, Fe-stained 10.32-10.33 m: EW, 15°, SM, PR, sand
			Is(50) D=1.1 MPa Is(50) A=1.4 MPa				Terminated at 11.60 m.					

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A Sharma

Start Date: 23/03/26

Checked by: C Jayasuriya

End Date: 24/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering

Hole Diameter: 90 mm

Easting: 313908.12

RL: 39.781 m

Drill Rig: Atlas Copco R TTe

Inclination: 60°

Northing: 6238705.62

Ver. Datum: AHD

Total Depth: 12 m

Bearing: 230°

Hor. Datum: MGA2020-56

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PT	PVC	D: 0.00-0.10 m PP=350 kPa D: 0.10-0.20 m PP=400 kPa D: 0.20-0.50 m			0.0		CH	TOPSOIL - Sandy CLAY: high plasticity, yellow-brown; sand, fine grained; trace root fibers.	w ≈ PL		TOPSOIL
					CI	Gravelly Sandy CLAY: medium plasticity, red-brown; gravel, fine to coarse, sub-rounded, sandstone; sand, fine to coarse grained.	w < PL	RESIDUAL SOIL			
					GP	Sandy GRAVEL: fine to coarse grained, sub-angular, red-brown mottled grey; sand, fine to coarse grained.	D	EXTREMELY WEATHERED MATERIAL			
		PP=550 kPa						Log continued on next page.			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A Sharma

Start Date: 23/03/26

Checked by: C Jayasuriya

End Date: 24/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering

Hole Diameter: 90 mm

Easting: 313908.12

RL: 39.781 m

Drill Rig: Atlas Copco R TTe

Inclination: 60°

Northing: 6238705.62

Ver. Datum: AHD

Total Depth: 12 m

Bearing: 230°

Hor. Datum: MGA2020-56

Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ Soil: VL L M H VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0							
							<i>Log continued from previous page.</i>					
							SANDSTONE: fine grained, red-brown mottled grey, indistinctly bedded, iron-stained.	XW	87 (62) [16]			HAWKSBUURY SANDSTONE 0.54-0.63 m: EW, RF, IR, CT clay
			Is(50) D=0.009 MPa Is(50) A=0.006 MPa	39.0								0.92-1.02 m: CS, RF, IR, SN, Fe
					1.0		SANDSTONE: fine to coarse grained, red-brown with pale grey bands, very thinly bedded at 15-20°, iron-stained.	HW				1.35 m: HB
			Is(50) D=0.02 MPa Is(50) A=0.06 MPa				SANDSTONE: fine to coarse grained, pale grey, indistinctly bedded, iron stained.	XW				1.52 m: P, 10°, RF, UN, SN, Fe 1.60-1.70 m: HB, x 2
					2.0		from 1.83 to 1.88 m, red-brown sandstone band					1.84 m: P, 30°, RF, PR, CN
							NO CORE: 2.05 - 2.23 m.					
							SANDSTONE: medium to coarse grained, red brown with pale grey bands, thinly bedded, iron-stained.	HW	100 (93) [68]			2.30 m: DB 2.34 m: P, 30°, RF, PR, CN 2.39 m: P, 20°, RF, PR, CN 2.45 m: P, 30°, RF, CU, SN sand
			Is(50) D=0.6 MPa Is(50) A=0.7 MPa				from 2.65m: SANDSTONE : medium to coarse grained, red-brown mottled grey, indistinctly thinly laminated					2.58 m: P, 30°, RF, PR, SN Fe staining 2.58-2.67 m: P, RF, PR, SN Fe, x 2 spaced 50mm 2.67 m: P, 20°, RF, PR, SN Fe staining
					3.0		from 2.90m: becoming pale grey					2.95 m: HB 3.00 m: EW, 5°, RF, UN, CN
							from 3.10m: becoming red-brown mottled grey					3.20 m: HB
							SANDSTONE: medium to coarse grained, pale grey mottled brown, indistinctly to cross-laminated at 10-30°; Fe staining.	MW	100 (100) [87]			3.51 m: DB
			Is(50) D=0.5 MPa Is(50) A=0.9 MPa				from 3.70m: becoming yellow brown mottled grey and massive					3.69 m: P, 35°, RF, UN, Fe staining 3.80 m: HB 3.92 m: HB 4.02 m: HB
					4.0							4.15 m: P, 35°, RF, PR, CN
												4.54 m: HB
			Is(50) D=0.5 MPa Is(50) A=0.8 MPa									
					5.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A Sharma

Start Date: 23/03/26

Checked by: C Jayasuriya

End Date: 24/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering

Hole Diameter: 90 mm

Easting: 313908.12

RL: 39.781 m

Drill Rig: Atlas Copco R TTe

Inclination: 60°

Northing: 6238705.62

Ver. Datum: AHD

Total Depth: 12 m

Bearing: 230°

Hor. Datum: MGA2020-56

Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS(50) A: • D: o UCS: □ SOIL: VL L M H VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC	4		Is(50) D=0.8 MPa Is(50) A=0.6 MPa	35.0			SANDSTONE: fine to coarse grained, yellow-brown, indistinctly laminated.	MW	99 (99) [80]			5.00 m: HB 5.12 m: DB 5.26 m: P, 10°, RF, PR, SN Fe 5.30 m: DB 5.43 m: P, 30°, RF, PR, SN clay 5.51 m: DB 5.53 m: IS, 15°, RF, PR, VN clay, 5 mm
							Cross laminations					
	5		C: 6.20-6.35 m UCS=16 MPa Is(50) D=0.4 MPa Is(50) A=0.8 MPa	34.0			5.53m: SANDSTONE : medium to coarse grained, pale grey, thinly laminated at 15-20°	SW				6.07 m: P, 40°, RF, PR, CN 6.22 m: DB
	6		Is(50) D=0.4 MPa Is(50) A=1.1 MPa	33.0			SANDSTONE: coarse grained, dark brown, thinly laminated at 15-20°	XW	100 (100) [87]			6.52 m: DB 6.57 m: P, 20°, RF, UN, VN clay 6.66 m: P, 15°, RF, PR, SN Fe 6.73 m: P, 25°, RF, PR, SN Fe 6.78-6.83 m: SZ, 25°, RF, IR, VN clay
							SANDSTONE: medium to coarse grained, pale grey and brown, laminated at 0-5°.	MW				7.54 m: DB 7.82 m: P, 20°, RF, PR, CN 7.87 m: P, 20°, SM, PR, CT clay
	6		Is(50) D=0.7 MPa Is(50) A=1.7 MPa	32.0			8.43m: SANDSTONE : medium to coarse grained, pale grey indistinctly cross laminated	HW SW	100 (100) [92]			8.45 m: HB 8.62 m: HB
			Is(50) D=0.5 MPa Is(50) A=0.6 MPa				from 8.88m: SANDSTONE : medium to coarse grained, yellow-brown pale grey thinly laminated at 0-5°	XW MW				9.07 m: P, 0°, RF, PR, CT X 9.45 m: HB 9.59 m: P, 25°, RF, PR, SN Fe 9.67 m: EW, 15°, RF, IR, SN Fe
								SW	100 (100) [85]			9.79 m: DB 9.94 m: HB

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A Sharma

Start Date: 23/03/26

Checked by: C Jayasuriya

End Date: 24/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering

Hole Diameter: 90 mm

Easting: 313908.12

RL: 39.781 m

Drill Rig: Atlas Copco R TTe

Inclination: 60°

Northing: 6238705.62

Ver. Datum: AHD

Total Depth: 12 m

Bearing: 230°

Hor. Datum: MGA2020-56

Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: o UCS: □ SOIL CLASSIFICATION VL L M H VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC	7		Is(50) D=0.3 MPa Is(50) A=0.7 MPa	31.0			SANDSTONE: coarse grained, pale grey and yellow brown, cross laminated.	SW				10.21 m: HB
	10.65						from 10.30m: becoming pale grey and red brown					
	8		Is(50) D=0.4 MPa Is(50) A=0.3 MPa	11.0			from 10.70m: becoming yellow brown, thinly laminated at 5-10°	MW	100 (96) [36]			10.71-10.86 m: P, 10-20°, RF, PR, SN Fe x 3
							indistinctly cross laminated	HW				11.00 m: HB 11.04 m: HB
				30.0			SANDSTONE: coarse grained, grey with brown bands, indistinctly bedded .	SW				11.21 m: 35°, RF, UN, SN Fe, black lamination 11.25-11.30 m: CS, RF, IR, CT, X and CA 11.34 m: IS, 30°, SM, UN, VN clay, X 11.38 m: P, 30°, RF, PR, VN sand, X 11.55 m: P, 30°, RF, CU, SN Fe 11.65 m: J, 40°, RF, UN, SN Fe 11.70 m: J, 40°, RF, PR, SN Fe 11.75-11.90 m: DB, x 3
				12.0			Terminated at 12.00 m. Target depth.					11.97 m: J, 80°, RF, PR, SN Fe
				29.0								
				13.0								
				28.0								
				14.0								
				27.0								
				15.0								

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Driller: Pointe Engineering Pty Ltd

Drill Rig: Ripamonti Skid Drill

Total Depth: 11.47 m

Hole Diameter: 76 mm

Inclination: 60°

Bearing: 38°

Project No: 60731324

Logged by: R Rees

Start Date: 24/02/26

Easting: 313937.02

Northing: 6238749.52

Hor. Datum: MGA2020-56

Checked by: J Riley

End Date: 24/02/26

Location Meth.: dGPS0.1

RL: 37.657 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
WS	U	D: 0.00-0.16 m			0.0			TOPSOIL - Silty SAND: fine to medium grained, pale brown mottled yellow; trace gravel, fine to coarse, angular; trace clay.	D	S	TOPSOIL
					37.0			Log continued on next page.			
					1.0						
					2.0						
					3.0						
					4.0						
					5.0						
					6.0						
					7.0						
					8.0						

Location: Heathcote Road, Holsworthy, NSW 2173

Start Date: 24/02/26

End Date: 24/02/26

RL: 37.657 m

Inclination: 60°

Northing: 6238749.52

Ver. Datum: AHD

Bearing: 38°

Hor. Datum: MGA2020-56

Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: ○ UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0					SOIL 0-4 VL 0-4 L 0-4 E 0-4 H 0-4 VH 0-4 EH 0-4		
							<i>Log continued from previous page.</i>					
			Is(50) A=0.35 MPa Is(50) D=0.39 MPa		37.0		SANDSTONE: medium to coarse grained, pale grey mottled pale yellow-brown and pale red-brown, iron stained. from 0.60 m: becoming very thinly bedded at 50°	MW - SW	96 (82) [82]			0.32-0.41 m: EW HAWKESBURY SANDSTONE 0.42 m: DB
			Is(50) A=0.65 MPa Is(50) D=0.59 MPa		1.0							1.00 m: HB
												1.23 m: J, 5°, RF, PR, CN, 1 - 2 mm aperture
												1.74 m: P, 50°, RF, PR, CI-CL, root fibres, 4 mm 1.84-1.93 m: EW, CI-CL, root fibres
	2.00				36.0		NO CORE: 1.93 - 2.00 m.	XW				
			Is(50) A=1.56 MPa Is(50) D=0.99 MPa		2.0		SANDSTONE: medium to coarse grained, pale grey mottled yellow-brown and red-brown, thinly to very thinly bedded at 0-50°. from 2.24 m: becoming pale grey with occasional red-brown iron staining.	MW - SW	100 (97) [83]			2.16 m: P, 40°, SM, PR, CN, 1 - 2 mm aperture 2.30 m: 25°, RF, PR, CN, 1 - 2 mm aperture
												3.00 m: HB 3.10 m: P, 30°, SM, UN, CN, 1 - 2 mm aperture 3.16 m: P, 30°, SM, UN, CI-CL, root fibres, 3 mm 3.27 m: P, 30°, SM, UN, CI-CL, 4 mm aperture
	3.50		Is(50) A=1.16 MPa Is(50) D=2.99 MPa C: 3.75-3.90 m UCS=27 MPa		35.0				100 (96) [83]			3.50 m: DB
												3.96-4.03 m: EW
												4.16 m: P, 40°, SM, IR, CI-CL, root fibres, 2mm 4.17 m: J, 35°, SM, UN, CN
			Is(50) A=2.09 MPa Is(50) D=1.79 MPa		4.0		from 4.50 m: becoming yellow-brown, thinly bedded at 0-50°, trace gravel inclusions, fine grained, sub-rounded, quartz					4.44-4.63 m: P, 25-35°, SM, UN, CN, spaced 20-40 mm, x 5
			Is(50) A=3.03 MPa Is(50) D=2.23 MPa		5.0				100 (100) [100]			4.93-5.13 m: HB, x 4
	5.13				33.0		from 5.48 m: becoming thinly bedded at 30-40° from 5.60 m: becoming fine to medium grained					5.91-6.00 m: HB, x 2
			Is(50) A=2.87 MPa Is(50) D=1.78 MPa		6.0							
							from 6.33 m: becoming thinly to very thinly bedded at 20-40° from 6.58 - 6.85 m: pale grey with black flecks		100 (100) [100]			6.65 m: J, 65°, RF, PR, CN 6.73 m: DB
	6.73				32.0							6.94 m: P, 40°, SM, UN, CN 7.00 m: HB 7.06 m: HB 7.27 m: P, 30°, SM, UN, CN
			Is(50) A=2.13 MPa Is(50) D=1.54 MPa		7.0		from 7.30 m: becoming pale grey and pale yellow-brown from 7.60 m: becoming pale grey and grey					7.80 m: P, 30°, SM, UN, CT , ML
					31.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: R Rees
Start Date: 24/02/26

Checked by: J Riley
End Date: 24/02/26
Location Meth.: dGPS0.1
RL: 37.657 m
Ver. Datum: AHD
Surface: Topsoil

Driller: Pointe Engineering Pty Ltd
Drill Rig: Ripamonti Skid Drill
Total Depth: 11.47 m
Hole Diameter: 76 mm
Inclination: 60°
Bearing: 38°

Easting: 313937.02
Northing: 6238749.52
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: ○ UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC	8.33	25/02/2026 08:29:13	Is(50) A=1.3 MPa Is(50) D=1.32 MPa C: 8.50-8.65 m UCS=22 MPa	30.0	9.0		SANDSTONE: medium to coarse grained, pale grey and grey, very thinly to thinly bedded at 10-30°; trace iron staining. from 8.33 to 8.69 m: trace gravel inclusions, fine, sub-angular, quartz; trace siltstone inclusions up to 20 mm from 9.00 m: becoming yellow-brown with black flecks, trace gravel inclusions; fine, sub-rounded, quartz	MW - SW	100 (98) [97]			7.95-8.33 m: HB, x 4 8.69-8.71 m: EW 8.73 m: P, 30°, SM, UN, CN 8.85 m: J, 75°, SM, UN, CN 8.86 m: P, 20°, SM, UN, CT silt and root fibres 8.97 m: P, 30°, SM, UN, CN 9.07 m: P, 25°, RF, PR, CN
	9.90		Is(50) A=2.71 MPa Is(50) D=2.39 MPa Is(50) A=1.11 MPa Is(50) D=2.22 MPa									
	6		Is(50) A=2.23 MPa Is(50) D=1.78 MPa	29.0	10.0		Terminated at 11.47 m.		100 (100) [100]			9.90 m: DB 10.00 m: HB 11.00 m: HB 11.33 m: HB

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A Liang/ A To

Start Date: 18/02/26

Checked by: J Riley

End Date: 20/02/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering Pty Ltd

Drill Rig: Ripamonti Speedydrill

Total Depth: 180

Total Depth: 9.3 m

Hole Diameter: 105 mm

Inclination: 90°

Bearing: N/A

Easting: 313876.92

Northing: 6238818.35

Hor. Datum: MGA2020-56

RL: 40.292 m

Ver. Datum: AHD

Surface: Grass

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
WS	PVC	D: 0.00-0.10 m D: 0.10-0.40 m D: 0.40-1.30 m	Not Observed	0.0	0.0		SC	TOPSOIL - Silty Clayey SAND: fine to medium grained, brown; trace gravel, sandstone; trace root fibers. Clayey SAND: fine to medium grained, yellow-brown; with gravel, fine to coarse, sub-angular, sandstone and ironstone.	D		TOPSOIL
		40.0			EXTREMELY WEATHERED MATERIAL						
				39.0				Log continued on next page.			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: A Liang/ A To
Start Date: 18/02/26
Checked by: J Riley
End Date: 20/02/26
Location Meth.: dGPS0.1
RL: 40.292 m
Ver. Datum: AHD
Surface: Grass

Driller: Pointe Engineering Pty Ltd
Drill Rig: Ripamonti Speedydrill
Total Depth: 9.3 m
Hole Diameter: 105 mm
Inclination: 90°
Bearing: N/A

Easting: 313876.92
Northing: 6238818.35
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ SOIL: 0-4 VL 2 L 6 M 20 H 60 VH 200 EH	DEFECT SPACING (mm) 20 60 200 600 2000	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0							
					40.0							
					1.0							
					39.0							
							<i>Log continued from previous page.</i>					
							Sandy CLAY: pale grey mottled yellow-brown and red-brown; sand, fine to medium grained; trace gravel, fine to medium grained.	XW	100 (0) [0]			EXTREMELY WEATHERED MATERIAL
							at 1.9 m: tree roots , 30 mm in size					
					2.0		NO CORE: 2.04 - 2.60 m.		38 (0) [0]			
					38.0							
							Sandy CLAY: pale grey mottled yellow-brown; sand, fine to medium grained; trace gravel, fine to medium grained.	XW				
					3.0		NO CORE: 2.94 - 3.08 m.		92 (82) [75]			
					37.0		SANDSTONE: fine to medium grained, pale grey and dark brown, indistinct bedding, iron stained.	MW - SW				HAWKESBURY SANDSTONE 3.13 m: J, 15°, VR, IR, CN 3.16-3.24 m: CS, 80 mm thick
							from 3.30 m: becoming pale brown-grey, very thinly bedded at 5-10°					3.50-4.40 m: MB, 100 - 300 mm spacing, x 6 3.54 m: J, 63°, RF, UN, Qz, 5 mm aperture 3.63 m: J, 75°, RF, CN
					4.0		from 4.00 m: becoming pale grey and pale yellow brown, very thinly to thickly laminated at 0-10°					4.02-4.05 m: EW, clay, 30 mm thick
					36.0							4.39-4.40 m: EW, clay, 10 mm thick 4.40-4.53 m: J, 10-20°, 30 - 60 mm spacing, healed, x 4 4.56-4.60 m: CS, 40 mm thick 4.64 m: DL 4.66-5.00 m: MB, 30 - 50 mm spacing, x 9
									96 (93) [72]			4.94 m: J, 7°, RF, CU, SN
					5.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: A Liang/ A To
Start Date: 18/02/26
Checked by: J Riley
End Date: 20/02/26
Location Meth.: dGPS0.1
RL: 40.292 m
Ver. Datum: AHD
Surface: Grass

Driller: Pointe Engineering Pty Ltd
Drill Rig: Ripamonti Speedydrill
Total Depth: 9.3 m
Hole Diameter: 105 mm
Inclination: 90°
Bearing: N/A

Easting: 313876.92
Northing: 6238818.35
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • Is(50) D: o UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)		
NMLC	4	Not Observed	C: 5.00-5.15 m Is(50) A=0.61 MPa Is(50) D=0.56 MPa UCS=2.5 MPa	35.0			SANDSTONE: fine to medium grained, pale grey and dark brown, indistinct bedding, iron stained.	MW - SW				5.40 m: J, 70°, VR, ST, from 5.38-5.43 m		
							NO CORE: 5.43 - 5.50 m.						5.60-5.63 m: MB, 30 mm spacing, x 2	
	5		6.30	Is(50) A=0.54 MPa Is(50) D=0.45 MPa	34.0	6.0		SANDSTONE: fine to medium grained, pale grey and grey with dark grey laminations, indistinct bedding, iron stained.	SW				5.78 m: J, 50-70°, RF, UN, SN, 30 mm spacing, x 2, from 5.63-5.92 m	
										6.00-6.03 m: MB, 30 mm spacing, x 2				
	6		7.91	Is(50) A=1.68 MPa Is(50) D=1.5 MPa	33.0	7.0		from 7.22 to 7.49: very thinly bedded at 0-15°	MW				6.18 m: J, 75°, VR, ST, from 6.16-6.22 m	
								from 7.36 m: becoming yellow-brown						6.24 m: J, 45-60°, RF, CN
	6			Is(50) A=2.11 MPa Is(50) D=1.78 MPa	32.0	8.0		from 7.50 m: becoming medium to coarse grained, brown-grey speckled dark brown	MW				6.30 m: DL	
														6.40-7.40 m: MB, 100 - 200 mm spacing, x 7
				Is(50) A=0.15 MPa Is(50) D=0.58 MPa	31.0	9.0		INTERBEDDED SANDSTONE AND SILTSTONE: SANDSTONE, (70%), medium to coarse grained, brown-grey speckled dark brown, indistinctly bedded; SILTSTONE, (30%), dark grey, indistinct bedding.	HW - MW	99 (96) [92]				6.47-6.50 m: CZ, 30 mm thick
										6.60-7.15 m: J, 60-80°, RF, SN clay, 5 mm aperture, 130 - 200 mm spacing, x 4				
								MW				7.48-7.51 m: CZ, 30 mm thick		
													7.62-8.00 m: MB, 80 mm spacing, x 8	
												8.05 m: J, 5°, RF, CN		
													8.29-8.33 m: J, 5-10°, RF, UN, SN, 10 mm spacing, x 4	
												8.48-9.30 m: MB, 200 - 300 mm spacing, x 7		
													8.70-8.75 m: J, 10-20°, RF, CN, 50 mm spacing, x 2	
								HW - MW				9.05 m: J, 85°, RF, CN, from 8.95-9.15 m		
													9.16-9.17 m: EW, clay, 10 mm thick	
							Terminated at 9.30 m. Target depth.							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A To

Start Date: 17/03/26

Checked by: J Riley

End Date: 17/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering Pty Ltd

Drill Rig: Ripamonti Speedydrill

180

Total Depth: 13.43 m

Hole Diameter: 90 mm

Inclination: 90°

Bearing: N/A

Easting: 314382.31

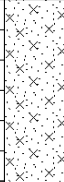
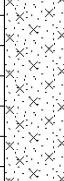
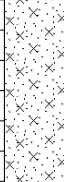
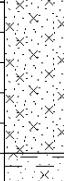
Northing: 6238275.25

Hor. Datum: MGA2020-56

RL: 1.062 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
WS		D: 0.00-0.20 m	17/03/2026 10:34:21 ▼	1.0	0.0			TOPSOIL - Silty CLAY: dark brown; trace root fibres.	W _p	PL	TOPSOIL
		D: 0.20-0.30 m				SC	Clayey SAND: fine to medium grained, dark brown-grey.	M		ALLUVIUM	
		D: 0.30-0.40 m D: 0.40-0.80 m				SM	Silty Clayey SAND: fine to medium grained, dark grey; trace root fibres; strong organic odour.				
WB	PVC	ES: 1.00-1.30 m		0.0	1.0				W	VL	
		D: 1.30-1.50 m									
		SPT: 1.50-1.95 m 0,0,1 N=1									
WB				2.0	-1.0				W		
WB				3.0	-2.0				W		
WB				4.0	-3.0		CI	Silty Sandy CLAY: medium plasticity, dark grey; sand, fine grained .		VS	
WB		SPT: 4.50-4.95 m 0,0,0 N=0									
		PP=10, 10, 10 kPa									

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A To

Start Date: 17/03/26

Checked by: J Riley

End Date: 17/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering Pty Ltd

Drill Rig: Ripamonti Speedydrill

Total Depth: 13.43 m

Hole Diameter: 90 mm

Inclination: 90°

Bearing: N/A

Easting: 314382.31

Northing: 6238275.25

Hor. Datum: MGA2020-56

RL: 1.062 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
WB	PVC	SPT: 6.00-6.45 m 0,0,0 N=0 PP=10, 10, 10 kPa SPT: 7.05-7.50 m 0,0,1 N=1 SPT: 7.50-7.95 m 0,0,1 N=1 PP=50 kPa		-4.0 -5.0 -6.0 -7.0 -8.0	4.0 5.0 6.0 7.0 8.0 9.0 10.0		CI	Silty Sandy CLAY: medium plasticity, dark grey; sand, fine grained .	w ≈ LL	VS	
							CI	Sandy CLAY: medium plasticity, pale brown and pale grey; sand, fine to medium grained. from 7.60-7.85 m: becoming dark grey and pale brown from 8.10 m, becoming pale grey and pale brown, trace iron staining	w ≈ PL	St	RESIDUAL
		Is(50) D=0.1 MPa						Log continued on next page.			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A To

Start Date: 17/03/26

Checked by: J Riley

End Date: 17/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering Pty Ltd	Hole Diameter: 90 mm
Drill Rig: Ripamonti Speedydrill	Inclination: 90°
Total Depth: 180	Bearing: N/A
Total Depth: 13.43 m	

Easting: 314382.31

Northings: 6238275.25

Hor. Datum: MGA2020-56

RL: 1.062 m

Ver. Datum: AHD

Surface: Topsoil

[illegible]

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: A To
Start Date: 17/03/26

Checked by: J Riley
End Date: 17/03/26
Location Meth.: dGPS0.1
RL: 1.062 m
Ver. Datum: AHD
Surface: Topsoil

Driller: Pointe Engineering Pty Ltd
Drill Rig: Ripamonti Speedydrill
Total Depth: 13.43 m
Hole Diameter: 90 mm
Inclination: 90°
Bearing: N/A

Easting: 314382.31
Northing: 6238275.25
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS(50) A: • IS(50) D: • D(50) UCS: □ UCS(50)	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC	10.10		Is(50) D=0.14 MPa Is(50) A=0.37 MPa	-9.0	10.0		SANDSTONE: medium grained, pale grey with dark grey laminations, thinly to very thinly bedded at 0-30.	FR	[100]			
	2		Is(50) D=0.11 MPa Is(50) A=0.21 MPa	-10.0	11.0		from 11.05 m: indistinct bedding	MW	100 (91) [91]			11.22-11.36 m: EW, clay, 140 mm thick
	11.73		Is(50) D=0.11 MPa Is(50) A=0.11 MPa	-11.0	12.0		from 11.4-11.73 m: becoming pale grey and yellow-brown At 11.64 m: iron stained from 11.73 m: becoming pale red-brown and yellow-brown		73 (56) [56]			11.44 m: MB 11.48 m: J, 65°, RF, UN, CN clay
	3		Is(50) D=0.11 MPa Is(50) A=0.11 MPa	-11.0	12.0		NO CORE: 12.25-12.48 m.					11.74-11.81 m: EW, clay, 70 mm thick 11.85 m: J, 20°, RF, IR, SN Fe 12.00 m: MB
	12.48		Is(50) D=0.07 MPa Is(50) A=0.11 MPa C: 13.10-13.25 m UCS=4.2 MPa	-12.0	13.0		SANDSTONE: medium grained, pale red-brown and yellow-brown, thinly bedded at 0-20, iron stained.	MW	100 (100) [100]			12.45-12.48 m: MB 12.66-12.90 m: P, 5-15°, RF, UN, 240 mm spacing, x 2
	4			-13.0	14.0		Terminated at 13.43 m. Target depth.					
				-13.0	15.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A Liang/A To

Start Date: 18/03/26

Checked by: J Riley

End Date: 18/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering Pty Ltd

Drill Rig: Ripamonti Speedydrill

Total Depth: 4

Hole Diameter: 100 mm

Inclination: 90°

Bearing: N/A

Easting: 313756.80

Northings: 6238938.18

Hor. Datum: MGA2020-56

RL: 36.097 m

Ver. Datum: AHD

Surface: Grass

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PT	PVC	D: 0.00-0.15 m	Not Observed	36.0	0.0			Silty SAND: fine to medium grained, dark brown; trace clay; trace root fibres.	D		TOPSOIL
		D: 0.15-0.72 m				SC	Clayey SAND: fine to medium grained, pale brown mottled orange; with silt; with gravel, fine to coarse, sub-angular, ironstone.	D - VD		EXTREMELY WEATHERED MATERIAL	
		SPT: 0.72-0.75 m 1/30 mm HB N=R			1.0			Log continued on next page.			
					35.0						
					2.0						
					34.0						
					3.0						
					33.0						
					4.0						
					32.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A Liang/A To

Start Date: 18/03/26

Checked by: J Riley

End Date: 18/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering Pty Ltd

Drill Rig: Ripamonti Speedydrill 120

Total Depth: 4

Hole Diameter:

Inclination: 90°

Bearing: N/A

Easting: 313756.80

Northing: 6238938.18

Hor. Datum: MGA2020-56

RL: 36.097 m

Ver. Datum: AHD

Surface: Grass

[illegible]

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Driller: Pointe Engineering Pty Ltd

Drill Rig: Ripamonti Speedydrill

Total Depth: 15 m

Hole Diameter: 100 mm

Inclination: 60°

Bearing: 230°

Project No: 60731324

Logged by: A Sharma

Start Date: 10/03/26

Easting: 314590.11

Northing: 6237928.61

Hor. Datum: MGA2020-56

Checked by: J Riley

End Date: 11/03/26

Location Meth.: dGPS0.1

RL: 20.559 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
NMLC	PVC	D: 0.00-0.10 m D: 0.10-0.60 m			0.0		GM	Sandy GRAVEL: fine to coarse grained, sub-angular, red-brown; sand, fine to coarse grained . Silty GRAVEL: fine to coarse grained, sub-angular, brown mottled pale yellow-brown; silt, low plasticity .	D		TOPSOIL EXTREMELY WEATHERED MATERIAL
								Log continued on next page.			
					20.0						
					1.0						
					19.0						
					2.0						
					18.0						
					3.0						
					4.0						
					17.0						
					5.0						

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: A Sharma
Start Date: 10/03/26

Checked by: J Riley
End Date: 11/03/26
Location Meth.: dGPS0.1
RL: 20.559 m
Ver. Datum: AHD
Surface: Topsoil

Driller: Pointe Engineering Pty Ltd
Drill Rig: Ripamonti Speedydrill
Total Depth: 15 m
Hole Diameter: 100 mm
Inclination: 60°
Bearing: 230°

Easting: 314590.11
Northing: 6237928.61
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: o UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0							
							<i>Log continued from previous page.</i>					
							SANDSTONE: fine to coarse grained, pale yellow-brown, indistinctly bedded.	HW	100 (61) [13]			HAWKSBUURY SANDSTONE 0.36 m: P, 60°, PR, CT Cl, 2 mm aperture 0.45 m: P, 45°, RF, PR, CN 0.59 m: DB 0.68 m: P, 50°, RF, IR, VN roots 0.75 m: HB 0.82 m: HB 0.97 m: J, 85°, RF, PR, CN 1.07-1.14 m: DB 1.19 m: P, 45°, RF, UN, CN 1.57-1.61 m: DB 1.91 m: P, 35°, RF, PR, CN 2.02 m: HB 2.05-2.20 m: J, 10-25°, RF, PR, VN root fibres, 30 spacing, x 4 2.10-2.16 m: EW, 15°, Cl 2.27 m: P, 50°, RF, PR, VN roots, Cl 2.37 m: DB 2.51 m: P, 60°, RF, PR, VN Cl, roots 2.56-2.88 m: J, 65°, RF, PR, VN roots, 50-200 spacing, x 5 2.92-2.93 m: EW, 15°, PR, Cl, roots, 10 mm aperture 2.99 m: HB 3.15 m: DB 3.18 m: J, 45°, RF, PR, VN roots 3.38 m: P, 40°, RF, PR, CN 3.76 m: J, 65°, RF, PR, CN SC, roots, 5 mm aperture 4.03 m: P, 5°, RF, PR, CN 4.11-4.31 m: P, 15°, RF, IR, CT Cl, roots, 40-100 spacing, x 4 4.57 m: P, 10°, RF, PR, VN SC 4.74 m: J, 45°, RF, PR, CN 4.94 m: J, 60°, RF, PR, SN clay
					20.0							
							from 0.83 m: becoming pale yellow-brown and pale grey					
					1.0							
							NO CORE: 1.30 - 1.57 m.		85 (66) [45]			
							SANDSTONE: fine to coarse grained, pale yellow and red-brown, medium bedded at 45°-45°, iron stained.	MW				
					19.0							
							from 2.0 m: becoming pale grey, with black flecks					
					2.0							
							from 2.29 to 2.45 m: red-brown, iron stained					
								MW - SW				
					18.0							
							from 3.33 to 3.66 m: laminated at 50°-65°, iron stained		100 (84) [69]			
					3.0							
							from 3.85 to 4.03 m: red-brown, iron stained					
					4.0							
							from 4.03 m: very thinly bedded at 15-25° with thin grey laminations					
					17.0							
							from 4.34 to 4.54 m: red-brown, iron stained					
							from 4.62 m: orange-brown, iron stained		100 (94) [94]			
					5.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A Sharma

Start Date: 10/03/26

Checked by: J Riley

End Date: 11/03/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering Pty Ltd

Drill Rig: Ripamonti Speedydrill

Total Depth: 180 m

Hole Diameter: 100 mm

Inclination: 60°

Bearing: 230°

Easting: 314590.11

Northing: 6237928.61

Hor. Datum: MGA2020-56

RL: 20.559 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS(50) A: • IS(50) D: • UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC	4		Is(50) A=1.62 MPa Is(50) D=1.65 MPa	16.0			SANDSTONE: fine to coarse grained, pale yellow and red-brown, medium bedded at 45°-45°, iron stained.	MW - SW				5.00 m: DB
												5.21 m: J, 70°, RF, PR, CN
												5.40 m: J, 20°, RF, PR, SN clay
5	6.30		Is(50) A=1.54 MPa Is(50) D=1.62 MPa C: 6.40-6.55 m UCS=37 MPa	15.0					100 (88) [88]			5.81 m: J, 60°, RF, UN, SN clay
												5.92 m: HB
												6.00-6.10 m: EW, SN clay
6	7.95		Is(50) A=1.7 MPa Is(50) D=1.69 MPa	7.0			from 7.12 to 7.77 m: pale grey					6.16 m: J, 80°, RF, PR, CN
												6.35 m: DB
												6.72 m: J, 15°, RF, PR, CN
8				14.0								7.00 m: HB
												7.06 m: J, 80°, RF, PR, SN Fe
												7.41 m: J, 0°, RF, PR, SN Fe
9												7.60-7.70 m: EW, 40°, clay
												7.88 m: J, 70°, RF, CU, SN Fe
												8.00-8.06 m: J, 70°, RF, PR, SN Fe, crystalline and coarse grained
10				8.0			SANDSTONE: fine to medium grained, pale grey and pale yellow-brown, indistinctly bedded, iron stained.	HW	100 (80) [75]			8.15-8.15 m: DB
								XW				8.28-8.38 m: SS, 25°, SL, PR, SN clayey SAND, Fe
							from 8.41 m: becoming very thinly bedded at 0-25°	MW				
11												8.79-8.79 m: HB
												9.00-9.00 m: DB, 0°
12	9.50		C: 9.50-15.00 m	13.0					100 (100) [91]			9.50-9.50 m: DB
												9.88-9.88 m: J, 40°, RF, PR, SN, Fe
			Is(50) A=1.08 MPa	12.0								
				10.0								

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: A Sharma
Start Date: 10/03/26

Checked by: J Riley
End Date: 11/03/26
Location Meth.: dGPS0.1
RL: 20.559 m
Ver. Datum: AHD
Surface: Topsoil

Driller: Pointe Engineering Pty Ltd
Drill Rig: Ripamonti Speedydrill
Total Depth: 15 m
Hole Diameter: 100 mm
Inclination: 60°
Bearing: 230°

Easting: 314590.11
Northing: 6237928.61
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • Is(50) D: o UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC	10.50		Is(50) D=1.29 MPa				SANDSTONE: fine to medium grained, pale grey and pale yellow-brown, indistinctly bedded, iron stained. from 10.0 to 10.77 m: becoming pale red-brown and pale yellow-brown, trace fine gravel inclusions	MW				9.97 m: HB 9.98 m: IS, 20°, RF, PR, SN clay 10.00 m: HB 10.02 m: J, 40°, RF, PR, SN, Fe
			Is(50) A=3.23 MPa Is(50) D=2.67 MPa						100 (92) [86]			10.43 m: J, 55°, RF, CU, SN, Fe 10.52 m: J, 30°, RF, PR, SN, Fe
	8		C: 11.20-11.35 m UCS=17 MPa Is(50) A=0.8 MPa Is(50) D=0.81 MPa				from 11.12 m: becoming pale grey with grey laminations	SW to FR				10.64-10.72 m: SS, 40°, SM, IR, SN clay and gravel, Fe 10.76 m: J, 40°, RF, PR, SN, Fe
	12.70		Is(50) A=1.3 MPa Is(50) D=1.28 MPa						100 (100) [92]			11.13 m: J, 45°, RF, PR, SN, Fe 11.23-11.32 m: SS, 45°, RF, IR, VN Clayey SAND 11.31-11.38 m: SZ, 45°, IR, VN Clayey SAND
9												12.44 m: J, 35°, RF, PR, SN, carbon stains
												12.70 m: DB
												12.84 m: J, 40°, RF, PR, CN
												12.97 m: J, 30°, RF, PR, clay
10	13.29		Is(50) A=1.39 MPa Is(50) D=1.46 MPa						100 (100) [99]			13.29 m: DB
												13.80 m: EW, 40°, SN clay, Fe
							from 13.80 m: becoming medium to thinly bedded at 0-25°					14.00 m: HB
			Is(50) A=2.4 MPa Is(50) D=2 MPa									14.35 m: J, 15°, RF, UN
												14.73 m: HB

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Driller: Pointe Engineering Pty Ltd

Drill Rig: Ripamonti Speedydrill

Total Depth: 15 m

Hole Diameter: 100 mm

Inclination: 60°

Bearing: 230°

Project No: 60731324

Logged by: A Sharma

Start Date: 10/03/26

Easting: 314590.11

Northing: 6237928.61

Hor. Datum: MGA2020-56

Checked by: J Riley

End Date: 11/03/26

Location Meth.: dGPS0.1

RL: 20.559 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: o UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
							Terminated at 15.00 m. Target depth.					14.97 m: MB
					7.0							
					16.0							
					6.0							
					17.0							
					5.0							
					18.0							
					19.0							
					4.0							
					20.0							

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Driller: Pointe Engineering

Drill Rig: Atlas Copco R Cobra TTe

Total Depth: 1.7 m

Hole Diameter: 50 mm

Inclination: 90°

Bearing: N/A

Project No: 60731324

Logged by: A Sharma

Start Date: 25/02/26

Easting: 314500.76

Northing: 6238105.33

Hor. Datum: MGA2020-56

Checked by: J Riley

End Date: 25/02/26

Location Meth.: dGPS0.1

RL: 0.464 m

Ver. Datum: AHD

Surface: TOPSOIL

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
WS	U	ES: 0.00-0.10 m D: 0.10-1.10 m	25/02/2026 09:14:05 ▼	0.0	0.0		SC	TOPSOIL - Silty SAND: fine to coarse grained, dark grey; with gravel, medium grained, angular. Clayey SAND: fine grained, dark grey; root fibres.	W		TOPSOIL
		ES: 1.10-1.20 m D: 1.20-1.50 m		-0.0	-0.0						ALLUVIUM
		D: 1.50-1.70 m		-1.0	-1.0		SP	SAND: coarse grained, dark brown mottled pale brown.			
					2.0			Terminated at 1.70 m. Target depth.			
					-2.0						
					-3.0						
					-4.0						
					-5.0						
					-6.0						
					-7.0						
					-8.0						

BH02
Sheet 1 of 2

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: V Tsambalas

Start Date: 15/02/26

Checked by: J Riley

End Date: 16/02/26

Driller: MacGeo

Drill Rig: Comacchio 300

Total Depth: 12.41 m

Hole Diameter: 110 mm

Inclination: 90°

Bearing: N/A

Easting: 314450.23

Northing: 6238177.51

Hor. Datum: MGA2020-56

Ground Level: 4.89 m

Ver. Datum: AHD

Top of Casing: m

METHOD	GROUND WATER	REMARKS	GRAPHIC LOG	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	RL (m AHD)	DEPTH (m)	WELL DIAGRAM
AD/T	16/02/2026 02:38:10			TOPSOIL - Silty SAND: fine to coarse grained, dark brown; trace gravel, fine, angular, igneous, trace rootlets, trace wood fragments. FILL - SAND: fine to medium grained, pale yellow-brown; trace gravel, rounded, sandstone. SP SAND: fine grained, dark brown; trace charcoal, trace wood fragments.	0.0		Grout Bentonite
HQ3	▼			SP SAND: fine grained, pale yellow; trace silt; trace gravel, fine to coarse, sandstone. SANDSTONE: medium to coarse grained, pale grey and yellow-brown, very thinly bedded to thickly laminated at 0-5°, trace fine gravel sized quartz inclusions, low strength, iron stained. from 2.84 to 3.23 m: pale grey mottled yellow-brown and pale brown from 3.32 m: becoming red-brown and pale grey NO CORE: 4.05 - 4.15 m. SANDSTONE: fine to medium grained, pale grey and red-brown, very thinly bedded to thickly laminated at 5-10°, iron stained. at 4.66m: wood fragments from 4.70 m: becoming pale grey mottled red-brown and yellow-brown NO CORE: 5.26 - 5.59 m. SANDSTONE: fine to medium grained, pale grey mottled yellow-brown, very thinly bedded to thickly laminated at 0-5°. SILTSTONE: grey and pale grey, indistinct bedding, multiple irregular healed defects, ironstone inclusions. NO CORE: 6.87 - 7.00 m. SANDSTONE: fine to coarse grained, yellow-brown, pale brown and pale grey, indistinctly bedded, with fine gravel quartz inclusions, iron stained . from 7.03 to 7.06 m: ironstone from 7.80 m: trace quartz inclusions SANDSTONE: medium to coarse grained, pale grey, very thickly bedded at 0-10°, iron stained . NO CORE: 9.28 - 9.38 m. SANDSTONE: medium to coarse grained, pale grey, iron stained .	4.0	1.0	
					3.0	2.0	
					2.0	3.0	
					1.0	4.0	
					0.0	5.0	
					-1.0	6.0	
					-2.0	7.0	
					-3.0	8.0	
					-4.0	9.0	
					-5.0	10.0	
							Sand Backfill

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

DBH15 Sheet 1 of 1

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 03/03/26

Checked by: J Riley

End Date: 04/03/26

Driller: Pointe Engineering Pty Ltd

Hole Diameter: 76 mm

Drill Rig: Ripamonti Skid Drill

Inclination: 90°

Total Depth: 7.99 m

Bearing: N/A

Easting: 314148.35

Northing: 6238530.59

Hor. Datum: MGA2020-56

Ground Level: 4.22 m

Ver. Datum: AHD

Top of Casing: m

METHOD	GROUND WATER	REMARKS	GRAPHIC LOG	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	RL (m AHD)	DEPTH (m)	WELL DIAGRAM
W	S			Silty SAND: fine to medium grained, brown; trace gravel, gravel is fine to coarse, sub-angular, sandstone. Gravelly SAND: fine to medium grained, orange-brown; gravel, fine to coarse, sub-angular, sandstone. NO CORE: 0.13 - 0.60 m (470mm).	4.0	0.0	Concrete
				SANDSTONE: fine to medium grained, pale grey with black flecks and mottled orange-brown, indistinctly bedded, iron stained. from 0.83 m: becoming laminated to very thinly bedded at 0-5deg from 1.07 m: becoming pale red-brown with black flecks from 1.37 m: becoming indistinctly bedded from 1.55 m: becoming red-brown; distinctly iron stained from 1.64 - 1.86 m: becoming pale grey; occasional carbonaceous material.	3.0	1.0	Bentonite
				NO CORE: 2.38 - 2.45 m (70 mm). SANDSTONE: fine to medium grained, pale red brown with black flecks, indistinctly bedded, occasional carbonaceous material; occasional iron staining. from 3.23 - 3.55 m: becoming cross-bedded at 5 - 10 mm at 50-60deg from 3.57 - 3.61 m: becoming pale grey	2.0	2.0	
				from 4.07 - 4.33 m: becoming red-brown with orange-brown banding; laminated to very thinly bedded at 20deg; iron stained from 4.45 to 4.53 m: cross bedding visible from 4.70 m: trace inclusions of silt, up to 5 mm	1.0	3.0	
				NO CORE: 5.11 - 5.31 m (200 mm). SANDSTONE: fine to medium grained, pale grey mottled brown with black flecks, medium to thickly bedded at 0-20°, trace siltstone inclusions up to 3 mm wide; trace iron staining. from 5.52 m: becoming pale grey, grey and red-brown NO CORE: 5.82 - 5.90 m (80 mm). SANDSTONE: fine to medium grained, pale grey, grey and red-brown, thinly to very thinly bedded at 0-20°, trace cross-bedding, iron stained. from 6.68 m: iron stained defects	0.0	4.0	Sand Backfill
				NO CORE: 7.34 - 7.43 m (90 mm). SANDSTONE: fine to medium grained, pale grey with black flecks and mottled red-brown, very thinly bedded at 15-20° and cross-bedded at 30-40°, trace gravel inclusions, fine, subrounded, quartz; occasional iron staining. from 7.48 - 7.78 m: trace carbonaceous material	-1.0	5.0	
				Terminated at 7.99 m. Target depth.	-2.0	6.0	
					-3.0	7.0	
					-4.0	8.0	Bentonite
					-5.0	9.0	
					-10.0	10.0	

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

DBH35B
Sheet 1 of 1

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Driller: Pointe Engineering

Drill Rig: Speedy Drill

Total Depth: 7 m

Hole Diameter: 50 mm

Inclination: 90°

Bearing: N/A

Project No: 60731324

Logged by: A Sharma

Start Date: 26/03/26

Checked by: C Jayasuriya

End Date: 26/03/26

Easting: 314046.32

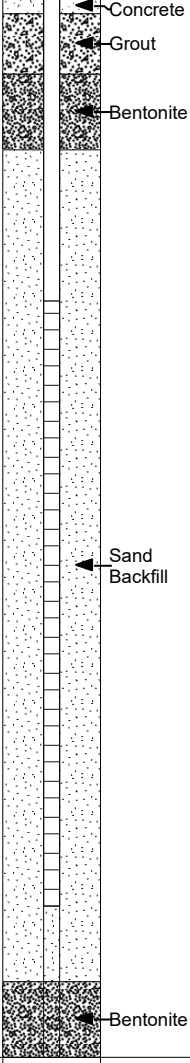
Northing: 6238574.15

Hor. Datum: MGA2020-56

Ground Level: 24.431 m

Ver. Datum: AHD

Top of Casing: m

METHOD	GROUND WATER	REMARKS	GRAPHIC LOG	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	RL (m AHD) DEPTH (m)	WELL DIAGRAM
WB	25/03/2026 15:00:00 ▼			TOPSOIL - Silty SAND: fine to coarse grained, brown; silt, low plasticity; with root fibres.	0.0	
				SP Gravelly SAND: fine to coarse grained, orange-brown; gravel, fine to coarse grained, angular. SANDSTONE: fine to coarse grained, red-brown mottled grey with red-brown bands, iron stained.	24.0 1.0 23.0 2.0 22.0 3.0 21.0 4.0 20.0 5.0 19.0 6.0 18.0 7.0	
				Terminated at 7.00 m. Standpipe piezometer installed.	17.0 8.0 16.0 9.0 15.0 10.0	

DBH38B
Sheet 1 of 2

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Driller: Pointe Engineering

Drill Rig: Ripamonti Speedydrill

Total Depth: 12.23 m

Hole Diameter: 76 mm

Inclination: 90°

Bearing: N/A

Project No: 60731324

Logged by: R Rees

Start Date: 25/02/26

Checked by: J Riley

End Date: 25/02/26

Easting: 313936.76

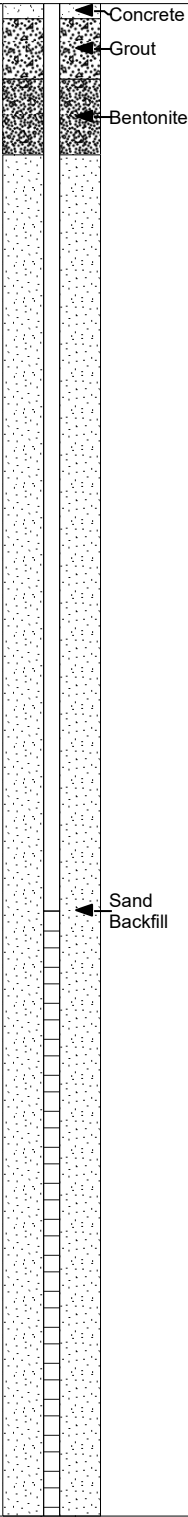
Northing: 6238749.77

Hor. Datum: MGA2020-56

Ground Level: 37.659 m

Ver. Datum: AHD

Top of Casing: m

METHOD	GROUND WATER	REMARKS	GRAPHIC LOG	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	RL (m AHD) DEPTH (m)	WELL DIAGRAM
WB				TOPSOIL - Silty SAND: fine to medium grained, pale brown; trace organic fibers.	0.0	
				SANDSTONE: fine to medium grained, pale brown, medium to high strength, moderately to slightly weathered.	37.0	
					1.0	
					2.0	
					3.0	
					4.0	
					5.0	
					6.0	
					7.0	
					8.0	
					9.0	
					10.0	

DBH38B
Sheet 2 of 2

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Driller: Pointe Engineering

Drill Rig: Ripamonti Speedydrill

Total Depth: 12.23 m

Hole Diameter: mm

Inclination: 90°

Bearing: N/A

Project No: 60731324

Logged by: R Rees

Start Date: 25/02/26

Easting: 313936.76

Northing: 6238749.77

Hor. Datum: MGA2020-56

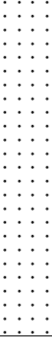
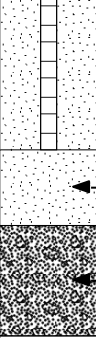
Checked by: J Riley

End Date: 25/02/26

Ground Level: 37.659 m

Ver. Datum: AHD

Top of Casing: m

METHOD	GROUND WATER	REMARKS	GRAPHIC LOG	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	RL (m AHD)	DEPTH (m)	WELL DIAGRAM
WB	04/03/2026 10:19:51			SANDSTONE: fine to medium grained, pale brown, medium to high strength, moderately to slightly weathered.	27.0 11.0 26.0 12.0		 Sand Backfill Bentonite
				Terminated at 12.23 m. Groundwater monitoring well installed.	25.0 13.0 24.0 14.0 23.0 15.0 22.0 16.0 21.0 17.0 20.0 18.0 19.0 19.0 18.0 20.0		

DBH67B
Sheet 1 of 1

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A Sharma

Start Date: 18/03/26

Checked by: C Jayasuriya

End Date: 18/03/26

Driller: Pointe Engineering

Drill Rig: Speedy Drill

Total Depth: 8 m

Hole Diameter: 50 mm

Inclination: 90°

Bearing: N/A

Easting: 314590.52

Northing: 6237926.85

Hor. Datum: MGA2020-56

Ground Level: 20.544 m

Ver. Datum: AHD

Top of Casing: m

METHOD	GROUND WATER	REMARKS	GRAPHIC LOG	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	RL (m AHD) DEPTH (m)	WELL DIAGRAM
WB	18/03/2026 10:00:00			TOPSOIL - Sandy GRAVEL: fine to coarse grained, sub-angular, red-brown; sand, fine to coarse grained. GM Silty GRAVEL: fine to coarse grained, sub-angular, yellow-brown. SANDSTONE: fine to medium grained, pale grey and yellow-brown, very thinly bedded, iron stained.	0.0 20.0 1.0 19.0 2.0 18.0 3.0 17.0 4.0 16.0 5.0 15.0 6.0 14.0 7.0 13.0 8.0 12.0 9.0 11.0 10.0	Concrete Bentonite Sand Backfill Bentonite
				Terminated at 8.00 m.		

Client: Transport for NSW

Project No: 60731324

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Logged by: A Sharma

Checked by: J Riley

Location: Heathcote Road, Holsworthy, NSW 2173

Start Date: 25/02/26

End Date: 25/02/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering

Hole Diameter: 50 mm

Easting: 314531.41

RL: 6.292 m

Drill Rig: Atlas Copco

Inclination: 90°

Northing: 6237986.35

Ver. Datum: AHD

Total Depth: 0.55 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Fill

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
WS	U	2	D: 0.00-0.05 m D: 0.05-0.50 m	Not Encountered	6.0	0.0		SM	GRAVEL: fine to coarse grained, sub-angular, brown mottled yellow; trace roots. Silty SAND: fine to coarse grained, dark brown.	D	VD	FILL COLLUVIUM
		13										
		23										
		16										
		17										
		12	D: 0.50-0.55 m					GP	GRAVEL: coarse grained, yellow-brown. Terminated at 0.55 m. Refusal on inferred boulder.			
		12										
		10										
		20										
		14										
		25										

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project No: 60731324

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Logged by: A Sharma

Checked by: J Riley

Location: Heathcote Road, Holsworthy, NSW 2173

Start Date: 25/02/26

End Date: 25/02/26

Driller: Pointe Engineering

Hole Diameter: 50 mm

Easting: 314520.41

Location Meth.: dGPS0.1

Drill Rig: Atlas Copco R Cobra

Inclination: 90°

Northing: 6238015.87

Ver. Datum: AHD

Total Depth: 1.3 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Topsoil

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
WS	U	1	D: 0.00-0.25 m	Not Encountered	4.0	0.0			TOPSOIL - Silty SAND: fine grained, grey brown; silt, low plasticity ; root fibres.	M		TOPSOIL
		4										
		7	D: 0.25-0.50 m						Silty SAND: fine to coarse grained, dark brown, low plasticity; root fibres.		D	COLLUVIUM
		9										
		10	D: 0.50-0.65 m					SP	Gravelly SAND: fine to coarse grained, brown with grey; gravel, coarse grained, sub-angular.			
		25	D: 0.65-1.10 m					SP	Gravelly SAND: fine to coarse grained, dark brown; gravel, fine to coarse grained, angular.		VD	
			D: 1.10-1.30 m		3.0			SM	Silty SAND: fine grained, dark brown and pale brown; silt, low plasticity .			core
									Terminated at 1.30 m. Refusal on inferred boulder.			
						2.0						
						3.0						
						4.0						
						5.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A Sharma

Start Date: 23/02/26

Checked by: J Riley

End Date: 23/02/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering

Hole Diameter: 50 mm

Easting: 314420.74

RL: 1.577 m

Drill Rig: Atlas Copo R Cobra TTe

Inclination: 90°

Northing: 6238227.76

Ver. Datum: AHD

Total Depth: 2 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Grass

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
WS	U	1	D: 0.00-0.20 m	23/02/2026 15:13:01	1.577	0.0		ML	TOPSOIL - SILT: low plasticity, dark brown; trace gravel, coarse grained, angular; roots.	W	S - F VL - L MD - D L	TOPSOIL
		1	D: 0.20-0.50 m			0.2		CI-CH	Sandy CLAY: medium to high plasticity, dark brown; sand, fine grained; roots.			ALLUVIUM
		1	D: 0.50-0.90 m			0.5		SP	SAND: coarse grained, orange-brown.			
		1	D: 0.90-2.00 m			0.9						
		1				1.0						
		1				1.1						
		1				1.2						
		1				1.3						
		1				1.4						
		1				1.5						
		1				1.6						
		1				1.7						
		1				1.8						
		1				1.9						
		1				2.0						
						2.0			Terminated at 2.00 m.			
						-1.0						
						-3.0						
						-2.0						
						-4.0						
						-5.0						
						-6.0						
						-7.0						
						-8.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A Sharma

Start Date: 20/02/26

Checked by: J Riley

End Date: 20/02/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering

Drill Rig: Atlas Copco R Cobra

Total Depth: 2.3 m

Hole Diameter: 50 mm

Inclination: 90°

Bearing: N/A

Easting: 314208.62

Northing: 6238465.65

Hor. Datum: MGA2020-56

RL: 1.624 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY/RELIATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)						
WS	U	1	D: 0.00-0.30 m	Not Observed	1.624	0.0			TOPSOIL - Clayey SAND: fine to coarse grained, brown; trace gravel, fine to medium, sub-angular, sandstone .	D		TOPSOIL						
		5	D: 0.30-0.65 m				FILL - Gravelly SAND: fine to coarse grained, black mottled red; gravel, fine to coarse grained, sub-angular, sandstone and ironstone ; trace organic fibres.		FILL									
		4				D: 0.65-1.00 m	CL	Sandy CLAY: medium plasticity, yellow-brown; sand, fine to coarse grained .	w ≈ PL	VS - S	ALLUVIUM							
		0										PP=60, 70, 60 kPa	D: 1.00-1.80 m	SC	Clayey SAND: fine to coarse grained, dark brown.		L	from 0.90 - 1.00 m, 100mm sample loss
		2																
		2																
		1																
		1																
		1																
		2										from 1.80 - 2.30 m 500mm sample loss, possible groundwater table						
		2																
		2																
		2																

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A Sharma

Start Date: 20/02/26

Checked by: J Riley

End Date: 20/02/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering

Drill Rig: Atlas Copco R Cobra
TTG

Total Depth: 2 m

Hole Diameter: 50 mm

Inclination: 90°

Bearing: N/A

Easting: 314171.85

Northing: 6238506.91

Hor. Datum: MGA2020-56

RL: 1.139 m

Ver. Datum: AHD

Surface: Topsoil

[illegible]

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A Sharma

Start Date: 19/02/26

Checked by: J Riley

End Date: 19/02/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering

Drill Rig: Atlas Copco R Cobra

Total Depth: 0.15 m

Hole Diameter: 50 mm

Inclination: 90°

Bearing: N/A

Easting: 314135.64

Northing: 6238543.59

Hor. Datum: MGA2020-56

RL: 7.715 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
WS	U	9	D: 0.00-0.10 m	Not		0.0			TOPSOIL - Silty SAND: fine to medium grained, brown; trace gravel, fine to medium, sub-rounded, sandstone; trace root fibres.	D		TOPSOIL
		25	D: 0.10-0.15 m	Encounter				SP	Gravelly SAND: fine to medium grained, yellow-brown mottled orange; gravel, fine to coarse, sub-angular, sandstone.		VD	COLLUVIUM
									Terminated at 0.15 m. Refusal on inferred top of rock.			
						7.0						
						1.0						
						6.0						
						2.0						
						5.0						
						3.0						
						4.0						
						4.0						
						3.0						
						5.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: R Rees
Start Date: 19/02/26

Checked by: J Riley
End Date: 19/02/26
Location Meth.: dGPS0.1
Ver. Datum: AHD
Surface: Pavement

Driller: Durkin Construction Pty Ltd
Drill Rig: Pavement Truck 37
Total Depth: 0.62 m

Hole Diameter: 300 mm
Inclination: 90°
Bearing: N/A

Easting: 314590.66
Northing: 6237707.39
Hor. Datum: MGA2020-56

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC				B: 0.00-0.16 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 25 mm, 160 mm thick.			PAVEMENT
AD/T	U		Not Encountered	B: 0.16-0.62 m					FILL - Gravelly Clayey SAND: fine to coarse grained, dark grey; gravel, fine to coarse, sub-rounded, basalt.	M		FILL
		22/20 mm							Terminated at 0.62 m. Refusal.			
						29.0						
						1.0						
						28.0						
						2.0						
						27.0						
						3.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A Sharma

Start Date: 23/02/26

Checked by: J Riley

End Date: 23/02/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering .

Drill Rig: Atlas Copco R Cobra

Drinking. TTe

Total Depth: 0.2 m

Hole Diameter: 50 mm

Inclination: 90°

Bearing: N/A

Easting: 314551.72

Northing: 6238011.42

Hor. Datum: MGA2020-56

RL: 9.001 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
WS	U	9	D: 0.00-0.20 m	Not Observed	9.0	0.0		GW	TOPSOIL - Silty SAND: fine grained, brown, poorly graded; silt, low plasticity; roots and grass.	D	D	TOPSOIL
		20									VD	RESIDUAL SOIL
		25										Inferred Sandstone
									Terminated at 0.20 m. Refusal.			
					8.0	1.0						
					7.0	2.0						
					6.0	3.0						
					5.0	4.0						
					4.0	5.0						
					3.0	6.0						
					2.0	7.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project No: 60731324

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Logged by: A Sharma

Checked by: J Riley

Location: Heathcote Road, Holsworthy, NSW 2173

Start Date: 19/02/26

End Date: 19/02/26

Driller: Pointe Engineering

Hole Diameter: 50 mm

Easting: 314000.24

Location Meth.: dGPS0.1

Drill Rig: Atlas Copco R Cobra

Inclination: 90°

Northing: 6238670.95

Ver. Datum: AHD

Total Depth: 0.17 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Topsoil

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
WS	U	8	D: 0.00-0.17 m	Not Encountered		0.0			TOPSOIL - Silty SAND: fine to medium grained, pale brown; with gravel, fine to medium, sub-angular to sub-rounded, sandstone ; trace root fibres.	D		TOPSOIL
		8										
		4				30.0			Terminated at 0.17 m. Refusal on inferred top of rock.			
		25										
						1.0						
						29.0						
						2.0						
						28.0						
						3.0						
						27.0						
						4.0						
						26.0						
						5.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project No: 60731324

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Logged by: A Sharma

Checked by: J Riley

Location: Heathcote Road, Holsworthy, NSW 2173

Start Date: 23/02/26

End Date: 23/02/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering

Hole Diameter: 50 mm

Easting: 314417.91

RL: 1.463 m

Drill Rig: Atlas Copco R Cobra

Inclination: 90°

Northing: 6238238.53

Ver. Datum: AHD

Total Depth: 2 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Topsoil

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)	
WS	U	1	D: 0.00-0.20 m	Not Observed	0.0	0.0			TOPSOIL - SILT: dark brown, with roots.	w < PL	VS	TOPSOIL	
		1	ES: 0.20-0.40 m						ML			Sandy SILT: dark brown; sand, fine grained.	
		1	D: 0.20-0.40 m										
		0	D: 0.40-0.70 m										
		1	D: 0.40-0.70 m										
		1	D: 0.40-0.70 m										
		2	D: 0.70-1.00 m						SC	Clayey SAND: coarse grained, grey-brown.	w ≈ PL	MD - D	ALLUVIUM
		3	D: 0.70-1.00 m										
		4	D: 0.70-1.00 m										
		5	ES: 1.00-1.20 m						SP	SAND: coarse grained, grey-brown.	W		
		6	D: 1.20-2.00 m										
		7	D: 1.20-2.00 m										
		4	D: 1.20-2.00 m										
		3	D: 1.20-2.00 m										
		4	D: 1.20-2.00 m										
Terminated at 2.00 m.													

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: A Sharma

Start Date: 20/02/26

Checked by: J Riley

End Date: 20/02/26

Location Meth.: dGPS0.1

Driller: Pointe Engineering
Drill Rig: Atlas Copco R Cobra
TTe

Total Depth: 2 m

Hole Diameter: 50 mm

Inclination: 90°

Bearing: N/A

Easting: 314280.05

Northing: 6238375.31

Hor. Datum: MGA2020-56

RL: 5.207 m

Ver. Datum: AHD

Surface: Topsoil

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)					
WS	U	0	D: 0.00-0.25 m			0.0			TOPSOIL - Sandy CLAY: medium plasticity, dark brown; sand, fine to medium grained .	w ≈ PL		TOPSOIL					
		1	PP=60, 80, 110 kPa D: 0.25-1.40 m			SM	Silty SAND: fine to medium grained, brown; with clay. From 0.75 m : becoming yellow-brown	M	VL - L	From 1.33 - 1.38m, sample loss							
		1															
		1															
		3															
		3															
		1															
		0															
		3															
		1															
		1															
		1															
		0															
		0															
		0															
		0															
		0															
		0															
											2.0			Terminated at 2.00 m. Target depth.			
											3.0						
		3.0															
		2.0															
		4.0															
		1.0															

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 19/02/26

Checked by: J Riley

End Date: 19/02/26

Location Meth.: dGPS0.1

Driller: Durkin Construction Pty Ltd

Hole Diameter: 300 mm

Easting: 314457.36

RL: 4 m

Drill Rig: Utilcor Pavement Truck

Inclination: 90°

Northing: 6238153.60

Ver. Datum: AHD

Total Depth: 2 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC				CONCC: 0.00-0.87 m	4.0	0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 20 mm, 180 mm thick.			PAVEMENT
									CONCRETE: grey, aggregates up to 25 mm, 180 mm thick.			
									Heavily Bound: fine to coarse grained, angular to sub-angular, brown-grey, 510 mm thick.			
AD/T	U	9 22	Not Encountered	D: 0.87-0.92 m B: 0.92-1.10 m D: 0.92-1.10 m B: 1.10-1.60 m D: 1.10-1.60 m B: 1.60-2.00 m D: 1.60-2.00 m	3.0	1.0			FILL - Gravelly Clayey SAND: fine to coarse grained, dark brown; gravel, fine to coarse, angular to sub-angular, basalt. FILL - Gravelly Clayey SAND: fine to medium grained, yellow-brown; gravel, fine to coarse, angular to sub-angular, sandstone and basalt.	M		FILL
									FILL - Gravelly SAND: fine to coarse grained, brown; gravel, sandstone and basalt; with clay; trace organics, wood fragments and root fibres.			
									SP SAND: fine to coarse grained, brown; with clay; trace organics, wood fragments and root fibres.			
		1 2 14 6									MD - D	ALLUVIUM
					2.0	2.0			Terminated at 2.00 m. Target depth.			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 19/02/26

Checked by: J Riley

End Date: 19/02/26

Location Meth.: dGPS0.1

Driller: Durkin Construction Pty Ltd

Hole Diameter: 300 mm

Easting: 314434.80

RL: 3.583 m

Drill Rig: Utilicor Pavement Truck

Inclination: 90°

Northing: 6238190.40

Ver. Datum: AHD

Total Depth: 2 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC				CONCC: 0.00-0.40 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 20 mm thick, 80 mm thick.			PAVEMENT
							ASPHALTIC CONCRETE: dark grey, aggregates up to 30 mm, 100 mm thick.					
AD/T	U	22/25 mm	Not Encountered	B: 0.40-0.61 m D: 0.40-0.61 m					ASPHALTIC CONCRETE: dark grey, aggregates up to 15 mm, 70 mm thick.	D		FILL
							ASPHALTIC CONCRETE: dark grey, aggregates up to 30 mm, 90 mm thick.					
				6				ASPHALTIC CONCRETE: dark grey, aggregates up to 20 mm, 70 mm thick.	M			
				11				FILL - Gravelly SAND: fine to medium grained, pale yellow-brown; gravel, fine to coarse, angular to sub-rounded, sandstone and basalt; with clay.				
				17				FILL - Gravelly SAND: fine to coarse grained, yellow-brown; gravel, fine to coarse, angular to sub-rounded, sandstone; with clay.				
				10								
				16				FILL - Gravelly SAND: fine to medium grained, yellow-brown; with gravel, fine to coarse, sub-angular to rounded, sandstone and basalt; with clay.				
				14								
				10								
				6								
				4								
				5								
				5								
				6								
				7								
												B: 1.40-2.00 m D: 1.40-2.00 m
						2.0			Terminated at 2.00 m. Target depth.			

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Driller: Durkin Construction Pty Ltd

Drill Rig: Utilicor Pavement Truck

Total Depth: 0.75 m

Hole Diameter: 300 mm

Inclination: 90°

Bearing: N/A

Project No: 60731324

Logged by: R Rees

Start Date: 19/02/26

Easting: 314593.27

Northing: 6237731.48

Hor. Datum: MGA2020-56

Checked by: J Riley

End Date: 19/02/26

Location Meth.: dGPS0.1

Ver. Datum: AHD

Surface: Pavement

METHOD	PC	AD/T	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
			U		Not Encountered	CONCC: 0.00-0.12 m	27.0	0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 25 mm, 110 mm thick.			PAVEMENT
				19		B: 0.12-0.51 m					FILL - Gravelly Clayey SAND: fine to coarse grained, dark brown; gravel, fine to coarse, angular to sub-angular, basalt.	D		FILL
				22		B: 0.51-0.75 m		SP	SAND: medium to coarse grained, yellow-brown; with gravel, fine to coarse, angular to sub-angular, sandstone; trace clay.	M	VD	RESIDUAL SOIL		
								1.0			Terminated at 0.75 m. Refusal.			
								26.0						
								2.0						
								25.0						
								3.0						

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: R Rees
Start Date: 19/02/26

Checked by: J Riley
End Date: 19/02/26
Location Meth.: dGPS0.1
Ver. Datum: AHD
Surface: Pavement

Driller: Durkin Construction Pty Ltd
Drill Rig: Utilicor Pavement Truck
Total Depth: 0.82 m

Hole Diameter: 300 mm
Inclination: 90°
Bearing: N/A

Easting: 314576.46
Northing: 6237921.30
Hor. Datum: MGA2020-56

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC	U	6	Not Encountered	CONCC: 0.00-0.26 m	0.00	0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 25 mm, 90 mm thick.	M		PAVEMENT
				B: 0.26-0.42 m					ASPHALTIC CONCRETE: dark grey, aggregates up to 20 mm, 120 mm thick.			FILL
				B: 0.42-0.71 m					FILL - Clayey SAND: medium to coarse grained, brown; with gravel, fine to coarse, sub-angular to angular, sandstone.			
									FILL - SAND: medium to coarse grained, brown; with gravel, fine to coarse, sub-angular to angular, sandstone; with clay.			
		25/30 mm							SANDSTONE: medium to coarse grained, brown and red-brown, highly weathered, low strength; iron staining.			HAWKESBURY SANDSTONE
									Terminated at 0.82 m. Refusal.			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: R Rees
Start Date: 18/02/26

Checked by: J Riley
End Date: 18/02/26
Location Meth.: dGPS0.1

Driller: Durkin Construction Pty Ltd
Drill Rig: Utilicor Pavement Truck
Total Depth: 1.1 m

Hole Diameter: 300 mm
Inclination: 90°
Bearing: N/A

Easting: 314520.56
Northing: 6238048.72
Hor. Datum: MGA2020-56

RL: 4.4981 m
Ver. Datum: AHD
Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC	U		Not Encountered	CONCC: 0.00-0.76 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 20 mm, 200 mm thick.			PAVEMENT
									FILL - CONCRETE: grey, aggregates up to 30 mm, 150 mm thick.			
AD/T						1.0			FILL: HEAVILY BOUND, brown-grey, aggregates up to 30 mm, 750 mm thick.			
						3.0			Terminated at 1.10 m. Refusal.			From 0.75 m: At maximum capacity of coring bit (900 mm), unable to recover remaining pavement, switched to auger

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 20/02/26

Checked by: J Riley

End Date: 20/02/26

Location Meth.: dGPS0.1

Driller: Durkin Construction Pty Ltd

Hole Diameter: 300 mm

Drill Rig: Utilicor Pavement Truck

Inclination: 90°

Total Depth: 0.58 m

Bearing: N/A

Easting: 314577.52

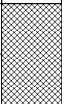
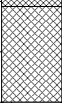
Northing: 6237607.36

Hor. Datum: MGA2020-56

RL: 36.914 m

Ver. Datum: AHD

Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
AD/T	PC	U	Not Encountered	CONCC: 0.00-0.21 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 30 mm, 210 mm thick.			PAVEMENT
				B: 0.21-0.40 m				FILL - Gravelly SAND: fine to coarse grained, brown; gravel, fine to coarse, angular to sub-rounded, sandstone and basalt; with clay.	D		FILL	
				B: 0.40-0.58 m			SP	SAND: medium to coarse grained, yellow-brown; with gravel, fine to medium, sub-angular, sandstone.	M	VD	EXTREMELY WEATHERED MATERIAL	
Terminated at 0.58 m. Refusal.												

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 09/02/26

Checked by: J Riley

End Date: 09/02/26

Location Meth.: dGPS0.1

Driller: Durkin Construction Pty Ltd

Hole Diameter: 300 mm

Drill Rig: Utilicor Pavement Truck

Inclination: 90°

Total Depth: 0.52 m

Bearing: N/A

Easting: 314578.19

Northing: 6237893.44

Hor. Datum: MGA2020-56

RL: 17.057 m

Ver. Datum: AHD

Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC	U		Not Encountered	CONCC: 0.00-0.21 m	17.0	0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 15 mm, 100 mm thick.	M	VD	PAVEMENT
						ASPHALTIC CONCRETE: dark grey, aggregates up to 15 mm, 60 mm thick.		at 0.10m: asphaltic concrete is degraded				
						ASPHALTIC CONCRETE: dark grey, aggregates up to 10 mm, 50 mm thick.						
						FILL - Gravelly SAND: fine to medium grained, yellow-brown; gravel, fine to medium, sub-angular to angular, ironstone and sandstone; with clay. Gravelly SAND: fine to medium grained, pale-brown; gravel, fine to coarse; trace silt.		FILL				
AD/T		18		B: 0.21-0.27 m B: 0.27-0.52 m								EXTREMELY WEATHERED MATERIAL
		20/20 mm							Terminated at 0.52 m. Refusal on inferred top of rock.			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Project No: 60731324

Logged by: R Rees

Start Date: 18/02/26

Checked by: J Riley

End Date: 18/02/26

Location Meth.: dGPS0.1

Hole Diameter: 300 mm

Easting: 314285.62

Inclination: 90°

Northing: 6238353.00

Ver. Datum: AHD

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC	U	5 22 11 3 22/25 mm	Not Encountered	CONCC: 0.00-0.18 m	0.0 3.0 <							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: R Rees
Start Date: 17/02/26

Checked by: J Riley
End Date: 17/02/26
Location Meth.: dGPS0.1
RL: 3.609 m
Ver. Datum: AHD
Surface: Pavement

Driller: Durkin Construction Pty Ltd
Drill Rig: Utilicor Pavement Truck
Total Depth: 0.96 m
Hole Diameter: 300 mm
Inclination: 90°
Bearing: N/A

Easting: 314235.79
Northing: 6238406.45
Hor. Datum: MGA2020-56

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC				CONCC: 0.00-0.09 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 30 mm, 90 mm thick.			PAVEMENT
AD/T	U	19 22/80 mm	Not Observed	B: 0.17-0.27 m					FILL - Gravelly SAND: fine to coarse grained, brown; gravel, fine to medium, sub-angular to angular, sandstone; with clay; trace organic matter, root fibres.	D		FILL
				B: 0.27-0.46 m					FILL - Gravelly SAND: fine to medium grained, pale brown; gravel, fine to coarse, sub-angular to angular, sandstone; with cobbles, sub-angular to angular, sandstone; with clay.			
				B: 0.46-0.96 m				SC	Clayey SAND: fine to coarse grained, pale yellow-brown; with gravel, fine to coarse, sub-angular to angular, sandstone.			RESIDUAL SOIL
						3.0				M	VD	
						1.0			Terminated at 0.96 m. Refusal on inferred top of rock.			
						2.0						
						2.0						
						1.0						
						3.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: R Rees
Start Date: 11/02/26

Checked by: J Riley
End Date: 11/02/26
Location Meth.: dGPS0.1
RL: 16.368 m
Ver. Datum: AHD
Surface: Pavement

Driller: Durkin Construction Pty Ltd
Drill Rig: Utilicor Pavement Truck
Total Depth: 0.72 m
Hole Diameter: 300 mm
Inclination: 90°
Bearing: N/A

Easting: 314087.04
Northing: 6238555.89
Hor. Datum: MGA2020-56

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC				CONCC: 0.00-0.24 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 40 mm, 240 mm thick.			PAVEMENT
AD/T	U		Not Encountered	B: 0.24-0.57 m		16.0			FILL - Gravelly SAND: fine to medium grained, brown; gravel, fine to medium, sub-angular to angular, ironstone and sandstone; trace clay.	D		FILL
				B: 0.57-0.72 m					SANDSTONE: fine to coarse grained, pale-brown, extremely weathered, very low to low strength.			HAWKESBURY SANDSTONE From 0.57 m : Jackhammer used.
									Terminated at 0.72 m. Refusal on inferred top of rock.			
						1.0						
						15.0						
						2.0						
						14.0						
						3.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 11/02/26

Checked by: J Riley

End Date: 11/02/26

Location Meth.: dGPS0.1

Driller: Durkin Construction Pty Ltd

Hole Diameter: 300 mm

Easting: 313888.49

RL: 39.538 m

Drill Rig: Utilicor Pavement Truck

Inclination: 90°

Northing: 6238758.29

Ver. Datum: AHD

Total Depth: 0.64 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Pavement

METHOD	PC	AD/T	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)		
			U		Not Encountered	CONCC: 0.00-0.34 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 30 mm, 210 mm thick.			PAVEMENT		
							B: 0.34-0.42 m					ASPHALTIC CONCRETE: grey-brown, aggregates up to 20 mm, 130 mm thick.				
						22/95 mm		B: 0.42-0.64 m					FILL - Gravelly SAND: fine to medium grained, yellow-brown; gravel, fine to medium, sub-angular to angular, sandstone and ironstone; trace clay.			FILL
											SP		Gravelly SAND: fine to medium grained, pale yellow-brown; gravel, fine to medium, angular, sandstone; trace clay; sandstone parent rock.	M	VD	EXTREMELY WEATHERED MATERIAL
							39.0									
								1.0								
							38.0									
								2.0								
							37.0									
											Terminated at 0.64 m. Refusal on inferred top of rock.					

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Driller: Durkin Contractor Pty Ltd

Drill Rig: Utilicor Pavement Truck

Total Depth: 0.5 m

Hole Diameter: 300 mm

Inclination: 90°

Bearing: N/A

Project No: 60731324

Logged by: R Rees

Start Date: 17/02/26

Easting: 313802.65

Northing: 6238863.92

Hor. Datum: MGA2020-56

Checked by: J Riley

End Date: 17/02/26

Location Meth.: dGPS0.1

RL: 41.1869 m

Ver. Datum: AHD

Surface: Pavement

METHOD	PC	AD/T	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)	
			U		Not Encountered	CONCC: 0.00-0.14 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 30 mm, 140 mm thick.			PAVEMENT	
							B: 0.14-0.30 m		41.0			FILL - Gravelly Clayey SAND: fine to medium grained, brown; gravel, fine to coarse, angular to sub-angular, sandstone and basalt.			FILL
						12		B: 0.30-0.50 m			SP	Gravelly SAND: fine to medium grained, pale yellow-brown; gravel, fine to medium, angular, sandstone; trace clay; trace organic matter, root fibres; sandstone parent rock.	M		EXTREMELY WEATHERED MATERIAL
						22HB								VD	
											Terminated at 0.50 m. Refusal on inferred top of rock.				
								1.0							
								40.0							
								2.0							
								39.0							
								3.0							

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 09/02/26

Checked by: J Riley

End Date: 09/02/26

Location Meth.: dGPS0.1

Driller: Durkin Construction Pty Ltd

Hole Diameter: 300 mm

Drill Rig: Utilicor Pavement Truck

Inclination: 90°

Total Depth: 0.87 m

Bearing: N/A

Easting: 314643.24

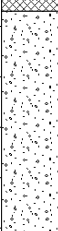
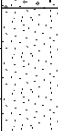
Northing: 6237879.75

Hor. Datum: MGA2020-56

RL: 25.959 m

Ver. Datum: AHD

Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC	AD/T U		Not Encountered	CONCC: 0.00-0.07 m B: 0.07-0.17 m	0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 15 mm, 70 mm thick.			PAVEMENT	
							FILL - Sandy GRAVEL: fine to coarse grained, yellow-brown, sub-angular to angular, ironstone and sandstone; sand, fine to coarse grained; trace clay.	D	FILL			
		18		B: 0.17-0.62 m			SP	Gravelly SAND: fine to coarse grained, pale brown; gravel, fine to medium, sub-angular to angular, ironstone and sandstone; trace clay; trace organic matter, root fibres.	M	VD	RESIDUAL SOIL	
		13										
		11										
		7										
		5		B: 0.62-0.87 m								
		9					SW	SAND: fine to coarse grained, yellow-brown; with gravel, fine to medium, angular, sandstone.		MD to VD		
	5											
		20/15 mm			Terminated at 0.87 m. Refusal on inferred top of rock.							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 11/02/26

Checked by: J Riley

End Date: 11/02/26

Location Meth.: dGPS0.1

Driller: Durkin Construction Pty Ltd

Hole Diameter: 300 mm

Drill Rig: Utilicor Pavement Truck

Inclination: 90°

Total Depth: 1.5 m

Bearing: N/A

Easting: 314155.56

Northing: 6238483.75

Hor. Datum: MGA2020-56

RL: 7.693 m

Ver. Datum: AHD

Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)			
PC	AD/T	U	Not Encountered	CONCC: 0.00-0.34 m	7.0	0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 15 mm, 70 mm thick. from 0.07 m: RECOVERED AS: Gravelly SAND: fine to coarse grained, dark grey; gravel, fine to medium, sub-angular, various lithologies.	M	VD	PAVEMENT From 0.07 m: Unable to recover pavement core using conventional methods due to degraded condition. Auger bit used to remove.			
B: 0.34-0.50 m				FILL - Gravelly SAND: fine to coarse grained, pale yellow-brown with black flecks; gravel, fine to medium, angular, sandstone; trace clay. from 0.50 m: becoming yellow-brown		FILL									
B: 0.50-0.78 m						SP	SAND: fine to medium grained, yellow-brown with black flecks; with clay; with gravel, fine to coarse, angular to sub-angular, sandstone.	VD							
B: 0.78-1.52 m				D					RESIDUAL SOIL						
22/80 mm															
4															
8															
11															
8/10 mm				Terminated at 1.50 m. Refusal on inferred top of rock.											

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 15/02/26

Checked by: J Riley

End Date: 15/02/26

Location Meth.: dGPS0.1

Driller: Durkin Construction Pty Ltd

Hole Diameter: 300 mm

Easting: 313706.59

RL: 40.857 m

Drill Rig: Utilicor Pavement Truck

Inclination: 90°

Northing: 6238962.58

Ver. Datum: AHD

Total Depth: 2 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
AD/T	PC	U	Not Encountered	CONCC: 0.00-0.21 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 30mm, 210mm thick.	D		PAVEMENT
				B: 0.21-0.61 m			FILL - Clayey Gravelly SAND: fine to medium grained, pale yellow-brown; gravel, fine to coarse, angular to sub-angular, sandstone and ironstone. .	FILL				
				B: 0.61-1.30 m			FILL - Gravelly SAND: fine to medium grained, orange-brown; gravel, fine to coarse, angular to sub-angular, ironstone and sandstone; with clay.					
				B: 1.30-2.00 m			CL	CLAY: low plasticity, pale brown; with gravel, fine to medium, angular, sandstone; with sand, fine to medium grained.	w < PL	St - VSt	RESIDUAL SOIL	
											Hole drilled to 2.0m, however material is collapsing to 1.7 m	
						2.0			Terminated at 2.00 m. Target depth.			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: R Rees
Start Date: 16/02/26

Checked by: J Riley
End Date: 16/02/26
Location Meth.: dGPS0.1

Driller: Durkin Construction Pty Ltd
Drill Rig: Utilicor Pavement Truck
Total Depth: 1.55 m

Hole Diameter: 300 mm
Inclination: 90°
Bearing: N/A

Easting: 313699.20
Northing: 6239036.72
Hor. Datum: MGA2020-56

Ver. Datum: AHD
Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
P. C				CONCC: 0.00-0.04 m B: 0.04-0.14 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 15 mm, 40 mm thick.			PAVEMENT
		6		B: 0.14-0.58 m					FILL - Gravelly Clayey SAND: fine to medium grained, yellow-brown; gravel, fine to medium, angular to subangular, basalt and sandstone.			FILL
		11							FILL - Gravelly SAND: fine to medium grained, yellow-brown; gravel, fine to coarse, angular to sub-angular, ironstone and sandstone; with clay.			
		7										
		10										
		9				38.0						
		7		B: 0.58-1.10 m				GP	Sandy GRAVEL: fine to coarse grained, angular to sub-angular, pale yellow-brown; sand, fine to medium grained; with clay.			RESIDUAL SOIL
		18										
		14/10 mm	Not Encountered			1.0						
				B: 1.10-1.55 m				SC	Clayey SAND: fine grained, pale yellow-brown; with gravel, fine to coarse, angular, sandstone.			EXTREMELY WEATHERED MATERIAL
						37.0						
									Terminated at 1.55 m. Refusal.			
						2.0						
						36.0						
						3.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 15/02/26

Checked by: J Riley

End Date: 15/02/26

Location Meth.: dGPS0.1

Driller: Durkin Construction Pty Ltd

Hole Diameter: 300 mm

Drill Rig: Utilicor Pavement Truck

Inclination: 90°

Total Depth: 2 m

Bearing: N/A

Easting: 313267.47

Northing: 6239253.01

Hor. Datum: MGA2020-56

RL: 30.99 m

Ver. Datum: AHD

Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC				CONCC: 0.00-0.18 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 20 mm, 50 mm thick.			PAVEMENT
									ASPHALTIC CONCRETE: dark grey, aggregates up to 10 mm, 40 mm thick.			
									ASPHALTIC CONCRETE: dark grey, aggregates up to 20 mm, 40 mm thick.			
									ASPHALTIC CONCRETE: dark grey, aggregates up to 10 mm, 50 mm thick.			
				B: 0.18-0.40 m					FILL - Gravelly SAND: fine to medium grained, brown; gravel, fine to medium, sub-angular, sandstone, charcoal and ironstone; with clay.			FILL
		11		B: 0.40-0.72 m					At 0.40 m: becoming orange-brown.			
		14										
		11										
		8		B: 0.72-1.70 m				SP	SAND: fine grained, yellow-brown; with clay; with gravel, fine to medium, sub-angular to sub-rounded, sandstone.			RESIDUAL SOIL
		6										
		5										
		6										
		3										
		4										
		7										
		12										
		14										
		12										
		4		B: 1.70-2.00 m				SC	Clayey SAND: fine to medium grained, brown; with gravel; with gravel, fine to medium, sub-angular to sub-rounded, sandstone; with organics, wood fragments.			EXTREMELY WEATHERED MATERIAL
		2										
		3										
						29.0			Terminated at 2.00 m. Target depth.			
						28.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Project No: 60731324

Logged by: R Rees

Start Date: 15/02/26

Checked by: J Riley

End Date: 15/02/26

Location Meth.: dGPS0.1

Hole Diameter: 300 mm

Easting: 312789.45

Inclination: 90°

Northing: 6239557.79

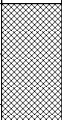
RL: 17.51 m

Ver. Datum: AHD

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
AD/T	PC	U	Not Encountered	CONCC: 0.00-0.38 m	0.00-0.38 m	0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 30 mm, 380 mm thick.			PAVEMENT
				Heavily Bound: brown-grey, aggregates up to 30 mm, 140 mm thick.								
				B: 0.38-0.44 m								
						17.0			Terminated at 0.44 m. Refusal.			
						1.0						
						16.0						
						2.0						
						15.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by:
Start Date: 08/02/26

Checked by: J Riley
End Date: 08/02/26
Location Meth.: dGPS0.1
RL: 39.44 m
Ver. Datum: AHD
Surface: Asphalt

Driller:
Drill Rig:
Total Depth: 1.09 m
Hole Diameter: 300 mm
Inclination: 90°
Bearing: N/A

Easting: 314568.08
Northing: 6237568.76
Hor. Datum: MGA2020-56

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC	U	19 22	Not Encountered	CONCC: 0.00-0.15 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 20 mm, 30 mm thick.	M	D to VD	PAVEMENT
									ASPHALTIC CONCRETE: dark grey, aggregates up to 10 mm, 50 mm thick.			
									ASPHALTIC CONCRETE: dark grey, aggregates up to 20 mm, 50 mm thick.			
				B: 0.15-0.35 m					FILL - Silty Gravelly SAND: fine to coarse grained, brown; gravel, fine to medium, sub-angular to angular, ironstone and sandstone.			FILL
AD/T	U		Not Encountered	B: 0.35-1.09 m		39.0		SW	Silty Gravelly SAND: fine to coarse grained, pale yellow-brown; gravel, fine to coarse, angular, sandstone. .	M	D to VD	EXTREMELY WEATHERED MATERIAL
						1.0						
									Terminated at 1.09 m. Refusal on inferred top of rock.			
						38.0						
						2.0						
						37.0						
						3.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 08/02/26

Checked by: J Riley

End Date: 08/02/26

Location Meth.: dGPS0.1

Driller: Durkin Construction Pty Ltd

Hole Diameter: 300 mm

Easting: 314577.85

RL: 33.57 m

Drill Rig: Utilicor Pavement Truck

Inclination: 90°

Northing: 6237656.98

Ver. Datum: AHD

Total Depth: 0.87 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC				CONCC: 0.00-0.06 m B: 0.06-0.14 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 20 mm, 60 mm thick.			PAVEMENT
				B: 0.14-0.48 m					ASPHALTIC CONCRETE: dark grey, degraded asphalt, aggregates up to 20 mm, 80 mm thick.			
									FILL - Silty Gravelly SAND: fine to medium grained, pale brown; gravel, fine to medium, sub-angular to angular, ironstone and sandstone.			FILL
AD/T	U		Not Encountered	B: 0.48-0.87 m		33.0		SW	Silty Gravelly SAND: fine to coarse grained, pale brown; gravel, fine to coarse, angular, sandstone; with cobbles, angular, <100 mm, sandstone.	M	D to VD	EXTREMELY WEATHERED MATERIAL From 0.48 m: Drillers noted difficulty drilling through cobbles.
									Terminated at 0.87 m. Refusal.			
						1.0						
						32.0						
						2.0						
						31.0						
						3.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: R Rees
Start Date: 08/02/26

Checked by: J Riley
End Date: 08/02/26
Location Meth.: dGPS0.1
RL: 25.31 m
Ver. Datum: AHD
Surface: Pavement

Driller: Durkin Construction Pty Ltd
Drill Rig: Utilicor Pavement Truck
Total Depth: 1.05 m
Hole Diameter: 300 mm
Inclination: 90°
Bearing: N/A

Easting: 314588.61
Northing: 6237777.37
Hor. Datum: MGA2020-56

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
AD/T	U	19 20/20 mm	Not Encountered	CONCC: 0.00-0.10 m	0.0				ASPHALTIC CONCRETE: dark grey, aggregates up to 15 mm, 100 mm thick.	D		PAVEMENT
				B: 0.10-0.18 m					ASPHALTIC CONCRETE: dark grey, degraded asphalt, aggregates up to 15 mm, 80 mm thick.			FILL
				B: 0.18-0.48 m					FILL - Silty Gravelly SAND: fine to medium grained, pale brown; gravel, fine to medium, sub-angular to angular, sandstone.			
				B: 0.48-0.70 m					FILL - Silty Gravelly SAND: fine to coarse grained, pale brown; gravel, fine to coarse, angular, sandstone; with cobbles, <100 mm, angular, sandstone.	M	VD	RESIDUAL SOIL
				B: 0.70-1.05 m					SAND: fine to medium grained, grey-brown; with gravel, fine to medium, angular to sub-angular, sandstone; trace silt.			
		11 11 14 20							at 0.85 m: roots, 5 mm thick.			
									Terminated at 1.05 m. Refusal.			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: R Rees
Start Date: 10/02/26

Checked by: J Riley
End Date: 10/02/26
Location Meth.: dGPS0.1

Driller: Durkin Construction Pty Ltd
Drill Rig: Utilicor Pavement Truck
Total Depth: 0.69 m

Hole Diameter: 300 mm
Inclination: 90°
Bearing: N/A

Easting: 314188.16
Northing: 6238449.54
Hor. Datum: MGA2020-56

RL: 5.38 m
Ver. Datum: AHD
Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
AD/T	U	6 10HB	Not Encountered	CONCC: 0.00-0.17 m B: 0.17-0.33 m	5.0	0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 15 mm, 170 mm thick.	D	D to VD	PAVEMENT
									FILL - Gravelly SAND: fine to medium grained, brown; gravel, fine to coarse, sub-angular to angular, ironstone and sandstone; trace clay.			FILL
								SP	Gravelly SAND: fine to medium grained, pale brown; gravel, fine to coarse, angular, sandstone.			EXTREMELY WEATHERED MATERIAL
									Terminated at 0.69 m. Refusal.			
						1.0						
						4.0						
						2.0						
						3.0						
						3.0						

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 17/02/26

Checked by: J Riley

End Date: 18/02/26

Location Meth.: dGPS0.1

Driller: Durkin Construction Pty Ltd

Hole Diameter: 300 mm

Drill Rig: Utilicor Pavement Truck

Inclination: 90°

Total Depth: 0.98 m

Bearing: N/A

Easting: 314105.35

Northing: 6238543.52

Hor. Datum: MGA2020-56

RL: 14.73 m

Ver. Datum: AHD

Surface: Pavement

[illegible]

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: R Rees
Start Date: 17/02/26

Checked by: J Riley
End Date: 17/02/26
Location Meth.: dGPS0.1
RL: 23.74 m
Ver. Datum: AHD
Surface: Pavement

Driller: Durkin Construction Pty Ltd
Drill Rig: Utilicor Pavement Truck
Total Depth: 1.05 m
Hole Diameter: 300-200 mm
Inclination: 90°
Bearing: N/A

Easting: 314033.69
Northing: 6238618.43
Hor. Datum: MGA2020-56

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC					B: 0.00-0.19 m	0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 30 mm, 190 mm thick.			PAVEMENT
					B: 0.19-0.44 m				FILL - Gravelly Clayey SAND: fine to medium grained, pale brown; gravel, fine to coarse, sub-angular to angular, ironstone and sandstone.			FILL
AD/T	U	22/30 mm	Not Encountered		B: 0.44-1.05 m			SP	Gravelly SAND: fine to coarse grained, pale brown; gravel, fine to coarse, angular to sub-angular, sandstone; with cobbles, >90 mm, angular, sandstone; with clay.	M		RESIDUAL SOIL
						23.0			From 0.8 m: increasing cobble content		VD	At 0.95 m: 300 mm auger refusal, changing to 200 mm. At 1.05 m: Auger refusal, large boulders or possible bedrock.
						1.0			Terminated at 1.05 m. Refusal.			
						22.0						
						2.0						
						21.0						
						3.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Project No: 60731324

Logged by: R Rees

Start Date: 11/02/26

Checked by: J Riley

End Date: 11/02/26

Location Meth.: dGPS0.1

Hole Diameter: 200-300 mm

Easting: 313738.03

RL: 40.75 m

Inclination: 90°

Northing: 6238913.42

Ver. Datum: AHD

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
AD/T	PC		Not Encountered	CONCC: 0.00-0.39 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 20 mm, 130 mm thick.			PAVEMENT
						ASPHALTIC CONCRETE: dark grey, aggregates up to 50 mm, 90 mm thick.						
						ASPHALTIC CONCRETE: grey, aggregates up to 30 mm, 170 mm thick.						
							FILL - Gravelly SAND: brown; gravel, fine to coarse, sub-angular to angular, sandstone, ironstone and shale; trace clay.	M	VD	FILL		
							FILL - Gravelly SAND: fine to medium grained, pale yellow-brown; gravel, fine to medium, sub-angular to angular, sandstone.					
	12			B: 0.39-0.70 m								From 0.70 - 0.85 m: possible reworked natural material into fill.
	16			B: 0.70-0.85 m								
	18			B: 0.85-1.00 m			SW	Gravelly SAND: fine to coarse grained, yellow-brown; gravel, fine to coarse, sub-angular to sub-rounded, sandstone; trace clay.		RESIDUAL SOIL		
	15HB			B: 1.00-2.00 m			SP	SAND: fine grained, yellow-brown; with cobbles, sandstone; with gravel, fine, angular to sub-angular, sandstone.		From 1.10 m: changed from 300 mm auger to 200 mm auger.		
									Terminated at 2.00 m. Target depth.			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 12/02/26

Checked by: J Riley

End Date: 12/02/26

Location Meth.: dGPS0.1

Driller: Durkin Construction Pty Ltd

Hole Diameter: 300 mm

Drill Rig: Utilicor Pavement Truck

Inclination: 90°

Total Depth: 2 m

Bearing: N/A

Easting: 313428.46

Northing: 6239141.77

Hor. Datum: MGA2020-56

RL: 32.02 m

Ver. Datum: AHD

Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)		
PC	U		Not Encountered	CONCC: 0.00-0.40 m	-32.0	0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 20 mm, 225 mm thick.	M		PAVEMENT		
							ASPHALTIC CONCRETE: grey, aggregates up to 15 mm, 170 mm thick.							
4				B: 0.40-0.92 m				FILL - Gravelly SAND: fine to medium grained, brown; gravel, fine to coarse, angular to sub-angular, shale; trace clay.	M		FILL			
16														
17														
9				B: 0.92-1.50 m			CL-CI	CLAY: low to medium plasticity, pale yellow-brown; trace sand; trace gravel, fine.						RESIDUAL SOIL
9														
10														
9														
10														
8														
22/55 mm		B: 1.50-2.00 m				CL-CI	Sandy CLAY: low to medium plasticity, pale red-brown; sand, fine grained; trace gravel, fine, angular, sandstone.	w < PL	VSt - H					
					-30.0	2.0			Terminated at 2.00 m. Target depth.					

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 08/02/26

Checked by: J Riley

End Date: 08/02/26

Location Meth.: Survey

Driller: Durkin Construction Pty Ltd

Hole Diameter: 300-200 mm

Easting: 314586.33

Drill Rig: Utilcor Pavement Truck

Inclination: 90°

Northing: 6237834.61

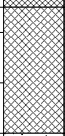
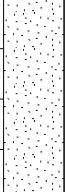
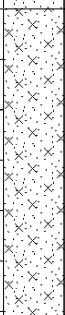
Total Depth: 1.42 m

Bearing: N/A

Hor. Datum: MGA2020-56

Ver. Datum: AHD

Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC				CONCC: 0.00-0.15 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 20 mm, 50 mm thick.			PAVEMENT
									ASPHALTIC CONCRETE: dark grey, aggregates up to 25 mm, 50 mm thick.			
									ASPHALTIC CONCRETE: dark grey, aggregates up to 15 mm, 50 mm thick.			
AD/T	U	22/40 mm	Not Encountered	B: 0.15-0.40 m B: 0.40-0.80 m B: 0.80-1.42 m		21.0			FILL - Clayey Gravelly SAND: fine to medium grained, brown; gravel, fine to coarse, sub-angular to angular, ironstone and sandstone.		D	FILL From 0.15 - 0.4 m: possibly reworked residual soil into fill
								SP	SAND: fine to coarse grained, grey brown; with gravel, fine to medium, angular, sandstone; trace silt.		D - VD	RESIDUAL SOIL
									from 0.6 - 0.9 m: minor cobbles, > 100 mm, sandstone.			From 0.6 - 0.9 m: drilling resistance increased due to presence of cobbles
								SM	Silty SAND: fine to coarse grained, pale brown; with gravel, fine, angular, sandstone.		M	From 0.82 m: 300 mm auger changed to 200 mm auger.
		22/50 mm							Terminated at 1.42 m. Refusal.			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: R Rees
Start Date: 09/02/26

Checked by: J Riley
End Date: 09/02/26
Location Meth.: Survey
RL: 18.43 m
Ver. Datum: AHD
Surface: Pavement

Driller: Durkin Construction Pty Ltd
Drill Rig: Pavement Truck 37
Total Depth: 0.62 m
Hole Diameter: 300 mm
Inclination: 90°
Bearing: N/A

Easting: 314582.32
Northing: 6237874.47
Hor. Datum: MGA2020-56

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC				CONCC: 0.00-0.19 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 15 mm, 190 mm thick.			PAVEMENT
AD/T	U	14	Not Encountered	B: 0.19-0.45 m					FILL - Gravelly SAND: fine to coarse grained, brown; gravel, fine to medium, sub-angular to angular, sandstone; with clay.	M		FILL From 0.19 - 0.45 m: possible reworked residual soil as fill
		20/15 mm		B: 0.45-0.62 m		18.0		SP	Gravelly SAND: medium to coarse grained, pale yellow-brown; gravel, fine to coarse, sub-angular to angular, sandstone; sandstone parent rock.		VD	EXTREMELY WEATHERED MATERIAL
									Terminated at 0.62 m. Refusal.			
						1.0						
						17.0						
						2.0						
						16.0						
						3.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 18/02/26

Checked by: J Riley

End Date: 18/02/26

Location Meth.: Survey

Driller: Durkin Constructikn Pty Ltd

Hole Diameter: 300 mm

Easting: 314255.20

RL: 3.28 m

Drill Rig: Pavement Truck Pty Ltd

Inclination: 90°

Northing: 6238386.13

Ver. Datum: AHD

Total Depth: 1.45 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
AD/T	U	7 20 16 10 9 10 13 6 4 14 12 22	Not Encountered	CONCC: 0.00-0.18 m	0.0				ASPHALTIC CONCRETE: dark grey, aggregates up to 20 mm, 90 mm thick.	D		PAVEMENT
				B: 0.18-0.32 m					ASPHALTIC CONCRETE: dark grey, aggregates up to 15 mm, 30 mm thick.			
				B: 0.32-0.61 m	3.0				ASPHALTIC CONCRETE: grey, aggregates up to 20 mm, 60 mm thick.			FILL
				B: 0.61-0.88 m				SC	FILL - Gravelly SAND: fine to medium grained, yellow-brown; gravel, fine to coarse, sub-angular to angular, sandstone; with clay.	M	D	RESIDUAL SOIL
				B: 0.88-1.45 m				SC	Clayey SAND: fine to medium grained, pale yellow-brown; with gravel, fine to coarse, sub-angular to angular, sandstone.			
								SC	Clayey SAND: fine to coarse grained, dark brown; with gravel, fine to medium, sub-angular to angular, sandstone; trace organic matter, root fibres.			
								SC	Clayey SAND: fine to coarse grained, brown; trace gravel, fine, sub-angular to sub-rounded, sandstone.			
					1.0							
					2.0							
									Terminated at 1.45 m. Refusal.			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 13/02/26

Checked by: J Riley

End Date: 13/02/26

Location Meth.: Survey

Driller: Durkin Construction Pty Ltd

Hole Diameter: 300 mm

Easting: 312716.80

RL: 14.17 m

Drill Rig: Pavement Truck 37

Inclination: 90°

Northing: 6239584.63

Ver. Datum: AHD

Total Depth: 1.65 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC				CONCC: 0.00-0.41 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 20 mm, 100 mm thick.			PAVEMENT
						14.0			ASPHALTIC CONCRETE: grey, aggregates up to 25 mm, 60 mm thick.			
									ASPHALTIC CONCRETE: brown grey, aggregates up to 20 mm, 250 mm thick.			
AD/T	U	11 11 11 22	Not Encountered	B: 0.41-0.75 m B: 0.75-1.35 m B: 1.35-1.60 m					FILL - Gravelly SAND: medium to coarse grained, grey; gravel, fine to coarse, angular, sandstone and shale; with clay.	M		FILL From 0.75 m: possible reworked natural material as fill From 1.60 m : drilling through a log, no progress - refusal
									FILL - Gravelly SAND: fine to coarse grained, dark brown; gravel, fine to coarse, sub-angular to angular, shale; trace clay.			
						1.0			FILL - Clayey Gravelly SAND: fine to coarse grained, brown; gravel, fine to medium, sub-angular, shale and sandstone.			
						13.0			at 1.52 m: with fibrous roots, 10 - 30 mm			
									Terminated at 1.65 m. Refusal.			
						2.0						
						12.0						
						3.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: R Rees
Start Date: 20/02/26

Checked by: P Waddel
End Date: 20/02/26
Location Meth.: Survey
RL: 35.2 m
Ver. Datum: AHD
Surface: Pavement

Driller: Durkin Construction Pty Ltd
Drill Rig: Pavement Truck 37
Total Depth: 1.38 m
Hole Diameter: 300 mm
Inclination: 90°
Bearing: N/A

Easting: 314580.41
Northing: 6237631.10
Hor. Datum: MGA2020-56

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC				CONCC: 0.00-0.16 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 25 mm, 160 mm thick.			PAVEMENT
AD/T	U			B: 0.16-0.62 m		35.0			FILL - Gravelly SAND: fine to coarse grained, brown; gravel, fine to coarse, angular to sub-angular, sandstone and basalt; trace clay; basecourse.	D		FILL
		8		B: 0.62-0.84 m					FILL - SAND: medium to coarse grained, yellow-brown; with gravel, fine to medium, sub-angular to sub-rounded, sandstone, charcoal and basalt.		D	
		8										EXTREMELY WEATHERED MATERIAL
		9		B: 0.84-1.38 m				SP	SAND: medium to coarse grained, yellow-brown; trace gravel, fine to medium, sub-angular, sandstone.	M	VD	
		22				1.0						
						34.0						
									Terminated at 1.38 m.			
						2.0						
						33.0						
						3.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 10/02/26

Checked by: J Riley

End Date: 10/02/26

Location Meth.: Survey

Driller: Durkin Construction Pty Ltd

Hole Diameter: 200-300 mm

Easting: 314346.14

Northing: 6238287.17

RL: 1.95 m

Ver. Datum: AHD

Total Depth: 2 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC				CONCC: 0.00-0.26 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 20 mm, 120 mm thick.			PAVEMENT
				B: 0.26-0.42 m					ASPHALTIC CONCRETE: dark grey, aggregates up to 30 mm, 70 mm thick.			
				B: 0.42-0.64 m					ASPHALTIC CONCRETE: dark grey, aggregates up to 20 mm, 70 mm thick.			
		9		B: 0.64-1.23 m					FILL - Gravelly SAND: fine to coarse grained, pale yellow-brown; gravel, fine to coarse, sub-angular to angular, ironstone and sandstone; trace clay.	D		FILL
		9										At 0.45 m: additional layer in photograph is actually just a slightly damper version of this layer of fill.
		22						SW	SAND: fine to coarse grained, pale yellow-brown; with gravel, fine to medium, angular to sub-angular, sandstone; trace clay.			RESIDUAL SOIL
						1.0						
						1.0						
				B: 1.23-2.00 m				SP	SAND: medium grained, dark brown; with gravel, fine to medium, angular to sub-angular, sandstone; trace clay; trace cobbles, >100 mm, sandstone.	M		
		3										From 1.2 m: drillers noted drilling through cobbles. From 1.3 m: changed from 300 mm auger to 200 mm auger
		6										
		20										
		21										
		22/90 mm										
						-0.0						
						2.0			Terminated at 2.00 m. Target depth.			
						-1.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Project No: 60731324

Logged by: R Rees

Start Date: 12/02/26

Checked by: J Riley

End Date: 12/02/26

Location Meth.: dGPS0.1

Hole Diameter: 300 mm

Easting: 313656.94

Inclination: 90°

Northing: 6238987.00

RL: 40.005 m

Ver. Datum: AHD

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)	
PC	U		Not Observed	CONCC: 0.00-0.36 m	40.0	0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 15 mm, 45 mm thick.			PAVEMENT	
									ASPHALTIC CONCRETE: dark grey, aggregates up to 30 mm, 165 mm thick.				
									Heavily Bound: brown-grey, aggregates up to 20 mm, 180 mm thick.				
									SANDSTONE: fine to medium grained, pale grey-white mottled pale red-brown, very low to low strength.				
Terminated at 0.39 m.									D		At 0.36 m: Jackhammer used HAWKESBURY SANDSTONE		
					39.0	1.0							
					38.0	2.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 12/02/26

Checked by: J Riley

End Date: 12/02/26

Location Meth.: dGPS0.1

Driller: Durkin Construction Pty Ltd

Hole Diameter: 300 mm

Easting: 313408.33

Drill Rig: Utilicor Pavement Truck

Inclination: 90°

Northing: 6239153.46

Total Depth: 2 m

Bearing: N/A

Hor. Datum: MGA2020-56

Ver. Datum: AHD

Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC				CONCC: 0.00-0.39 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 40 mm, 225 mm thick..			PAVEMENT
									Heavily Bound: brown-grey, aggregates up to 30 mm, 165 mm thick.			
AD/T	U	3 8 15 22/60 mm	Not Observed	B: 0.39-0.52 m B: 0.52-0.80 m B: 0.80-1.03 m B: 1.03-1.49 m B: 1.49-2.00 m					FILL - Gravelly SAND: fine grained, orange-brown; gravel, fine to medium, subangular, ironstone, sandstone and shale; with clay; trace roots.	D		FILL
									Gravelly SAND: fine to medium grained, orange-brown; gravel, fine to coarse, angular to subangular, sandstone; with clay.			
									Sandy Gravelly CLAY: low to medium plasticity, orange-brown; sand, fine grained; gravel, fine to medium, sub angular, sandstone.			
									Gravelly Clayey SAND: fine grained, pale orange-brown; gravel, fine to coarse, angular to subangular, ironstone.			
									Sandy CLAY: low plasticity, pale grey-brown; sand, fine grained.			
									Terminated at 2.00 m.			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: R Rees
Start Date: 12/02/26

Checked by: J Riley
End Date: 12/02/26
Location Meth.: dGPS0.1
Ver. Datum: AHD
Surface: Pavement

Driller: Durkin Construction Pty Ltd
Drill Rig: Utilicor Pavement Truck
Total Depth: 2 m

Hole Diameter: 300 mm
Inclination: 90°
Bearing: N/A

Easting: 313236.94
Northing: 6239264.53
Hor. Datum: MGA2020-56

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC	U	6 7 8 2 3 7 10 4 8 14 4 22	Not Observed	CONCC: 0.00-0.20 m	0.00	0.0		CI	ASPHALTIC CONCRETE: dark grey, aggregates up to 20 mm, 200 mm thick.	M	St - VSt	PAVEMENT
				B: 0.20-0.40 m					FILL - Clayey Gravelly SAND: fine to medium grained, brown; gravel, fine to medium, subangular to angular, ironstone, sandstone and shale.			FILL
				B: 0.40-0.75 m					FILL - Clayey SAND: fine to medium grained, brown mottled orange; with gravel, fine to coarse, angular, ironstone and sandstone.			
				B: 0.75-1.50 m				CI	CLAY: medium plasticity, orange-brown; with sand, fine grained; with gravel, fine to coarse, angular to subangular, ironstone.	w ≈ PL	St - VSt	RESIDUAL SOIL
AD/T	U	Not Observed	Not Observed	B: 1.50-2.00 m				SC	Clayey SAND: fine to medium grained, pale orange-brown mottled grey; trace gravel, fine to medium, angular, ironstone.	M	VD	
									Terminated at 2.00 m.			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: R Rees
Start Date: 15/02/26

Checked by: J Riley
End Date: 15/02/26
Location Meth.: dGPS0.1
RL: 18.626 m
Ver. Datum: AHD
Surface: Pavement

Driller: Durkin Construction Pty Ltd
Drill Rig: Utilicor Pavement Truck
Total Depth: 0.51 m
Hole Diameter: 300 mm
Inclination: 90°
Bearing: N/A

Easting: 312805.18
Northing: 6239546.24
Hor. Datum: MGA2020-56

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC	U		Not Observed	CONCC: 0.00-0.39 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 35 mm, 220 mm thick.			PAVEMENT
									Heavily Bound: grey, aggregates up to 20 mm, 190 mm thick.			At 0.39 m: Jackhammer used to recover remaining pavement
AD/T				B: 0.41-0.51 m					FILL - Gravelly SAND: fine to coarse grained, orange-brown; gravel, fine to coarse, angular to sub-angular, sandstone and basalt.	M		FILL
									Terminated at 0.51 m.			
						18.0						
						1.0						
						17.0						
						2.0						
						16.0						
						3.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: R Rees
Start Date: 13/02/26

Checked by: J Riley
End Date: 13/02/26
Location Meth.: dGPS0.1
Ver. Datum: AHD
Surface: Pavement

Driller: Durkin Construction Pty Ltd
Drill Rig: Utilicor Pavement Truck
Total Depth: 1.7 m

Hole Diameter: 300 mm
Inclination: 90°
Bearing: N/A

Easting: 312635.78
Northing: 6239630.49
Hor. Datum: MGA2020-56

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC				CONCC: 0.00-0.39 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 25 mm, 150 mm thick..			PAVEMENT
									Heavily Bound: brown-grey, aggregates up to 30 mm, 240 mm thick..			
AD/T	U	6 10 12 22 21 18 11 18HB	Not Observed	B: 0.39-0.72 m					FILL - Gravelly SAND: fine to coarse grained, grey; gravel, fine to coarse, sub-angular to angular, sandstone and shale; with clay.	M		FILL
				B: 0.72-1.30 m					FILL - Gravelly SAND: fine to medium grained, brown mottled grey; gravel, fine to coarse, angular to sub-angular, ironstone; with clay.			
				B: 1.30-1.45 m			SC		Clayey SAND: fine to medium grained, red-brown; trace gravel, fine to medium.		VD	RESIDUAL SOIL
				B: 1.45-1.70 m					from 1.45 m: becoming red-brown mottled grey			
									Terminated at 1.70 m.			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW

Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design

Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324

Logged by: R Rees

Start Date: 16/02/26

Checked by: J Riley

End Date: 16/02/26

Location Meth.: dGPS0.1

Driller: Durkin Construction Pty Ltd

Hole Diameter: 300 mm

Easting: 313707.55

RL: 36.166 m

Drill Rig: Utilicor Pavement Truck

Inclination: 90°

Northing: 6239091.01

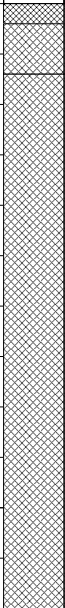
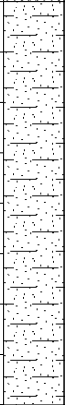
Ver. Datum: AHD

Total Depth: 2 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Pavement

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
PC	U	6	Not Observed	CONCC: 0.00-0.04 m B: 0.04-0.14 m B: 0.14-1.20 m	36.0	0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 15 mm, 40 mm thick.	M		PAVEMENT
									Clayey Gravelly SAND: fine to medium grained, brown; gravel, fine to coarse, angular to subangular, sandstone and basalt.			FILL
ADT	U	11	Not Observed	B: 1.20-2.00 m	35.0			SC	Clayey SAND: fine to medium grained, pale brown.	F to St		EXTREMELY WEATHERED MATERIAL
		14										
		7										
		14										
		11										
		18HB										
						2.0			Terminated at 2.00 m. Target depth.			
						34.0						
						3.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Transport for NSW
Project: Heathcote Road, New Illawarra Road to the Avenue - Concept Design
Location: Heathcote Road, Holsworthy, NSW 2173

Project No: 60731324
Logged by: R Rees
Start Date: 10/02/26
Checked by: J Riley
End Date:
Location Meth.: dGPS0.1
RL: 25.553 m
Ver. Datum: AHD
Surface: Pavement

Driller: Durkin Construction Pty Ltd
Drill Rig: Utilicor Pavement Truck
Total Depth: 0.19 m
Hole Diameter: 300 mm
Inclination: 90°
Bearing: N/A

Easting: 314629.28
Northing: 6237897.59
Hor. Datum: MGA2020-56

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
CC				CONCC: 0.00-0.06 m B: 0.06-0.19 m		0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 30 mm.			PAVEMENT
AD/T	U		Not Observed						FILL - Gravelly SAND: fine to coarse grained, brown; gravel, angular to sub-angular, sandstone and ironstone; trace clay.	M		FILL At 0.12 m: Jackhammer used.
									Terminated at 0.19 m. Refusal on inferred top of rock.			
						25.0						
						1.0						
						24.0						
						2.0						
						23.0						
						3.0						